

Project Data Page:

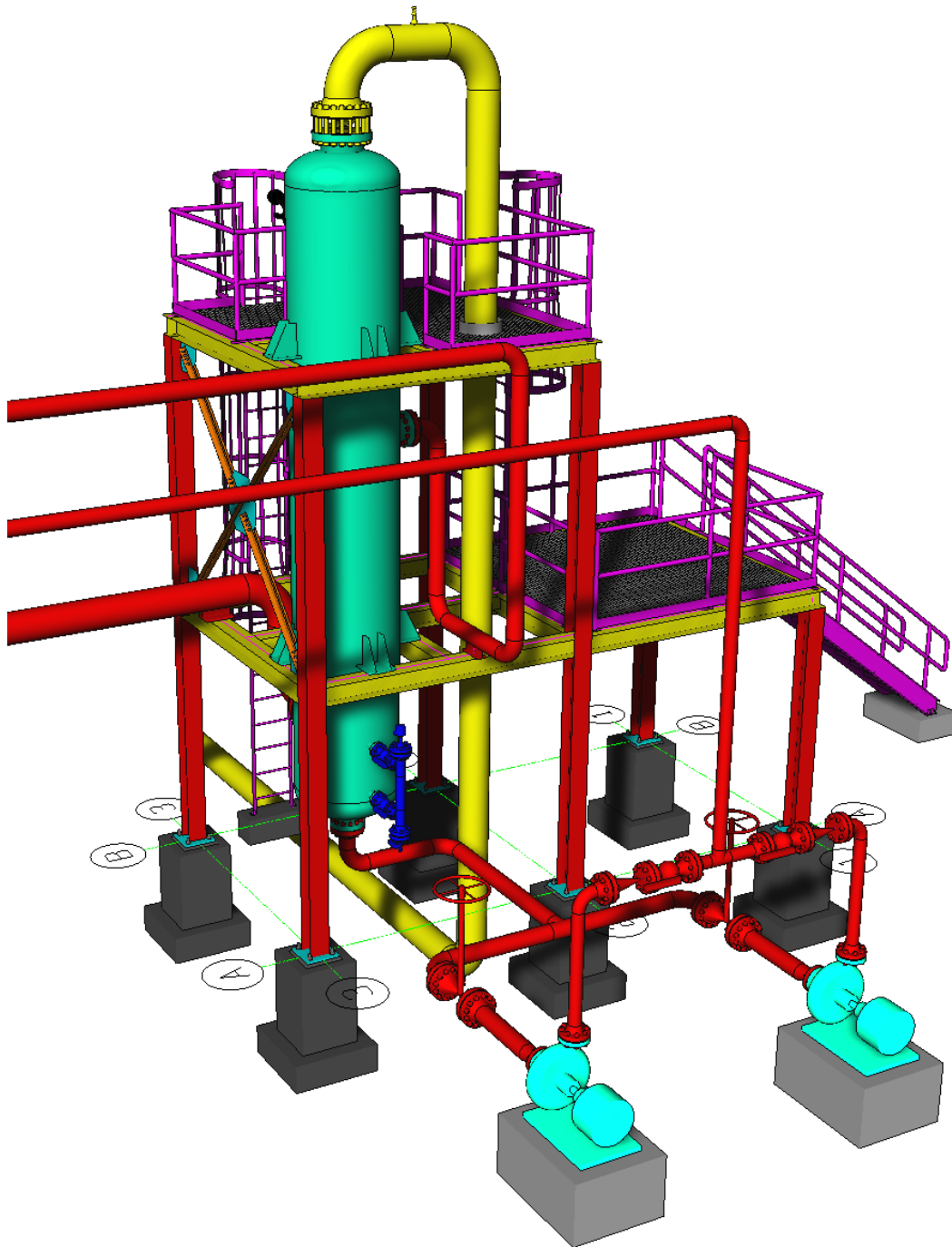




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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\198-CChannel side MOC= CS.pvdb

Date of Analysis : Sep 9, 2026 10:19am

PV Elite 28, January 2026



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 FileName : 198-C_Channel side MOC= CS Page 2 of 265
 Vessel Design Summary: Step: 35 10:19am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 2720.000 mm ID
 Vessel Design Length, Tangent to Tangent 8770.05 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 230 °C
 Channel Side Design Temperature 82 °C
 Shell Side Design Pressure 2500.000 KPa.
 Channel Side Design Pressure 1000.000 KPa.
 Shop Shell Side Test Pressure 3368.346 KPa.
 Field Shell Side Test Pressure 3250.000 KPa.
 Shop Channel Side Test Pressure 1352.241 KPa.
 Field Channel Side Test Pressure 1300.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 70	...	...	K02700	
Head	SA-516 70	...	...	K02700	
Flange	SA-182 F60	...	...	S32205	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Nozzle Flg	SA-182 F22	3	...	K21590	
Tubes	SA-789	...	...	S32205	
Tubesheet	SA-182 F60	...	...	S32205	
Flo. Head	SA-240	...	...	S32205	
Bk. Bolting	SA-193 B7M	...	<= 100	G41400	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Ch. cover	1029.602	1.01	1135.000	3.0000	N/A	N/S
LHS CH FL	1026.965	1.01	1175.000	0.0000	N/A	No
Ch.shell	1026.691	1.01	1082.000	3.0000	N/A	No
RHS CH FL	1026.965	1.01	1081.000	3.0000	N/A	No
Shell flg-1	2526.966	0.00	2594.000	1.0000	N/A	No
Shell	2526.671	0.00	2530.000	1.0000	N/A	No
shell flg-3	2526.966	0.00	2587.000	1.0000	N/A	No
shell flg-2	2526.966	0.00	2587.000	1.0000	N/A	No
Shell -2	2526.671	0.00	2530.000	1.0000	N/A	No



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head- front	2526.671	0.00	2513.000	1.0000	No	No
-------------	----------	------	----------	--------	----	----

N/S in the Creep Range column specifies that the Creep Range Temperature was not specified in the ASME Code Section II Part D for this material.

Liquid Level: 3020.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	136.3	130.0	138.0	130.0	122.4	45.5	1.00	1.00
Body Flg	396.3	260.0	218.0	210.0	196.9	191.3	1.00	1.00
Cylinder	1896.3	1500.0	14.0	14.0	13.2	4.1	1.00	1.00
Body Flg	2230.7	328.0	308.0	300.0	297.0	293.6	1.00	1.00
Body Flg	2741.7	336.0	288.0	280.0	271.1	264.0	1.00	1.00
Cylinder	7341.7	4600.0	28.0	26.5	26.2	...	1.00	1.00
Body Flg	7684.1	336.0	288.0	280.0	267.7	258.0	1.00	1.00
Body Flg	8020.1	336.0	288.0	280.0	268.9	257.1	1.00	1.00
Cylinder	8720.1	700.0	28.0	26.5	26.2	...	1.00	1.00
Ellipse	8770.1	50.0	36.0	30.5	30.4	...	0.85	0.85

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
Tube inlet	WRC-297	1.283	Shell	Failed
Tube outlet	WRC-297	1.290	Shell	Failed
Shell inlet	WRC-297	0.971	Shell	Passed
Shell outlet	WRC-297	0.986	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	625.000 mm
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	2100.000 mm
Wear Pad Width	950.000 mm
Wear Pad Thickness	24.000 mm
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	920.000 mm
Baseplate Length	2350.000 mm
Baseplate Thickness	56.000 mm
Baseplate Width	650.000 mm
Number of Ribs (including outside ribs)	6
Rib Thickness	24.000 mm
Web Thickness	24.000 mm
Height of Center Web	700.000 mm
Number of Bolts in Baseplate	10

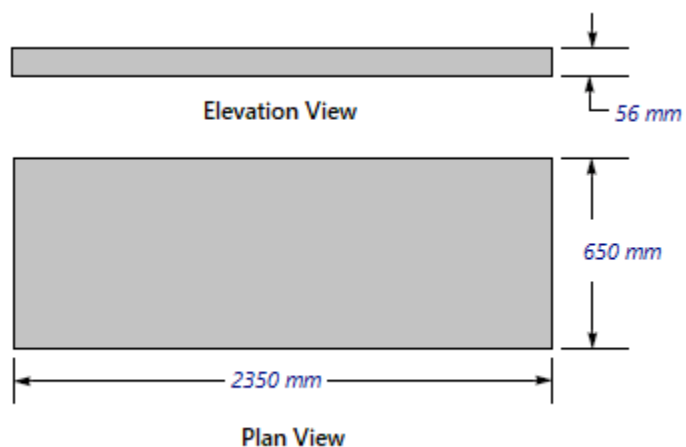
Baseplate Sketch

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 FileName : 198-C_Channel side MOC= CS
 Vessel Design Summary:

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Maximum Tensile Bolt Load 996. Kgf

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	116876.18	Kgf
Maximum Transverse Saddle Shear Load	5735.96	Kgf
Maximum Longitudinal Saddle Shear Load	9045.91	Kgf

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	102836.30	Kgf
Maximum Transverse Saddle Shear Load	314.27	Kgf
Maximum Longitudinal Saddle Shear Load	373.66	Kgf

Weights:

Fabricated	- Bare w/o removable internals	108635.8	kg.
Shop Test	- Fabricated + water (full)	156621.5	kg.
Shipping	- Fab. + rem. intls.+ shipping app.	108635.8	kg.
Erected	- Fab. + rem. intls.+ insul. (etc)	108635.8	kg.
Empty	- Fab. + intls. + details + wghts.	110997.7	kg.
Operating	- Empty + operating liquid (no ca)	159203.8	kg.
Field Test	- Empty weight + test medium (full)	158131.2	kg.

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 Nozzle Summary: Step: 33 10:19am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
Tube inlet	1082	OK	...	OK	8.33	OK	Passed
Tube outlet	1082	OK	...	OK	8.33	OK	Passed
Shell inlet	2530	...	...	OK	11.33	OK	Passed
Shell outlet	2530	...	...	OK	11.33	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2530 Nozzle : Shell inlet [Shellside]
 Minimum MAWP Nozzles : 1082 Nozzle : Tube inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	Tube inlet	1146.350	0.000	635.000
30	Tube outlet	1146.350	180.000	635.000
60	Shell inlet	5041.700	0.000	1536.000
60	Shell outlet	5041.700	180.000	1161.401

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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 Nozzle Schedule: Step: 32 10:19am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
Tube inlet	14.000 in	80	WNF	14.000	19.050	755.60	16.00	375.67	150
Tube outlet	14.000 in	80	WNF	14.000	19.050	755.60	16.00	375.67	150
Shell outlet	24.000 in	40	WNF	24.000	17.450	1109.60	28.00	411.10	300
Shell inlet	30.000 in Actual		WNF	31.969	25.000	1382.00	30.00	438.52	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Tube inlet	SA-106 B	14.000	15.000	12.000	16.000	...
Tube outlet	SA-106 B	14.000	15.000	12.000	16.000	...
Shell outle	SA-106 B	28.500	20.000	22.000	28.000	...
Shell inlet	SA-106 B	28.500	20.000	22.000	30.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Tube inlet	1146.350	0.0	350.00	0.00	Ch.shell
Tube outlet	1146.350	180.0	350.00	0.00	Ch.shell
Shell outlet	5041.700	180.0	350.00	0.00	Shell
Shell inlet	5041.700	0.0	350.00	0.00	Shell

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 Bill of Material: Step: 31 10:19am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 138.0mm THK X 3000.0mm OD	SA-182 F60
1	WELD NECK FLANGE: 218.0mm THK X 3000.0mm OD	SA-266 2
1	CYLINDER: 14.0mm THK X 2720.0mm ID X 1500.0mm	SA-516 70
1	WELD NECK FLANGE: 308.0mm THK X 3140.0mm OD	SA-266 2
1	WELD NECK FLANGE: 288.0mm THK X 3144.0mm OD	SA-266 2
2	WELD NECK FLANGE: 288.0mm THK X 3125.0mm OD	SA-266 2
1	CYLINDER: 28.0mm THK X 2720.0mm ID X 700.0mm	SA-516 70
1	ELLIPTICAL HEAD: 2.0 X 1, 36.0mm THK X 2720.0mm ID X 50.0mm	SA-516 70
1	TUBESHEET: 2806.0mm X 175.0mm THK	SA-182 F60
5600	TUBES: 5350.0mm X 25.4mm DIA X 2.1mm THK	SA-789
1	FLOATING TUBESHEET: 2710.0mm X 175.0mm THK	SA-182 F60
52	FLOATING HEAD FLANGE BOLTS: 38.1mm DIA	SA-193 B7M
104	NUTS FOR FLOATING HEAD FLANGE BOLTS: 38.1mm DIA	...
1	FLOATING HEAD FLANGE GASKET: 2597.0mm OD X 2571.0mm ID	...
1	INSULATION: 345mm X 100mm THK	Insulation
1	LINING: 356.000mm X 6.000mm THK	...
2	SADDLE: 625.0mm X 120 DEG	SA-516 70
1	INSULATION: 4600mm X 100mm THK	Hot Insulation
1	LINING: 4600.000mm X 3.000mm THK	...
2	INSULATION: 350mm X 100mm THK	Insulation
1	LINING: 396.000mm X 6.000mm THK	...
1	LINING: 360.000mm X 6.000mm THK	...
1	INSULATION: 700mm X 100mm THK	Hot Insulation
1	LINING: 700.000mm X 3.000mm THK	...
1	INSULATION: 619mm X 100mm THK	Hot Insulation
1	LINING: 730.000mm X 3.000mm THK	...
1	GASKET: 2796.0mm OD X 2752.0mm ID	...
32	BODY FLANGE BOLTS: 47.6mm DIA	SA-193 B7
64	NUTS FOR BODY FLANGE BOLTS: 47.6mm DIA	...
1	GASKET: 2794.0mm OD X 2752.0mm ID	...



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Bill of Material:

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64	BODY FLANGE BOLTS: 69.8mm DIA	SA-193 B7
128	NUTS FOR BODY FLANGE BOLTS: 69.8mm DIA	...
1	GASKET: 2800.0mm OD X 2756.0mm ID	...
1	CYLINDRICAL SEGMENT 28.0mm THK X 3370.5mm X 4600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 3346.8mm X 4600.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 28.0mm THK X 1827.7mm X 4600.0mm	SA-516 70
1	NAMEPLATE	...

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 MDMT Summary:

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Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Shell flg-1	[11]	B	2	-2		0.929	28.000	1.00	No
shell flg-3	[11]	B	2	-2		0.929	28.000	1.00	No
Bolting	[21]		-40						
shell flg-2	[11]	B	2	-2		0.929	28.000	1.00	No
Shell	[8]	B	1	0		0.983	26.500	1.00	No
Shell -2	[8]	B	1	0		0.983	26.500	1.00	No
head- front	[10]	B	4	-4		0.841	30.500	0.85	No
head- front	[7]	B	9	-7		0.717	36.000	0.85	No
Left Saddle f	[24]	B	-2	-3	-29	0.983	24.000	1.00	No
RH floating	[24]	B	-2	-3	-29	0.983	24.000	1.00	No
Shell inlet	[1]	B	1	-1		0.978	26.500	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Shell outlet	[1]	B	1	0		0.988	26.500	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Warmest MDMT:			9	0					
LHS CH FL	[11]	B	-18	-26	-29	0.869	14.000	1.00	No
RHS CH FL	[11]	B	-18	-32	-29	0.753	14.000	1.00	No
Ch.shell	[8]	B	-18	-20	-29	0.959	14.000	1.00	No
Tube inlet	[1]	B	-14	-29	-29	0.902	16.000	1.00	No
Nozzle Flg	[5]	A	-18	-48					
Tube outlet	[1]	B	-14	-29	-29	0.926	16.000	1.00	No
Nozzle Flg	[5]	B	-18	-47					
Bolting	[21]		-48						
Bolting	[21]		-40						
Warmest MDMT:			-14	-20					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	0.0	-12.0	Fail
Channel/Tube	-29.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld



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MDMT Summary:

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UG-84(b)(2) was not considered.
UCS-66(g) was not considered.
UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
UCS-66(i) applies to impact tested materials not by specification and
UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

Warning: RHS CH FL and Shell flg-1

There are mating flanges in this model that have different design pressures. This will generate different bolt loads. If a tubesheet is in between these flanges, PV Elite will compute the mating flange loads for you. If there is not a tubesheet physically separating these flanges, you must enter in the mating flange loads into the flange dialog.

Warning:

There is a flange in this model connected to an element whose temperature is different. The bolt allowable stress is based on it's parent design temperature. You may need to manually adjust the operating bolt allowable stress to a value based upon a representative maximum temperature the bolts will be subject to.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.



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Warnings and Errors:

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 Input Echo: Step: 1 10:19am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	2500	KPa.
Channel Side Design Pressure	999.99	KPa.
Shell Side Design Temperature	230.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Use ASME VIII-1 Appendix 1-9 N
 Perform Blast Load Analysis No
 Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes
 Do not use Nozzle MDMT Interpretation VIII-1 01-37 No
 Use Table G instead of exact equation for "A" Yes
 Shell Head Joints are Tapered Yes
 Compute "K" in corroded condition Yes
 Use Code Case 2286 No
 Use the MAWP to compute the MDMT Yes
 For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes
 For PWHT & P1 Materials the MDMT can be < -55F (-48C) No
 Using Metric Material Databases, ASME II D Yes
 Calculate B31.3 type stress for Nozzles with Loads Yes
 Reduce the MDMT due to lower membrane stress Yes
 Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/10
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Ch. cover
Distance "FROM" to "TO"	130 mm
Flange Outside Diameter	3000 mm
Element Thickness	130 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	138 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	1.013 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-182 F60
Allowable Stress, Ambient	187 N./mm ²
Allowable Stress, Operating	187 N./mm ²
Allowable Stress, Hydrotest	431.99 N./mm ²
Material Density	0.007806 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	437.34 N./mm ²
External Pressure Chart Name	HA-5
UNS Number	S32205
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	3020 mm
Liquid Density	0.0009984 kg./cm ³

Element#	2/10
Element From Node	20
Element To Node	30



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Element Type	Flange
Description	LHS CH FL
Distance "FROM" to "TO"	260 mm
Flange Inside Diameter	2720 mm
Element Thickness	210 mm
Internal Corrosion Allowance	0 mm
Nominal Thickness	218 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	1.013 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2751 mm
Liquid Density	0.0009984 kg./cm ³

Element#	3/10
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Ch.shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	2720 mm
Element Thickness	14 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	1.013 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	242.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0



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Weld is pre-Heated	No
Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2720 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	30
Detail Type	Nozzle
Detail ID	Tube inlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	14 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	153.3 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	30
Detail Type	Nozzle
Detail ID	Tube outlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	14 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	153.3 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B
Element From Node	30
Detail Type	Weight
Detail ID	Wght:[1 of 1]
Dist. from "FROM" Node / Offset dist	750 mm
Miscellaneous Weight	330 Kgf
Offset from Element Centerline	0 mm

Element#	4/10
Element From Node	40
Element To Node	50
Element Type	Flange
Description	RHS CH FL
Distance "FROM" to "TO"	328 mm
Flange Inside Diameter	2720 mm
Element Thickness	300 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	308 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1000 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	1.013 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³



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P Number Thickness	32.004	mm
Yield Stress, Operating	230.08	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K03506	
Product Form	Forgings	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2751	mm
Liquid Density	0.0009984	kg./cm ³

Element#	5/10	
Element From Node	50	
Element To Node	60	
Element Type	Flange	
Description	Shell flg-1	
Distance "FROM" to "TO"	336	mm
Flange Inside Diameter	2724	mm
Element Thickness	280	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	288	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2500	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2751	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	345	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Lining	
Detail ID	ss clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	356	mm
Thickness of Lining	6	mm
Density	0.008	kg./cm ³

Element#	6/10
Element From Node	60



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Element To Node	70
Element Type	Cylinder
Description	Shell
Distance "FROM" to "TO"	4600 mm
Inside Diameter	2720 mm
Element Thickness	26.5 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	28 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2500 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	220 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	60
Detail Type	Saddle
Detail ID	Left Saddle fix
Dist. from "FROM" Node / Offset dist	920 mm
Width of Saddle	625 mm
Height of Saddle at Bottom	2100 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	950 mm
Thickness of Wear Plate	24 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm

Element From Node	60
Detail Type	Saddle
Detail ID	RH floating
Dist. from "FROM" Node / Offset dist	3680 mm
Width of Saddle	625 mm
Height of Saddle at Bottom	2100 mm
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	950 mm
Thickness of Wear Plate	24 mm
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.30000001
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm



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Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2720 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	60
Detail Type	Insulation
Detail ID	Ins: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	4600 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	60
Detail Type	Lining
Detail ID	SS clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	4600 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

Element From Node	60
Detail Type	Nozzle
Detail ID	Shell inlet
Dist. from "FROM" Node / Offset dist	2300 mm
Nozzle Diameter	30 in.
Nozzle Schedule	None
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	659.84 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Nozzle
Detail ID	Shell outlet
Dist. from "FROM" Node / Offset dist	2300 mm
Nozzle Diameter	24 in.
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	499.38 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	60
Detail Type	Weight
Detail ID	Fireproofing
Dist. from "FROM" Node / Offset dist	920 mm
Miscellaneous Weight	520 Kg
Offset from Element Centerline	2100 mm

Element From Node	60
Detail Type	Weight
Detail ID	Fire proofing
Dist. from "FROM" Node / Offset dist	3680 mm
Miscellaneous Weight	520 Kg
Offset from Element Centerline	2100 mm



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Element#	7/10
Element From Node	70
Element To Node	80
Element Type	Flange
Description	shell flg-3
Distance "FROM" to "TO"	336 mm
Flange Inside Diameter	2724 mm
Element Thickness	280 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	288 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2500 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	136.8 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	208 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2751 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	350 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	70
Detail Type	Lining
Detail ID	ss Clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	396 mm
Thickness of Lining	6 mm
Density	0.008 kg./cm ³

Element#	8/10
Element From Node	80
Element To Node	90
Element Type	Flange
Description	shell flg-2
Distance "FROM" to "TO"	336 mm
Flange Inside Diameter	2720 mm
Element Thickness	280 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	288 mm



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External Corrosion Allowance	0	mm
Design Internal Pressure	2500	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-266 2	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	

Element From Node	80	
Detail Type	Liquid	
Detail ID	Liquid: 80	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	2751	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	80	
Detail Type	Insulation	
Detail ID	Insulation	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	350	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	80	
Detail Type	Lining	
Detail ID	ss clad	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Lining	360	mm
Thickness of Lining	6	mm
Density	0.008	kg./cm ³

Element#	9/10	
Element From Node	90	
Element To Node	100	
Element Type	Cylinder	
Description	Shell -2	
Distance "FROM" to "TO"	700	mm
Inside Diameter	2720	mm
Element Thickness	26.5	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	28	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	2500	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	0	KPa.
Design Temperature External Pressure	25	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	220	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Weld is pre-Heated	No	



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Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2720 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	90
Detail Type	Insulation
Detail ID	Ins: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	700 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	90
Detail Type	Lining
Detail ID	Ss clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	700 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

Element#	10/10
Element From Node	100
Element To Node	110
Element Type	Elliptical
Description	head- front
Distance "FROM" to "TO"	50 mm
Inside Diameter	2720 mm
Element Thickness	30.5 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	36 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	2500 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	25 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	100
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2720 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	100
Detail Type	Insulation
Detail ID	Ins: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	620 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element From Node	100
Detail Type	Lining



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Detail ID	SS clad
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	730 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³

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XY Coordinate Calculations:

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Ch. cover		136.35	...	130	...
LHS CH FL		396.35	...	260	...
Ch.shell		1896.35	...	1500	...
RHS CH FL		2230.7	...	328	...
Shell flg-1		2741.7	...	336	...
Shell		7341.7	...	4600	...
shell flg-3		7684.05	...	336	...
shell flg-2		8020.05	...	336	...
Shell -2		8720.05	...	700	...
head- front		8770.05	...	50	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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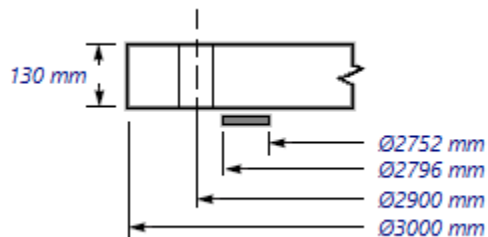
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 Flg Calc [Int P]: New Flange Flng: 31 10:19am Sep 9,2026

Flange Input Data Values

Description: New Flange :

Ch. cover

Flange Type		Blind	
Design Pressure	P	1029.60	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	130.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS number		S32205	
Flange Allowable Stress At Temperature	Sfo	187.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	187.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	9110.619	mm
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		32	
Flange Face Outside Diameter	Fod	2807.000	mm
Flange Face Inside Diameter	Fid	2720.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2796.000	mm
Gasket Inside Diameter	Gi	2752.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2725.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Gasket Contact Width,	$N = (Go - Gi)/2$	22.000 mm
Basic Gasket Width,	$bo = N/2$	11.000 mm



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Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 8.358 mm
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 2779.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 2779.2847^2 \cdot 1029.602$
 $= 636974.438 \text{ Kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot lp \cdot bPart \cdot mPart \cdot P$
 $= 2 \cdot 8.3576 \cdot \pi \cdot 2779.2847 \cdot 3.0 \cdot 1029.6$
 $+ 2 \cdot 2725.0 \cdot 5.0 \cdot 3.0 \cdot 1029.6022$
 $= 54554.242 \text{ Kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(636974 + 54554 + 0, 0)$
 $= 691528.688 \text{ Kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 68.95 \cdot 8.3576 \cdot \pi \cdot 2779.285 + 68.95 \cdot 5.0 \cdot 2725.0$
 $= 608859.188 \text{ Kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$
 $= \max((691529 + 0 + 4 \cdot 0 / 2779.28) / 172, 608859 / 172)$
 $= 394.292 \text{ cm}^2$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	394.292	475.664	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	284.250	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$
 $= 475.664 \cdot 172.0 / (68.95 \cdot \pi (2796.0 + 2752.0))$
 $= 6.808 \text{ mm}$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 $= Sa \cdot Ab$
 $= 172.0 \cdot 475.6635$
 $= 834241.56 \text{ Kgf}$

Gasket Load for the Operating Condition [HG]:
 $= Wo$
 $= 691528.69 \text{ Kgf}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2900.0 - 2779.2847) / 2$
 $= 60.3577 \text{ mm}$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:
 $= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$



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$$= 1.9 \cdot 691529 \cdot 60.3577 \cdot 1.0 / (127.0^2 \cdot 2779.2847) + 0.3 \cdot 1.0 \cdot 1029.6 \cdot 2779.2847^2 / 127.0^2$$

$$= 165.29 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 834242 \cdot 60.3577 \cdot 1.0 / (127.0^2 \cdot 2779.2847)$$

$$= 20.93 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 691529 / 475.6635$$

$$= 142.57 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (608859 / 475.6635)$$

$$= 125.53 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	165.29	187.00	20.93	187.00
Bolting	142.57	172.00	125.53	172.00

Reqd. Blind Flange Thickness at Center 122.400 mm
 Reqd. Blind Flange Thickness at Gasket 45.488 mm

Estimated M.A.W.P. (Operating) 1165 KPa.
 Estimated Finished Weight of Flange at given Thk. 7172.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 7172.9 kg.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	157.75	187.00	19.97	187.00
Bolting	142.57	172.00	125.53	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 119.405 mm

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Flange Input Data Values

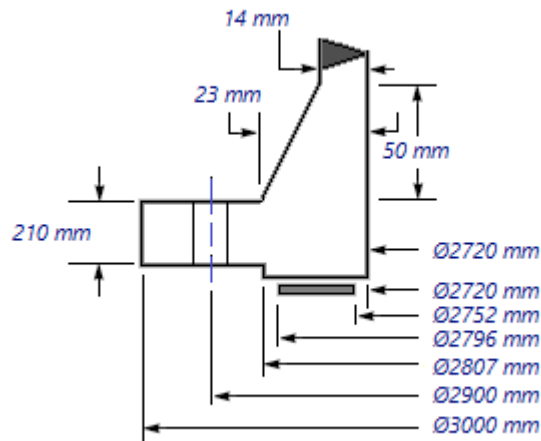
Description: LHS CH FL :

LHS CH FL

Flange Type	Integral Weld Neck		
Design Pressure	P	1026.97	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2720.000	mm
Flange Outside Diameter	A	3000.000	mm
Flange Thickness	t	210.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	g1	23.0000	mm
Length of Hub	h	50.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2900.000	mm
Nominal Bolt Diameter	a	47.6250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	2807.000	mm
Flange Face Inside Diameter	Fid	2720.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2796.000	mm
Gasket Inside Diameter	Gi	2752.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2725.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026.97 \cdot (2720.0/2 + 0.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1026.97) + C_a$$

$$= 10.1670 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (1360.0 + 0.6 \cdot 14.0)$$

$$= 1411.760 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (1360.0 + 0.6 \cdot 14.0)$$

$$= 1411.760 \text{ KPa.}$$

Corroded Flange ID,	Bcor = B + 2 * ci	2720.000 mm
Corroded Large Hub,	g1Cor = g1 - ci	23.000 mm
Corroded Small Hub,	g0Cor = g0 - ci	14.000 mm
Code R Dimension,	R = (C - B) / 2 - g1	67.000 mm
Gasket Contact Width,	N = (Go - Gi) / 2	22.000 mm
Basic Gasket Width,	bo = N / 2	11.000 mm
Effective Gasket Width,	b = Cb * sqrt(bo)	8.358 mm
Gasket Reaction Diameter,	G = Go - 2 * b	2779.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi / 4 \cdot G^2 \cdot P$$

$$= \pi / 4 \cdot 2779.2847^2 \cdot 1026.965$$

$$= 635343.188 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2779.2847 \cdot 3.0 \cdot 1026.97$$

$$+ 2 \cdot 2725.0 \cdot 5.0 \cdot 3.0 \cdot 1026.9653$$

$$= 54414.531 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi / 4 \cdot B_{cor}^2 \cdot P$$

$$= \pi / 4 \cdot 2720.0^2 \cdot 1026.9653$$

$$= 608527.312 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 635343 - 608527$$

$$= 26815.867 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(635343 + 54415 + 0, 0)$$

$$= 689757.688 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2779.285 + 68.95 \cdot 5.0 \cdot 2725.0$$

$$= 608859.188 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((689758 + 0 + 4 \cdot 0 / 2779.28) / 172, 608859 / 172)$$

$$= 393.283 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 47.625 + 6 \cdot 210.0 / (3.0 + 0.5)$$

$$= 455.250 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2900.0 \cdot \sin(\pi / 32)$$

$$= 284.250 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ B_s / (2a + t) }, 1)$$

$$= \max(\sqrt{ 284.25 / (2 \cdot 47.625 + 210.0) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	393.283	475.664	
Radial Distance between Hub and Bolts:	60.325	67.000	
Radial Distance between Bolts and Edge:	47.625	50.000	
Circ. Spacing between the Bolts:	101.600	284.250	455.250

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 475.664 \cdot 172.0 / (68.95 \cdot \pi (2796.0 + 2752.0))$$

$$= 6.808 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 475.6635$$

$$= 834241.56 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 689758 - 635343$$

$$= 54414.54 \text{ Kgf}$$

Moment Arm Calculations:



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Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2900.0 - 2779.2847) / 2$$

$$= 60.3577 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5((C - B) / 2 + hg)$$

$$= 0.5((2900.0 - 2720.0) / 2 + 60.3577)$$

$$= 75.1788 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g1) / 2$$

$$= (2900.0 - 2720.0 - 23.0) / 2$$

$$= 78.5000 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	608527.	78.5000	1.0000	47770.
Face Pressure, Mt	26816.	75.1788	1.0000	2016.
Gasket Load, Mg	54415.	60.3577	1.0000	3284.
Gasket Seating, Matm	834242.	60.3577	1.0000	50354.

Total Moment for Operation, Mo 53071. Kg-m.

Total Moment for Gasket seating, Mg 50354. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 195.141 mm

Hub Ratio, h/h0 = HL / H0 0.256

Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.643

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.887

Factor V per 4.16.5 0.365

Factor f per 4.16.5 1.570

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 3000.0 / 2720.0$$

$$= 1.103$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169(K^2 \cdot \log_{10}(K)) / (K^2 - 1)) / (K - 1))$$

$$= (0.66845 + (5.7169(1.103^2 \cdot \log_{10}(1.103)) / (1.103^2 - 1)) / (1.103 - 1))$$

$$= 19.773$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.103^2(1 + 8.55246 \cdot \log_{10}(1.103)) - 1) / ((1.0472 + 1.9448 \cdot 1.103^2) \cdot (1.103 - 1))$$

$$= 1.876$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2(1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136(K^2 - 1) \cdot (K - 1))$$

$$= (1.103^2(1 + 8.55246 \cdot \log_{10}(1.103)) - 1) / (1.36136(1.103^2 - 1) \cdot (1.103 - 1))$$

$$= 21.729$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$

$$= (1.103^2 + 1) / (1.103^2 - 1)$$

$$= 10.239$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g0^2 \cdot h0 / V$$

$$= 21.729 \cdot 14.0^2 \cdot 195.141 / 0.365$$



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$$= 2275485.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$\begin{aligned} &= F / h_0 \\ &= 0.887/195.141 \\ &= 0.005 \text{ 1/mm} \end{aligned}$$

Stress factor from Table 4.16.4 [L]:

$$\begin{aligned} &= (t \cdot e + 1) / T + t^3/d \\ &= (210.0 \cdot 0.005 + 1) / 1.876 + 210.0^3/2275485 \\ &= 5.112 \end{aligned}$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$\begin{aligned} &= (f \cdot Mop / Bcor) / (L \cdot g^{1^2}) \\ &= (1.5699 \cdot 53071 / 2720.0) / (5.1115 \cdot 23.0^2) \\ &= 111.09 \text{ N./mm}^2 \end{aligned}$$

Longitudinal Hub Stress, Seating [SHa]:

$$\begin{aligned} &= (f \cdot Matm / Bcor) / (L \cdot g^{1^2}) \\ &= (1.5699 \cdot 50354 / 2720.0) / (5.1115 \cdot 23.0^2) \\ &= 105.40 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Operating [SRo]:

$$\begin{aligned} &= (\beta \cdot Mop / Bcor) / (L \cdot t^2) \\ &= (2.2725 \cdot 53071 / 2720.0) / (5.1115 \cdot 210.0^2) \\ &= 1.93 \text{ N./mm}^2 \end{aligned}$$

Radial Flange Stress, Seating [SRa]:

$$\begin{aligned} &= (\beta \cdot Matm / Bcor) / (L \cdot t^2) \\ &= (2.2725 \cdot 50354 / 2720.0) / (5.1115 \cdot 210.0^2) \\ &= 1.83 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Operating [STo]:

$$\begin{aligned} &= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo \\ &= (19.7731 \cdot 53071 / (210.0^2 \cdot 2720.0)) - 10.2388 \cdot 2 \\ &= 66.04 \text{ N./mm}^2 \end{aligned}$$

Tangential Flange Stress, Seating [STa]:

$$\begin{aligned} &= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa \\ &= (19.7731 \cdot 50354 / (210.0^2 \cdot 2720.0)) - 10.2388 \cdot 2 \\ &= 62.66 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Operating [SAo]:

$$\begin{aligned} &= (SHo + \max(SRo, STo)) / 2 \\ &= (111 + \max(2.66)) / 2 \\ &= 88.57 \text{ N./mm}^2 \end{aligned}$$

Average Flange Stress, Seating [SAa]:

$$\begin{aligned} &= (SHa + \max(SRa, STa)) / 2 \\ &= (105 + \max(2.63)) / 2 \\ &= 84.03 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Operating [BSo]:

$$\begin{aligned} &= Wm1 / Ab \\ &= 689758 / 475.6635 \\ &= 142.21 \text{ N./mm}^2 \end{aligned}$$

Bolt Stress, Seating [BSa]:

$$\begin{aligned} &= (Wm2 / Ab) \\ &= (608859 / 475.6635) \\ &= 125.53 \text{ N./mm}^2 \end{aligned}$$

Flange Stress Analysis Results: N./mm²



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	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	111.09	207.00	105.40	207.00
Radial Flange	1.93	138.00	1.83	138.00
Tangential Flange	66.04	138.00	62.66	138.00
Maximum Average	88.57	138.00	84.03	138.00
Bolting	142.21	172.00	125.53	172.00

Minimum Required Flange Thickness [Rigidity] 196.875 mm

Estimated M.A.W.P. (Operating) 1202 KPa.
 Estimated Finished Weight of Flange at given Thk. 2109.0 kg.
 Estimated Unfinished Weight of Forging at given Thk 2534.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	298.39	374.99	0.796	298.39	374.99	0.796
Tangential Flange	Sto	117.99	250.00	0.472	117.99	250.00	0.472
Combined	Sh/f + St	618.11	749.99	0.824	618.11	749.99	0.824
Combined	Sr + St	307.10	749.99	0.409	307.10	749.99	0.409
Combined	Sh + Sto	617.99	749.99	0.824	617.99	749.99	0.824
Radial Flange	Sr	8.72	499.99	0.017	8.72	499.99	0.017
Maximum Stress Ratio				1.000			1.000



Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.8063/1.155)
 = 1.5639

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/h0 < 0.35, else max (0.25; fn3)
 = max(0.25; 0.3864/1.0469) if h/h0 < 0.35, else max (0.25; 0.3125/0.7735)
 = 0.3691

In this case, the hub ratio h/h0, 0.2562 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	819.1 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	14.8645 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	1.138 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2796.0^2 - 2752.0^2) + (0.5 \cdot 2725.0 \cdot 5.0)$
 $= 1985.4 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1985.38 / (32 \cdot 14.864)$
 $= 864.0 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(864.001; 507.5)$
 $= 507.5 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(507.5; 290.0)$
 $= 507.5 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(507.5; 819.051)$
 $= 507.5 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$
 $:: 507.5 \geq 140.0(1985.38 / (32 \cdot 14.864))$
 $:: 507.5 \geq 584.3 \text{ N./mm}^2 \text{ Check Fails}$

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 507.5 \geq (97.0 \cdot 1985.38 + \pi/4 \cdot 1026.965 \cdot 2752.0^2) / (0.7 \cdot 14.864 \cdot 32)$
 $:: 507.5 \geq 761.9 \text{ N./mm}^2 \text{ Check Fails}$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 507.5 \leq 276.0(1985.38 / (32 \cdot 14.864))$
 $:: 507.5 \leq 1152.0 \text{ N./mm}^2 \text{ Check Passes}$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 507.5 \leq 819.051(1.0/1.138)$
 $:: 507.5 \leq 720.0 \text{ N./mm}^2 \text{ Check Passes}$

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 507.5 \cdot 0.2 \cdot 14.864 \cdot 47.625$
 $= 732.7 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118 N./mm ²
Bolt Allowable Stress for: SA-193 B7	Sb	171 N./mm ²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 50353.9 / 1.0 \cdot 9806.637 \cdot 0.365 / (4.363 \cdot 201334 \cdot 14.0^2 \cdot 195.141 \cdot 0.3)$
 $= 0.933$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 53070.8 / 1.0 \cdot 9806.637 \cdot 0.365 / (4.363 \cdot 197847 \cdot 14.0^2 \cdot 195.141 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 50353.9 / 1.0 \cdot 9806.637 \cdot 0.365 / (5.112 \cdot 201334 \cdot 14.0^2 \cdot 195.141 \cdot 0.3)$
 $= 0.796$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 53070.8 / 1.0 \cdot 9806.637 \cdot 0.365 / (5.112 \cdot 197847 \cdot 14.0^2 \cdot 195.141 \cdot 0.3)$
 $= 0.854$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.869, Temperature Reduction per Fig. UCS 66.1 = 7 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	111.09	207.00	105.40	207.00
Radial Flange	1.93	138.00	1.83	138.00
Tangential Flange	66.04	138.00	62.66	138.00
Maximum Average	88.57	138.00	84.03	138.00
Bolting	142.21	172.00	125.53	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 195.529 mm

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Flange Input Data Values

Description: New Flange :

RHS CH FL

Flange Type	Integral Weld Neck, Integral type		
Design Pressure	P	1026.97	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2720.000	mm
Flange Outside Diameter	A	3140.000	mm
Flange Thickness	t	300.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	gl	21.0000	mm
Length of Hub	h	28.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

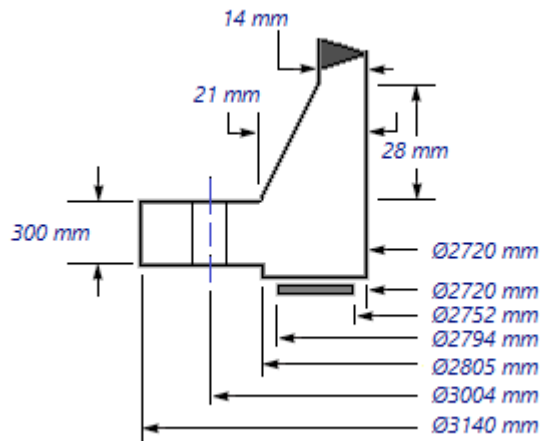
Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	3004.000	mm
Nominal Bolt Diameter	a	69.8500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	2805.000	mm
Flange Face Inside Diameter	Fid	2720.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2794.000	mm
Gasket Inside Diameter	Gi	2752.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2725.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (1026.97 \cdot (2720.0/2 + 3.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 1026.97) + C_a$$

$$= 13.1894 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 11.0) / (1363.0 + 0.6 \cdot 11.0)$$

$$= 1108.269 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 14.0) / (1360.0 + 0.6 \cdot 14.0)$$

$$= 1411.760 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	297.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2726.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	18.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	121.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	21.000 mm
Basic Gasket Width, $b_o = N/2$	10.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.165 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2777.669 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2777.6689^2 \cdot 1026.965$$

$$= 634604.500 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.1655 \cdot \pi \cdot 2777.6689 \cdot 3.0 \cdot 1026.97$$

$$+ 2 \cdot 2725.0 \cdot 5.0 \cdot 3.0 \cdot 1026.9653$$

$$= 53334.207 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2726.0^2 \cdot 1026.9653$$

$$= 611214.875 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

= H - Hd
 = 634605 - 611215
 = 23389.656 Kgf

Design Bolt Load for the Operating Condition [Wo]:

= max(Wo,1 + Wo,2 + H'p, 0)
 = max(634605 + 53334 + 0, 0)
 = 687938.750 Kgf
 = 1697411.125 Kgf, Mating Flange Load Governs

Design Bolt Load for the Gasket Seating Condition [Wg]:

= y • b • π • G + yPart • bPart • lp
 = 68.95 • 8.1655 • π • 2777.669 + 68.95 • 5.0 • 2725.0
 = 596771.250 Kgf
 = 597492.625 Kgf, Mating Flange Load Governs

Required Bolt Area [Am]:

= max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)
 = max((1697411 + 0 + 4 • 0/2777.67)/159, 597493/159)
 = 1046.951 cm²

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:

= 2a + 6t/(m + 0.5)
 = 2 • 69.85 + 6 • 297.0/(3.0 + 0.5)
 = 648.843 mm

Actual Circumferential Bolt Spacing [Bs]:

= C • sin(π / n)
 = 3004.0 • sin(π/32)
 = 294.444 mm

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

= max(√(Bs/(2a + t)), 1)
 = max(√(294.444/(2 • 69.85 + 297.0)), 1)
 = 1.0000

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1046.951	1085.727	
Radial Distance between Hub and Bolts:	85.725	121.000	
Radial Distance between Bolts and Edge:	66.675	68.000	
Circ. Spacing between the Bolts:	146.050	294.444	648.843

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

= Ab • Sa/(y • π(Go + Gi))
 = 1085.727 • 159.0/(68.95 • π(2794.0 + 2752.0))
 = 14.370 mm

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

= Sa • Ab
 = 159.0 • 1085.7268
 = 1760277.62 Kgf

Gasket Load for the Operating Condition [HG]:

= Wo - H
 = 1697411 - 634605



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= 1062806.50 Kgf

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (3004.0 - 2777.6689) / 2$
 $= 113.1656 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((3004.0 - 2726.0) / 2 + 113.1656)$
 $= 126.0828 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (3004.0 - 2726.0 - 18.0) / 2$
 $= 130.0000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	611215.	130.0000	1.0000	79460.
Face Pressure, Mt	23390.	126.0828	1.0000	2949.
Gasket Load, Mg	1062806.	113.1656	1.0000	120275.
Gasket Seating, Matm	1760278.	113.1656	1.0000	199207.

Total Moment for Operation, Mo 202684. Kg-m.
 Total Moment for Gasket seating, Mg 199207. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 173.165 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.162
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.636

Flange Factors for Integral Flange:
 Factor F per 4.16.5 0.903
 Factor V per 4.16.5 0.418
 Factor f per 4.16.5 1.939

Stress factor from Table 4.16.4 [K]:
 $= A / B$
 $= 3140.0 / 2726.0$
 $= 1.152$

Stress factor from Table 4.16.4 [Y]:
 $= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.152^2 \cdot \log_{10}(1.152))) / (1.152^2 - 1)) / (1.152 - 1)$
 $= 13.786$

Stress factor from Table 4.16.4 [T]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.152^2 (1 + 8.55246 \cdot \log_{10}(1.152)) - 1) / ((1.0472 + 1.9448 \cdot 1.152^2) \cdot (1.152 - 1))$
 $= 1.858$

Stress factor from Table 4.16.4 [U]:
 $= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.152^2 (1 + 8.55246 \cdot \log_{10}(1.152)) - 1) / (1.36136 (1.152^2 - 1) \cdot (1.152 - 1))$
 $= 15.149$

Stress factor from Table 4.16.4 [Z]:
 $= (K^2 + 1) / (K^2 - 1)$
 $= (1.152^2 + 1) / (1.152^2 - 1)$
 $= 7.120$



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Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$

$$= 15.149 \cdot 11.0^2 \cdot 173.165 / 0.418$$

$$= 758961.875 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.903 / 173.165$$

$$= 0.005 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (297.0 \cdot 0.005 + 1) / 1.858 + 297.0^3 / 758962$$

$$= 35.890$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.9395 \cdot 202684 / 2726.0) / (35.8904 \cdot 18.0^2)$$

$$= 121.61 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.9395 \cdot 199207 / 2726.0) / (35.8904 \cdot 18.0^2)$$

$$= 119.53 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (3.0657 \cdot 202684 / 2726.0) / (35.8904 \cdot 297.0^2)$$

$$= 0.71 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (3.0657 \cdot 199207 / 2726.0) / (35.8904 \cdot 297.0^2)$$

$$= 0.69 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.7857 \cdot 202684 / (297.0^2 \cdot 2726.0)) - 7.1198 \cdot 0.7$$

$$= 108.93 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.7857 \cdot 199207 / (297.0^2 \cdot 2726.0)) - 7.1198 \cdot 0.7$$

$$= 107.06 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (122 + \max(0.7, 109)) / 2$$

$$= 115.27 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (120 + \max(0.7, 107)) / 2$$

$$= 113.29 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$

$$= 1697411 / 1085.7268$$

$$= 153.32 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (597493 / 1085.7268)$$



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= 53.97 N./mm²

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	121.61	207.00	119.53	207.00
Radial Flange	0.71	138.00	0.69	138.00
Tangential Flange	108.93	138.00	107.06	138.00
Maximum Average	115.27	138.00	113.29	138.00
Bolting	153.32	159.00	53.97	159.00

Minimum Required Flange Thickness [Rigidity] 296.977 mm

Estimated M.A.W.P. (Operating) 1108 KPa.

Estimated Finished Weight of Flange at given Thk. 4527.2 kg.

Estimated Unfinished Weight of Forging at given Thk 4914.1 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	417.59	499.99	0.835	417.59	499.99	0.835
Tangential Flange	St	374.99	374.99	1.000	374.99	374.99	1.000
Tangential Flange	Sto	63.58	250.00	0.254	63.58	250.00	0.254
Combined	Sh/f + St	590.86	749.99	0.788	590.86	749.99	0.788
Combined	Sr + St	377.42	749.99	0.503	377.42	749.99	0.503
Combined	Sh + Sto	481.17	749.99	0.642	481.17	749.99	0.642
Radial Flange	Sr	2.43	499.99	0.005	2.43	499.99	0.005
Maximum Stress Ratio				1.000			1.000



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.9521/1.0091)
 = 1.9345

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.2513/0.8532) if $h/h0 < 0.35$, else max (0.25; 0.3168/0.8026)
 = 0.2945

In this case, the hub ratio $h/h0$, 0.1617 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	556.9 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	33.9290 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.540 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2794.0^2 - 2752.0^2) + (0.5 \cdot 2725.0 \cdot 5.0)$
 $= 1897.6 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1897.567 / (32 \cdot 33.929)$
 $= 361.8 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(361.782; 507.5)$
 $= 361.8 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(361.782; 290.0)$
 $= 361.8 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(361.782; 556.922)$
 $= 361.8 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin - S(Ag / (nb \cdot Abs))$
 $:: 361.782 \geq 140.0(1897.567 / (32 \cdot 33.929))$
 $:: 361.8 \geq 244.7 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin - O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 361.782 \geq (97.0 \cdot 1897.567 + \pi/4 \cdot 1026.965 \cdot 2752.0^2) / (0.7 \cdot 33.929 \cdot 32)$
 $:: 361.8 \geq 322.6 \text{ N./mm}^2$ Check Passes

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 361.782 \leq 276.0(1897.567 / (32 \cdot 33.929))$
 $:: 361.8 \leq 482.4 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 361.782 \leq 556.922(1.0/0.54)$
 $:: 361.8 \leq 1031.5 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 361.782 \cdot 0.2 \cdot 33.929 \cdot 69.85$
 $= 1748.6 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2	SbB8	118 N./mm ²
Bolt Allowable Stress for: SA-193 B7	Sb	158 N./mm ²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 199206.8 / 1.0 \cdot 9806.637 \cdot 0.418 / (34.832 \cdot 201334 \cdot 11.0^2 \cdot 173.165 \cdot 0.3)$
 $= 0.966$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 202684.1 / 1.0 \cdot 9806.637 \cdot 0.418 / (34.832 \cdot 197847 \cdot 11.0^2 \cdot 173.165 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 199206.8 / 1.0 \cdot 9806.637 \cdot 0.418 / (35.89 \cdot 201334 \cdot 11.0^2 \cdot 173.165 \cdot 0.3)$
 $= 0.937$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 202684.1 / 1.0 \cdot 9806.637 \cdot 0.418 / (35.89 \cdot 197847 \cdot 11.0^2 \cdot 173.165 \cdot 0.3)$
 $= 0.971$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.753, Temperature Reduction per Fig. UCS 66.1 = 14 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-32 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Longitudinal Hub	127.13	207.00	124.37	207.00
Radial Flange	1.04	138.00	1.02	138.00
Tangential Flange	103.43	138.00	101.19	138.00
Maximum Average	115.28	138.00	112.78	138.00
Bolting	153.32	159.00	53.97	159.00

Minimum Req'd Thickness, ambient, no Corrosion: 288.900 mm

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Flange Input Data Values

Description: Shell flg-1 :

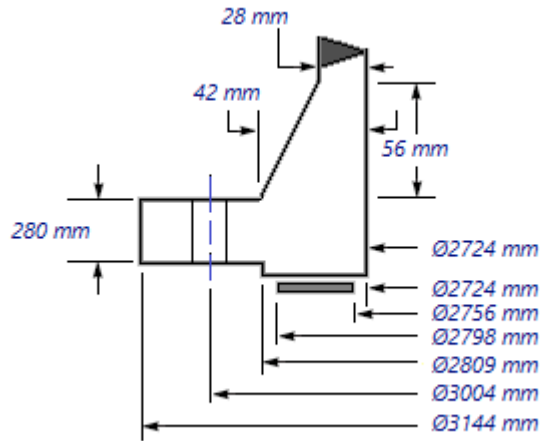
Shell flg-1

Flange Type	Integral Weld Neck		
Design Pressure	P	2526.97	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2724.000	mm
Flange Outside Diameter	A	3144.000	mm
Flange Thickness	t	280.0000	mm
Thickness of Hub at Small End	go	28.0000	mm
Thickness of Hub at Large End	gl	42.0000	mm
Length of Hub	h	56.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	3004.000	mm
Nominal Bolt Diameter	a	69.8500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	2809.000	mm
Flange Face Inside Diameter	Fid	2724.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2798.000	mm
Gasket Inside Diameter	Gi	2756.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2725.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2526.97 \cdot (2724.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 2526.97) + C_a$$

$$= 26.2376 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 27.0) / (1363.0 + 0.6 \cdot 27.0)$$

$$= 2701.361 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1362.0 + 0.6 \cdot 28.0)$$

$$= 2802.224 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	279.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2726.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	41.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	27.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	98.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	21.000 mm
Basic Gasket Width, $b_o = N/2$	10.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.165 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2781.669 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2781.6689^2 \cdot 2526.966$$

$$= 1566017.500 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{part} \cdot m_{part} \cdot P$$

$$= 2 \cdot 8.1655 \cdot \pi \cdot 2781.6689 \cdot 3.0 \cdot 2526.97$$

$$+ 2 \cdot 2725.0 \cdot 5.0 \cdot 3.0 \cdot 2526.9656$$

$$= 131393.562 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2726.0^2 \cdot 2526.9656$$

$$= 1503963.875 \text{ Kgf}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 1566018 - 1503964$$

$$= 62053.727 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(1566018 + 131394 + 0, 0)$$

$$= 1697411.125 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.1655 \cdot \pi \cdot 2781.669 + 68.95 \cdot 5.0 \cdot 2725.0$$

$$= 597492.625 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1697411 + 0 + 4 \cdot 0 / 2781.67) / 159, 597493 / 159)$$

$$= 1046.951 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 69.85 + 6 \cdot 279.0 / (3.0 + 0.5)$$

$$= 617.986 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 3004.0 \cdot \sin(\pi / 32)$$

$$= 294.444 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (294.444 / (2 \cdot 69.85 + 279.0)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1046.951	1085.727	
Radial Distance between Hub and Bolts:	85.725	98.000	
Radial Distance between Bolts and Edge:	66.675	70.000	
Circ. Spacing between the Bolts:	146.050	294.444	617.986

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 1085.727 \cdot 159.0 / (68.95 \cdot \pi (2798.0 + 2756.0))$$

$$= 14.349 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159.0 \cdot 1085.7268$$

$$= 1760277.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1697411 - 1566018$$

$$= 131393.52 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (3004.0 - 2781.6689) / 2$
 $= 111.1656 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((3004.0 - 2726.0) / 2 + 111.1656)$
 $= 125.0829 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (3004.0 - 2726.0 - 41.0) / 2$
 $= 118.5001 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1503964.	118.5001	1.0000	178223.
Face Pressure, Mt	62054.	125.0829	1.0000	7762.
Gasket Load, Mg	131394.	111.1656	1.0000	14607.
Gasket Seating, Matm	1760278.	111.1656	1.0000	195686.

Total Moment for Operation, Mo 200592. Kg-m.
 Total Moment for Gasket seating, Mg 195686. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 271.297 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.206
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.519

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.899
 Factor V per 4.16.5 0.407
 Factor f per 4.16.5 1.515

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 3144.0 / 2726.0$
 $= 1.153$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.153^2 \cdot \log_{10}(1.153))) / (1.153^2 - 1)) / (1.153 - 1)$
 $= 13.665$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.153^2 (1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / ((1.0472 + 1.9448 \cdot 1.153^2) \cdot (1.153 - 1))$
 $= 1.857$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.153^2 (1 + 8.55246 \cdot \log_{10}(1.153)) - 1) / (1.36136 (1.153^2 - 1) \cdot (1.153 - 1))$
 $= 15.016$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.153^2 + 1) / (1.153^2 - 1)$
 $= 7.057$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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$$= 15.016 \cdot 27.0^2 \cdot 271.297 / 0.407$$

$$= 7297556.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.899 / 271.297$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (279.0 \cdot 0.003 + 1) / 1.857 + 279.0^3 / 7297557$$

$$= 4.012$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g1^2)$$

$$= (1.5149 \cdot 200592 / 2726.0) / (4.012 \cdot 41.0^2)$$

$$= 162.10 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g1^2)$$

$$= (1.5149 \cdot 195686 / 2726.0) / (4.012 \cdot 41.0^2)$$

$$= 158.13 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2322 \cdot 200592 / 2726.0) / (4.012 \cdot 279.0^2)$$

$$= 5.16 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2322 \cdot 195686 / 2726.0) / (4.012 \cdot 279.0^2)$$

$$= 5.03 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (13.665 \cdot 200592 / (279.0^2 \cdot 2726.0)) - 7.0571 \cdot 5$$

$$= 90.28 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (13.665 \cdot 195686 / (279.0^2 \cdot 2726.0)) - 7.0571 \cdot 5$$

$$= 88.07 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (162 + \max(5, 90)) / 2$$

$$= 126.19 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (158 + \max(5, 88)) / 2$$

$$= 123.10 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 1697411 / 1085.7268$$

$$= 153.32 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (597493 / 1085.7268)$$

$$= 53.97 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	162.10	205.20	158.13	207.00
Radial Flange	5.16	136.80	5.03	138.00
Tangential Flange	90.28	136.80	88.07	138.00
Maximum Average	126.19	136.80	123.10	138.00
Bolting	153.32	159.00	53.97	159.00

Minimum Required Flange Thickness [Rigidity] 271.120 mm

Estimated M.A.W.P. (Operating) 2621 KPa.
 Estimated Finished Weight of Flange at given Thk. 4332.3 kg.
 Estimated Unfinished Weight of Forging at given Thk 5040.8 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	475.43	475.43	1.000	475.43	499.99	0.951
Tangential Flange	St	267.34	356.57	0.750	267.34	374.99	0.713
Tangential Flange	Sto	93.34	237.71	0.393	93.34	250.00	0.373
Combined	Sh/f + St	584.19	713.14	0.819	584.19	749.99	0.779
Combined	Sr + St	282.61	713.14	0.396	282.61	749.99	0.377
Combined	Sh + Sto	568.77	713.14	0.798	568.77	749.99	0.758
Radial Flange	Sr	15.27	475.43	0.032	15.27	499.99	0.031
Maximum Stress Ratio				1.000			0.951



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.6314/1.0872)
 = 1.5005

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.2851/0.9677) if $h/h0 < 0.35$, else max (0.25; 0.2777/0.7921)
 = 0.2946

In this case, the hub ratio $h/h0$, 0.2064 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	482.6 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	33.9290 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.707 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2798.0^2 - 2756.0^2) + (0.5 \cdot 2725.0 \cdot 5.0)$
 $= 1900.2 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1900.204 / (32 \cdot 33.929)$
 $= 362.3 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(362.285; 507.5)$
 $= 362.3 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(362.285; 290.0)$
 $= 362.3 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(362.285; 482.631)$
 $= 362.3 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 362.285 \geq 140.0(1900.204 / (32 \cdot 33.929))$
 $:: 362.3 \geq 245.0 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 362.285 \geq (97.0 \cdot 1900.204 + \pi/4 \cdot 2526.966 \cdot 2756.0^2) / (0.7 \cdot 33.929 \cdot 32)$
 $:: 362.3 \geq 440.9 \text{ N./mm}^2$ Check Fails

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 362.285 \leq 276.0(1900.204 / (32 \cdot 33.929))$
 $:: 362.3 \leq 483.0 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 362.285 \leq 482.631(1.0/0.707)$
 $:: 362.3 \leq 682.9 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 362.285 \cdot 0.2 \cdot 33.929 \cdot 69.85$
 $= 1751.0 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 92 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 195686.2 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.721 \cdot 201334 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.916$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 200592.2 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.721 \cdot 189090 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 195686.2 / 1.0 \cdot 9806.637 \cdot 0.407 / (4.012 \cdot 201334 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.849$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 200592.2 / 1.0 \cdot 9806.637 \cdot 0.407 / (4.012 \cdot 189090 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.927$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.929, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	160.41	207.00	155.87 207.00
Radial Flange	5.36	138.00	5.21 138.00
Tangential Flange	88.09	138.00	85.59 138.00
Maximum Average	124.25	138.00	120.73 138.00
Bolting	153.32	159.00	53.97 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 265.024 mm

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Flange Input Data Values

Description: New Flange :

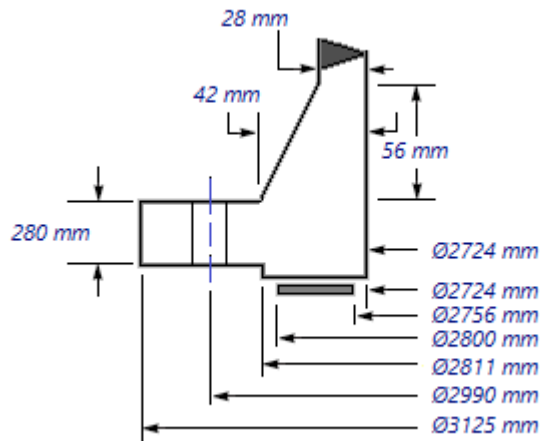
shell flg-3

Flange Type	Integral Weld Neck		
Design Pressure	P	2526.97	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2724.000	mm
Flange Outside Diameter	A	3125.000	mm
Flange Thickness	t	280.0000	mm
Thickness of Hub at Small End	go	28.0000	mm
Thickness of Hub at Large End	gl	42.0000	mm
Length of Hub	h	56.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2990.000	mm
Nominal Bolt Diameter	a	69.8500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	2811.000	mm
Flange Face Inside Diameter	Fid	2724.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2800.000	mm
Gasket Inside Diameter	Gi	2756.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2725.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2526.97 \cdot (2724.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 2526.97) + C_a$$

$$= 26.2376 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 27.0) / (1363.0 + 0.6 \cdot 27.0)$$

$$= 2701.361 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1362.0 + 0.6 \cdot 28.0)$$

$$= 2802.224 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	279.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2726.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	41.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	27.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	91.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2783.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2783.2847^2 \cdot 2526.966$$

$$= 1567837.375 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2783.2847 \cdot 3.0 \cdot 2526.97$$

$$+ 2 \cdot 2725.0 \cdot 5.0 \cdot 3.0 \cdot 2526.9656$$

$$= 134055.547 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2726.0^2 \cdot 2526.9656$$

$$= 1503963.875 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 1567837 - 1503964$$

$$= 63873.570 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(1567837 + 134056 + 0, 0)$$

$$= 1701893.000 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2783.285 + 68.95 \cdot 5.0 \cdot 2725.0$$

$$= 609597.562 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1701893 + 0 + 4 \cdot 0 / 2783.28) / 159, 609598 / 159)$$

$$= 1049.716 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 69.85 + 6 \cdot 279.0 / (3.0 + 0.5)$$

$$= 617.986 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2990.0 \cdot \sin(\pi / 32)$$

$$= 293.071 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (293.071 / (2 \cdot 69.85 + 279.0)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1049.716	1085.727	
Radial Distance between Hub and Bolts:	85.725	91.000	
Radial Distance between Bolts and Edge:	66.675	67.500	
Circ. Spacing between the Bolts:	146.050	293.071	617.986

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 1085.727 \cdot 159.0 / (68.95 \cdot \pi (2800.0 + 2756.0))$$

$$= 14.344 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159.0 \cdot 1085.7268$$

$$= 1760277.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1701893 - 1567837$$

$$= 134055.58 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2990.0 - 2783.2847) / 2$
 $= 103.3576 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2990.0 - 2726.0) / 2 + 103.3576)$
 $= 117.6789 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2990.0 - 2726.0 - 41.0) / 2$
 $= 111.5001 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1503964.	111.5001	1.0000	167695.
Face Pressure, Mt	63874.	117.6789	1.0000	7517.
Gasket Load, Mg	134056.	103.3576	1.0000	13856.
Gasket Seating, Matm	1760278.	103.3576	1.0000	181942.

Total Moment for Operation, Mo 189068. Kg-m.
 Total Moment for Gasket seating, Mg 181942. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 271.297 mm
 Hub Ratio, $h/h_o = H_L / H_o$ 0.206
 Thickness Ratio, $g1/g_o = (g1_{cor}/g_{oCor})$ 1.519

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.899
 Factor V per 4.16.5 0.407
 Factor f per 4.16.5 1.515

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 3125.0 / 2726.0$
 $= 1.146$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.146^2 \cdot \log_{10}(1.146))) / (1.146^2 - 1)) / (1.146 - 1)$
 $= 14.260$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.146^2 (1 + 8.55246 \cdot \log_{10}(1.146)) - 1) / ((1.0472 + 1.9448 \cdot 1.146^2) \cdot (1.146 - 1))$
 $= 1.860$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.146^2 (1 + 8.55246 \cdot \log_{10}(1.146)) - 1) / (1.36136 (1.146^2 - 1) \cdot (1.146 - 1))$
 $= 15.670$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.146^2 + 1) / (1.146^2 - 1)$
 $= 7.366$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_o^2 \cdot h_o / V$



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$$= 15.67 \cdot 27.0^2 \cdot 271.297 / 0.407$$

$$= 7615059.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.899 / 271.297$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (279.0 \cdot 0.003 + 1) / 1.86 + 279.0^3 / 7615060$$

$$= 3.886$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g^{1^2})$$

$$= (1.5149 \cdot 189068 / 2726.0) / (3.8864 \cdot 41.0^2)$$

$$= 157.72 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g^{1^2})$$

$$= (1.5149 \cdot 181942 / 2726.0) / (3.8864 \cdot 41.0^2)$$

$$= 151.77 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2322 \cdot 189068 / 2726.0) / (3.8864 \cdot 279.0^2)$$

$$= 5.02 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2322 \cdot 181942 / 2726.0) / (3.8864 \cdot 279.0^2)$$

$$= 4.83 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (14.2596 \cdot 189068 / (279.0^2 \cdot 2726.0)) - 7.3662 \cdot 5$$

$$= 87.63 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (14.2596 \cdot 181942 / (279.0^2 \cdot 2726.0)) - 7.3662 \cdot 5$$

$$= 84.33 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (158 + \max(5, 88)) / 2$$

$$= 122.67 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (152 + \max(5, 84)) / 2$$

$$= 118.05 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 1701893 / 1085.7268$$

$$= 153.72 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (609598 / 1085.7268)$$

$$= 55.06 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	157.72	205.20	151.77	207.00
Radial Flange	5.02	136.80	4.83	138.00
Tangential Flange	87.63	136.80	84.33	138.00
Maximum Average	122.67	136.80	118.05	138.00
Bolting	153.72	159.00	55.06	159.00

Minimum Required Flange Thickness [Rigidity] 267.716 mm

Estimated M.A.W.P. (Operating) 2614 KPa.
 Estimated Finished Weight of Flange at given Thk. 4129.3 kg.
 Estimated Unfinished Weight of Forging at given Thk 4797.2 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	475.43	475.43	1.000	475.43	499.99	0.951
Tangential Flange	St	266.69	356.57	0.748	266.69	374.99	0.711
Tangential Flange	Sto	93.34	237.71	0.393	93.34	250.00	0.373
Combined	Sh/f + St	583.54	713.14	0.818	583.54	749.99	0.778
Combined	Sr + St	281.96	713.14	0.395	281.96	749.99	0.376
Combined	Sh + Sto	568.77	713.14	0.798	568.77	749.99	0.758
Radial Flange	Sr	15.27	475.43	0.032	15.27	499.99	0.031
Maximum Stress Ratio				1.000			0.951



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Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)
 = max(1; 1.6314/1.0872)
 = 1.5005

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h0 < 0.35$, else max (0.25; fn3)
 = max(0.25; 0.2851/0.9677) if $h/h0 < 0.35$, else max (0.25; 0.2777/0.7921)
 = 0.2946

In this case, the hub ratio $h/h0$, 0.2064 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	502.8 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700



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Root Area of a Single Bolt	Abs	33.9290 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.707 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2800.0^2 - 2756.0^2) + (0.5 \cdot 2725.0 \cdot 5.0)$
 $= 1988.1 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 207.0 \cdot 1988.141 / (32 \cdot 33.929)$
 $= 379.1 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(379.05; 507.5)$
 $= 379.1 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(379.05; 290.0)$
 $= 379.1 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(379.05; 502.843)$
 $= 379.1 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 379.05 \geq 140.0(1988.141 / (32 \cdot 33.929))$
 $:: 379.1 \geq 256.4 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 379.05 \geq (97.0 \cdot 1988.141 + \pi/4 \cdot 2526.966 \cdot 2756.0^2) / (0.7 \cdot 33.929 \cdot 32)$
 $:: 379.1 \geq 452.1 \text{ N./mm}^2$ Check Fails

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 379.05 \leq 276.0(1988.141 / (32 \cdot 33.929))$
 $:: 379.1 \leq 505.4 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(\text{thetagmax} / \text{thetafmax})$
 $:: 379.05 \leq 502.843(1.0/0.707)$
 $:: 379.1 \leq 711.5 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 379.05 \cdot 0.2 \cdot 33.929 \cdot 69.85$
 $= 1832.1 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 92 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 181941.8 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.504 \cdot 201334 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.904$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 189068.0 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.504 \cdot 189090 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 1.000$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 181941.8 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.886 \cdot 201334 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.815$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 189068.0 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.886 \cdot 189090 \cdot 27.0^2 \cdot 271.297 \cdot 0.3)$
 $= 0.902$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.929, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	156.02	207.00	149.51 207.00
Radial Flange	5.22	138.00	5.00 138.00
Tangential Flange	85.46	138.00	81.89 138.00
Maximum Average	120.74	138.00	115.70 138.00
Bolting	153.72	159.00	55.06 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 260.756 mm

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Flange Input Data Values

Description: New Flange :

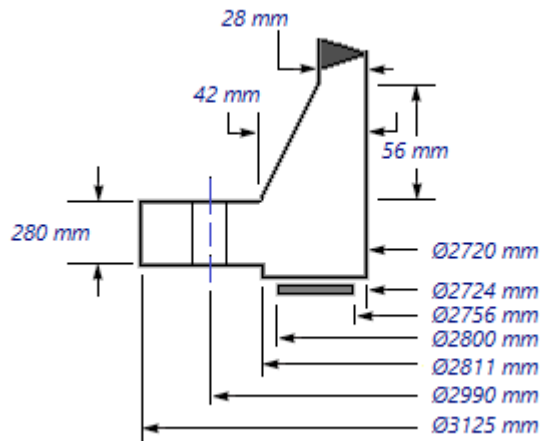
shell flg-2

Flange Type	Integral Weld Neck		
Design Pressure	P	2526.97	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	1.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	2720.000	mm
Flange Outside Diameter	A	3125.000	mm
Flange Thickness	t	280.0000	mm
Thickness of Hub at Small End	go	28.0000	mm
Thickness of Hub at Large End	gl	42.0000	mm
Length of Hub	h	56.0000	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	159.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00	N./mm ²

Warning:

It seems as if the selected bolt material does not match the bolt diameter range allowed per the material specification. Please check the material selection for the given nominal bolt diameter.

Diameter of Bolt Circle	C	2990.000	mm
Nominal Bolt Diameter	a	69.8500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		32	
Flange Face Outside Diameter	Fod	2811.000	mm
Flange Face Inside Diameter	Fid	2724.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	2800.000	mm
Gasket Inside Diameter	Gi	2756.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	6.3500	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	2725.0000	mm
Width of Partition Gasket	tp	10.0000	mm
Partition Gasket Factor	mPart	3.0000	
Partition Gasket Design Seating Stress	yPart	68.95	N./mm ²



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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (2526.97 \cdot (2720.0/2 + 1.0)) / (138.0 \cdot 1.0 - 0.6 \cdot 2526.97) + C_a$$

$$= 26.2005 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 27.0) / (1361.0 + 0.6 \cdot 27.0)$$

$$= 2705.283 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 28.0) / (1360.0 + 0.6 \cdot 28.0)$$

$$= 2806.294 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	279.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	2722.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	41.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	27.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	93.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	22.000 mm
Basic Gasket Width, $b_o = N/2$	11.000 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	8.358 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	2783.285 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 2783.2847^2 \cdot 2526.966$$

$$= 1567837.375 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P + 2 \cdot l_p \cdot b_{Part} \cdot m_{Part} \cdot P$$

$$= 2 \cdot 8.3576 \cdot \pi \cdot 2783.2847 \cdot 3.0 \cdot 2526.97$$

$$+ 2 \cdot 2725.0 \cdot 5.0 \cdot 3.0 \cdot 2526.9656$$

$$= 134055.547 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 2722.0^2 \cdot 2526.9656$$

$$= 1499553.750 \text{ Kg}$$



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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 1567837 - 1499554$$

$$= 68283.695 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(1567837 + 134056 + 0, 0)$$

$$= 1701893.000 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 8.3576 \cdot \pi \cdot 2783.285 + 68.95 \cdot 5.0 \cdot 2725.0$$

$$= 609597.562 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((1701893 + 0 + 4 \cdot 0 / 2783.28) / 159, 609598 / 159)$$

$$= 1049.716 \text{ cm}^2$$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmx]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 \cdot 69.85 + 6 \cdot 279.0 / (3.0 + 0.5)$$

$$= 617.986 \text{ mm}$$

Actual Circumferential Bolt Spacing [Bs]:

$$= C \cdot \sin(\pi / n)$$

$$= 2990.0 \cdot \sin(\pi / 32)$$

$$= 293.071 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:

$$= \max(\sqrt{ (Bs / (2a + t)) }, 1)$$

$$= \max(\sqrt{ (293.071 / (2 \cdot 69.85 + 279.0)) }, 1)$$

$$= 1.0000$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	1049.716	1085.727	
Radial Distance between Hub and Bolts:	85.725	93.000	
Radial Distance between Bolts and Edge:	66.675	67.500	
Circ. Spacing between the Bolts:	146.050	293.071	617.986

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 1085.727 \cdot 159.0 / (68.95 \cdot \pi (2800.0 + 2756.0))$$

$$= 14.344 \text{ mm}$$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 159.0 \cdot 1085.7268$$

$$= 1760277.62 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 1701893 - 1567837$$

$$= 134055.58 \text{ Kgf}$$



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Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2990.0 - 2783.2847) / 2$
 $= 103.3576 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5 ((C - B) / 2 + hg)$
 $= 0.5 ((2990.0 - 2722.0) / 2 + 103.3576)$
 $= 118.6788 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2990.0 - 2722.0 - 41.0) / 2$
 $= 113.4999 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1499554.	113.4999	1.0000	170203.
Face Pressure, Mt	68284.	118.6788	1.0000	8104.
Gasket Load, Mg	134056.	103.3576	1.0000	13856.
Gasket Seating, Matm	1760278.	103.3576	1.0000	181942.

Total Moment for Operation, Mo 192163. Kg-m.
 Total Moment for Gasket seating, Mg 181942. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_{oCor}}$ 271.098 mm
 Hub Ratio, $h/h_0 = HL / H_0$ 0.207
 Thickness Ratio, $g1/g_0 = (g1_{cor}/g_{oCor})$ 1.519

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.899
 Factor V per 4.16.5 0.407
 Factor f per 4.16.5 1.514

Stress factor from Table 4.16.4 [K]:

$= A / B$
 $= 3125.0 / 2722.0$
 $= 1.148$

Stress factor from Table 4.16.4 [Y]:

$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$
 $= (0.66845 + (5.7169 (1.148^2 \cdot \log_{10}(1.148))) / (1.148^2 - 1)) / (1.148 - 1)$
 $= 14.111$

Stress factor from Table 4.16.4 [T]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$
 $= (1.148^2 (1 + 8.55246 \cdot \log_{10}(1.148)) - 1) / ((1.0472 + 1.9448 \cdot 1.148^2) \cdot (1.148 - 1))$
 $= 1.859$

Stress factor from Table 4.16.4 [U]:

$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$
 $= (1.148^2 (1 + 8.55246 \cdot \log_{10}(1.148)) - 1) / (1.36136 (1.148^2 - 1) \cdot (1.148 - 1))$
 $= 15.506$

Stress factor from Table 4.16.4 [Z]:

$= (K^2 + 1) / (K^2 - 1)$
 $= (1.148^2 + 1) / (1.148^2 - 1)$
 $= 7.289$

Stress factor from Table 4.16.4 for integral flanges [d]:

$= U \cdot g_0^2 \cdot h_0 / V$



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$$= 15.506 \cdot 27.0^2 \cdot 271.098 / 0.407$$

$$= 7531458.500 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$

$$= 0.899 / 271.098$$

$$= 0.003 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3/d$$

$$= (279.0 \cdot 0.003 + 1) / 1.859 + 279.0^3 / 7531459$$

$$= 3.919$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$

$$= (1.5144 \cdot 192163 / 2722.0) / (3.9188 \cdot 41.0^2)$$

$$= 159.15 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$

$$= (1.5144 \cdot 181942 / 2722.0) / (3.9188 \cdot 41.0^2)$$

$$= 150.69 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$

$$= (2.2331 \cdot 192163 / 2722.0) / (3.9188 \cdot 279.0^2)$$

$$= 5.07 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$

$$= (2.2331 \cdot 181942 / 2722.0) / (3.9188 \cdot 279.0^2)$$

$$= 4.80 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$

$$= (14.1107 \cdot 192163 / (279.0^2 \cdot 2722.0)) - 7.2888 \cdot 5$$

$$= 88.56 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$

$$= (14.1107 \cdot 181942 / (279.0^2 \cdot 2722.0)) - 7.2888 \cdot 5$$

$$= 83.85 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$

$$= (159 + \max(5, 89)) / 2$$

$$= 123.86 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$

$$= (151 + \max(5, 84)) / 2$$

$$= 117.27 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 1701893 / 1085.7268$$

$$= 153.72 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (609598 / 1085.7268)$$

$$= 55.06 \text{ N./mm}^2$$



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Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	159.15	205.20	150.69	207.00
Radial Flange	5.07	136.80	4.80	138.00
Tangential Flange	88.56	136.80	83.85	138.00
Maximum Average	123.86	136.80	117.27	138.00
Bolting	153.72	159.00	55.06	159.00

Minimum Required Flange Thickness [Rigidity] 268.910 mm

Estimated M.A.W.P. (Operating) 2614 KPa.
 Estimated Finished Weight of Flange at given Thk. 4166.2 kg.
 Estimated Unfinished Weight of Forging at given Thk 4841.7 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	475.43	475.43	1.000	475.43	499.99	0.951
Tangential Flange	St	267.09	356.57	0.749	267.09	374.99	0.712
Tangential Flange	Sto	93.41	237.71	0.393	93.41	250.00	0.374
Combined	Sh/f + St	584.05	713.14	0.819	584.05	749.99	0.779
Combined	Sr + St	282.38	713.14	0.396	282.38	749.99	0.377
Combined	Sh + Sto	568.84	713.14	0.798	568.84	749.99	0.758
Radial Flange	Sr	15.29	475.43	0.032	15.29	499.99	0.031
Maximum Stress Ratio				1.000			0.951



Coefficients for f and $f2$

	f	$f2$	$f3$
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

$$= \max(1; F1_{num} / F1_{den})$$

$$= \max(1; 1.6311/1.0875)$$

$$= 1.4999$$

Hub Stress Correction Factor [f2]:

$$= \max(0.25; f_{n2}) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; f_{n3})$$

$$= \max(0.25; 0.2853/0.968) \text{ if } h/h0 < 0.35, \text{ else } \max(0.25; 0.2777/0.7921)$$

$$= 0.2947$$

In this case, the hub ratio $h/h0$, 0.2066 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sf_{max} , until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sb_{max}	507.5 N./mm ²
Minimum permissible bolt stress	Sb_{min}	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sf_{max}	506.5 N./mm ²
Target assembly gasket stress	SgT	208.0 N./mm ²
Maximum permissible gasket stress	Sg_{max}	276.0 N./mm ²
Minimum gasket operating stress	Sg_{min-O}	97.0 N./mm ²
Minimum gasket seating stress	Sg_{min-S}	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	$Phig$	0.700



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Root Area of a Single Bolt	Abs	33.9290 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.706 deg

Gasket Contact Area [Ag]:
 $= \pi/4 (GO.D.^2 - GI.D.^2) + (PartFactor \cdot Ip \cdot bPart)$
 $= \pi/4 (2800.0^2 - 2756.0^2) + (0.5 \cdot 2725.0 \cdot 5.0)$
 $= 1988.1 \text{ cm}^2$

Target Bolt Stress [Sbsel]: (Eq. O-1)
 $= SgT \cdot Ag / (nb \cdot Ab)$
 $= 208.0 \cdot 1988.141 / (32 \cdot 33.929)$
 $= 380.9 \text{ N./mm}^2$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)
 $= \min(Sbsel; Sbmax)$
 $= \min(380.882; 507.5)$
 $= 380.9 \text{ N./mm}^2$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)
 $= \max(Sbsel; Sbmin)$
 $= \max(380.882; 290.0)$
 $= 380.9 \text{ N./mm}^2$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)
 $= \min(Sbsel; Sfmax)$
 $= \min(380.882; 506.467)$
 $= 380.9 \text{ N./mm}^2$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)
 $:: Sbsel \geq Sgmin-S(Ag / (nb \cdot Abs))$
 $:: 380.882 \geq 140.0(1988.141 / (32 \cdot 33.929))$
 $:: 380.9 \geq 256.4 \text{ N./mm}^2$ Check Passes

Check if Gasket Operating Stress is maintained. (Eq. O-8)
 $:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$
 $:: 380.882 \geq (97.0 \cdot 1988.141 + \pi/4 \cdot 2526.966 \cdot 2756.0^2) / (0.7 \cdot 33.929 \cdot 32)$
 $:: 380.9 \geq 452.1 \text{ N./mm}^2$ Check Fails

Check if Gasket Maximum Stress is achieved. (Eq. O-9)
 $:: Sbsel \leq Sgmax(Ag / (nb \cdot Abs))$
 $:: 380.882 \leq 276.0(1988.141 / (32 \cdot 33.929))$
 $:: 380.9 \leq 505.4 \text{ N./mm}^2$ Check Passes

Check if Flange Rotation Limit is exceeded. (Eq. O-10)
 $:: Sbsel \leq Sfmax(thetagmax / thetafmax)$
 $:: 380.882 \leq 506.467(1.0/0.706)$
 $:: 380.9 \leq 717.0 \text{ N./mm}^2$ Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)
 $= Sbsel \cdot K \cdot Abs \cdot \phi b$
 $= 380.882 \cdot 0.2 \cdot 33.929 \cdot 69.85$
 $= 1840.9 \text{ Kg-m.}$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 92 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 158 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:



Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 181941.8 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.568 \cdot 201334 \cdot 27.0^2 \cdot 271.098 \cdot 0.3)$
 $= 0.888$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 192162.5 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.568 \cdot 189090 \cdot 27.0^2 \cdot 271.098 \cdot 0.3)$
 $= 0.999$ (should be ≤ 1)

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 181941.8 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.919 \cdot 201334 \cdot 27.0^2 \cdot 271.098 \cdot 0.3)$
 $= 0.809$ (should be ≤ 1)

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 192162.5 / 1.0 \cdot 9806.637 \cdot 0.407 / (3.919 \cdot 189090 \cdot 27.0^2 \cdot 271.098 \cdot 0.3)$
 $= 0.910$ (should be ≤ 1)

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.929, Temperature Reduction per Fig. UCS 66.1 = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 2 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -2 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual Allowed
Longitudinal Hub	157.45	207.00	148.47 207.00
Radial Flange	5.27	138.00	4.97 138.00
Tangential Flange	86.37	138.00	81.44 138.00
Maximum Average	121.91	138.00	114.95 138.00
Bolting	153.72	159.00	55.06 159.00

Minimum Req'd Thickness, ambient, no Corrosion: 262.179 mm

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 FileName : 198-C_Channel side MOC= CS Page 74 of 265
 Internal Pressure Calculations: Step: 9 10:19am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Ch. cover		1029.6	138	3	3000	187	1
LHS CH FL		1027	218	...	2720	138	2
Ch.shell		1026.7	14	3	2720	138	3
RHS CH FL		1027	308	3	2720	138	4
Shell flg-1		2527	288	1	2724	136.8	5
Shell		2526.7	28	1	2720	138	6
shell flg-3		2527	288	1	2724	136.8	7
shell flg-2		2527	288	1	2720	136.8	8
Shell -2		2526.7	28	1	2720	138	9
head- front		2526.7	36	1	2720	117.3	10

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Ch. cover		1000	1135.24	1220.53	130	122.4	1
LHS CH FL		1000	1175.45	1223.61	210	196.875	2
Ch.shell		1000	1081.58	1411.76	14	13.1867	3
RHS CH FL		1000	1081.3	1411.76	300	296.977	4
Shell flg-1		2500	2593.59	2620.56	280	271.12	5
Shell		2500	2529.99	2657.69	26.5	26.1976	6
shell flg-3		2500	2586.69	2613.66	280	267.716	7
shell flg-2		2500	2586.69	2613.66	280	268.91	8
Shell -2		2500	2529.99	2657.69	26.5	26.1976	9
head- front		2500	2512.62	2624.54	30.5	30.3531	10

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 2513 KPa.
 Shell Side MAPnc = 2614 KPa.
 Channel Side MAWP = 1063 KPa.
 Channel Side MAPnc = 1063 KPa.

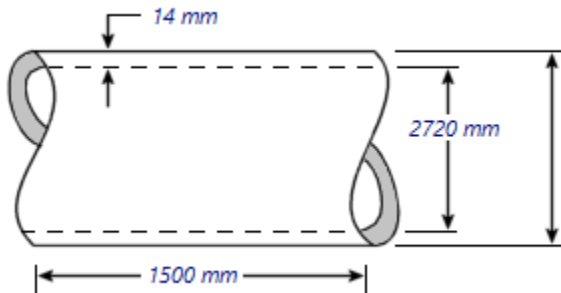
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Ch.shell

Cylindrical Shell Design Information:					
Design Pressure	P	1026.691 KPa.	Int. Design Temperature	82 °C	
Material		SA-516 70	Ext. Design Temperature	82 °C	
Length		1500.000 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.0000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	2720.000 mm	Actual Stress	127.8 N./mm ²	
Spec. Min. Thk t,e, or sa		14.0000 mm	Spec. Nominal Thk.	14.000 mm	
Surface Area		129496.5 cm ²	SG of Contents	1.000	
Element Volume		8717597.0 cm ³	Weight of Contents	8712.3 Kgf	
Empty Weight		1398.0 Kgf	Operating Weight	10110.2 Kgf	
			Comp. All. Ope. Stress	102.4 N./mm ²	
Radiography Information:					
User Defined			User Defined		
Circ. Joint Efficiency	100 %		Long. Joint Eff.	E or v	100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1026.691 \cdot 1363.0) / (138.0 \cdot 1.0 - 0.6 \cdot 1026.691) \\
 &= 10.1867 + 3.0000 = 13.1867 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.691 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 11.0) / (1363.0 + 0.6 \cdot 11.0) \\
 &= 1108 - 27 = 1082 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 14.0) / (1360.0 + 0.6 \cdot 14.0) \\
 &= 1412 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1026.691 \cdot (1363.0 + 0.6 \cdot 11.0)) / (1.0 \cdot 11.0) \\
 &= 127.840 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Ch.shell Step: 10 10:19am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.512 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 10.547$, $c = 3.0$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.959$, Temp. Reduction = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -20 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1062.805 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.959	Temperature Difference	2 °C
UCS-66(c) red.	No	Computed Min. Temp.	-20 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Ch.shell

Cylindrical Shell From 30 to 40 Ext. Chart: CS-2 at 82 °C

Elastic Modulus from Chart: CS-2 at 82 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	2748.00	1500.00	249.82	0.5459	0.0006335	63.33

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 63.3309) / (3 \cdot 249.8182) = 337.9909 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
1.085	2748.00	1500.00	2533.26	0.5459	0.0000193	1.92

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 1.9249) / (3 \cdot 2533.2627) = 1.013 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
11.000	2748.00	107881.41	249.82	39.2582	0.0000177	1.76

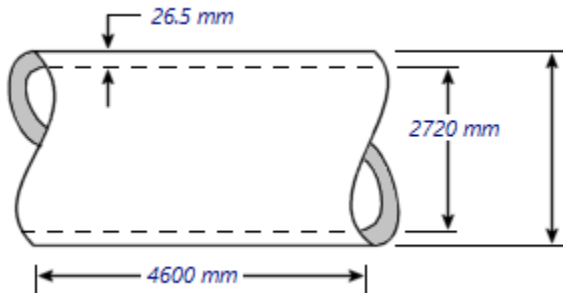
$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 1.7648) / (3 \cdot 249.8182) = 9.4186 \text{ KPa.}$$



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 FileName : 198-C_Channel side MOC= CS
 Element: Shell Step: 11 10:19am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell

Cylindrical Shell Design Information:			
Design Pressure P	2526.671 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	4600.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2720.000 mm	Actual Stress	136.4 N./mm ²
Spec. Min. Thk t _e , or sa	26.5000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	401168.9 cm ²	SG of Contents	1.000
Element Volume	13678840.0 cm ³	Weight of Contents	13670.5 Kg
Empty Weight	8618.1 Kg	Operating Weight	22288.6 Kg
		Comp. All. Ope. Stress	105.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2526.671 \cdot 1361.0) / (138.0 \cdot 1.0 - 0.6 \cdot 2526.671) \\
 &= 25.1976 + 1.0000 = 26.1976 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.671 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 25.5) / (1361.0 + 0.6 \cdot 25.5) \\
 &= 2557 - 27 = 2530 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 26.5) / (1360.0 + 0.6 \cdot 26.5) \\
 &= 2658 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2526.671 \cdot (1361.0 + 0.6 \cdot 25.5)) / (1.0 \cdot 25.5) \\
 &= 136.379 \text{ N./mm}^2
 \end{aligned}$$



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FileName : 198-C_Channel side MOC= CS Page 78 of 265
Element: Shell Step: 11 10:19am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.019 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 26.5$, $t_r = 25.056$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.983$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	0 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2512.621 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	26.500 mm	Basic MDMT	1 °C
Stress/Thk. Ratio	0.983	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E^*	100 %

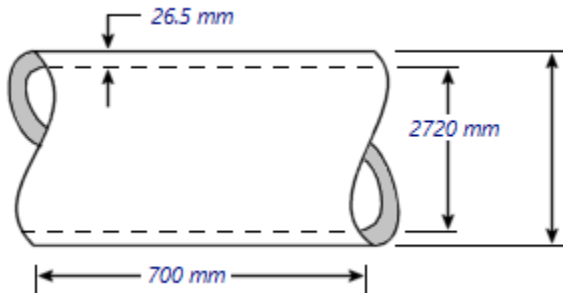
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 FileName : 198-C_Channel side MOC= CS Page 79 of 265
 Element: Shell -2 Step: 12 10:19am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Shell -2

Cylindrical Shell Design Information:			
Design Pressure P	2526.671 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2720.000 mm	Actual Stress	136.4 N./mm ²
Spec. Min. Thk t _e , or sa	26.5000 mm	Spec. Nominal Thk.	28.000 mm
Surface Area	61047.4 cm ²	SG of Contents	1.000
Element Volume	2821302.2 cm ³	Weight of Contents	2819.6 Kg
Empty Weight	1311.5 Kg	Operating Weight	4131.0 Kg
		Comp. All. Ope. Stress	105.6 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (2526.671 \cdot 1361.0) / (138.0 \cdot 1.0 - 0.6 \cdot 2526.671) \\
 &= 25.1976 + 1.0000 = 26.1976 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.671 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 25.5) / (1361.0 + 0.6 \cdot 25.5) \\
 &= 2557 - 27 = 2530 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 1.0 \cdot 26.5) / (1360.0 + 0.6 \cdot 26.5) \\
 &= 2658 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (2526.671 \cdot (1361.0 + 0.6 \cdot 25.5)) / (1.0 \cdot 25.5) \\
 &= 136.379 \text{ N./mm}^2
 \end{aligned}$$



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Element: Shell -2 Step: 12 10:19am Sep 9,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.019 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 26.5$, $t_r = 25.056$, $c = 1.0$ mm, $E^* = 1.0$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.983$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	0 °C

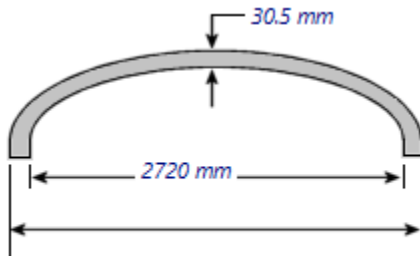
MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2512.621 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	26.500 mm	Basic MDMT	1 °C
Stress/Thk. Ratio	0.983	Temperature Difference	1 °C
UCS-66(c) red.	No	Computed Min. Temp.	0 °C
		Joint Efficiency E^*	100 %

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Internal Pressure Calculation Results:

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Internal Pressure Results for: head- front

Elliptical Head Design Information:			
Design Pressure P	2526.671 KPa.	Int. Design Temperature	230 °C
Material	SA-516 70	Ext. Design Temperature	25 °C
Length	50.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	2720.000 mm	Actual Stress	137.3 N./mm ²
Spec. Min. Thk t,e, or sa	30.5000 mm	Spec. Nominal Thk.	36.000 mm
Surface Area	88885.2 cm ²	SG of Contents	1.000
Element Volume	2925238.0 cm ³	Weight of Contents	2923.5 Kg
Empty Weight	2657.3 Kg	Operating Weight	5580.7 Kg
		Comp. All. Ope. Stress	105.6 N./mm ²
		Aspect Ratio	2.00

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff. E or v	85 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.000 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (2526.671 \cdot 2722.0 \cdot 0.999) / (2 \cdot 138.0 \cdot 0.85 - 0.2 \cdot 2526.671)$$

$$= 29.3531 + 1.0000 = 30.3531 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 26.671 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 29.5) / (0.999 \cdot 2722.0 + 0.2 \cdot 29.5)$$

$$= 2539 - 27 = 2513 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 0.85 \cdot 30.5) / (1.0 \cdot 2720.0 + 0.2 \cdot 30.5)$$

$$= 2625 \text{ KPa.}$$



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 Element: head- front Step: 13 10:19am Sep 9,2026

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (2526.671 \cdot (0.999 \cdot 2722.0 + 0.2 \cdot 29.5)) / (2 \cdot 0.85 \cdot 29.5)$$

$$= 137.312 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (2526.671 \cdot 1361.0) / (138.0 \cdot 0.85 - 0.6 \cdot 2526.671) + 1.0$$

$$= 30.702 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 26.671 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 0.85 \cdot 35.0) / (1361.0 + 0.6 \cdot 35.0)$$

$$= 2970 - 27 = 2944 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (2722.0 / (2 \cdot 681.0))^2) / 6$$

$$= 0.999022$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 5.620 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 30.5, tr = 29.189, c = 1.0 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.841$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 4 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -4 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 36.0, tr = 29.535, c = 1.0 mm, E* = 0.85
 Thickness Ratio = $tr \cdot E* / (tg - c) = 0.717$, Temp. Reduction = 16 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 9 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -7 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	2512.621 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	36.000 mm	Basic MDMT	9 °C
Stress/Thk. Ratio	0.717	Temperature Difference	16 °C
UCS-66(c) red.	No	Computed Min. Temp.	-7 °C
		Joint Efficiency E*	85 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 34.000 mm
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 17.000 mm

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3266.407 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3250.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3368.346 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2763.883 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3575.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3575.000 KPa.
App 27-4	= M.A.W.P.	2512.621 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1381.647 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1352.241 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1169.086 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1062.805 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell	171.7	234.0	0.734	1378.90
Shell	183.2	234.0	0.783	3395.01
Shell -2	183.2	234.0	0.783	3395.01
head- front	184.5	234.0	0.788	3395.01

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	138.00	1.000
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	138.00	1.000
Shell inlet	Nozzle	118.00	118.00	1.000
Shell inlet	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Ch. cover	187.00	187.00	1.000
LHS CH FL	138.00	138.00	1.000
Ch.shell	138.00	138.00	1.000
RHS CH FL	138.00	138.00	1.000
Shell flg-1	138.00	136.80	1.009
Shell	138.00	138.00	1.000
shell flg-3	138.00	136.80	1.009



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shell flg-2	138.00	136.80	1.009
Shell -2	138.00	138.00	1.000
head- front	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	187.00	171.60	1.090
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	187.00	171.60	1.090
Tube Material	187.00	187.00	1.000
Tubesheet Material	187.00	187.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	12.32	216.00	0.057
Tube outlet	12.32	216.00	0.057
Shell inlet	61.30	216.00	0.284
Shell outlet	70.26	153.40	0.458

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	3266.407 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	3250.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	3368.346 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	2763.883 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	3575.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	3575.000 KPa.
App 27-4	= M.A.W.P.	2512.621 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1381.647 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1300.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	1352.241 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1169.086 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1430.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1430.000 KPa.
App 27-4	= M.A.W.P.	1062.805 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 2500.0 \cdot 1.0$$

$$= 3250.000 \text{ KPa.}$$

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

$$= \text{Test Factor} \cdot \text{Design Pressure} \cdot \text{Stress Ratio}$$

$$= 1.3 \cdot 1000.0 \cdot 1.0$$

$$= 1300.000 \text{ KPa.}$$

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Ch.shell	165.2	234.0	0.706	1326.66
Shell	176.9	234.0	0.756	3276.66
Shell -2	176.9	234.0	0.756	3276.66
head- front	178.1	234.0	0.761	3276.66

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Tube inlet	Nozzle	118.00	118.00	1.000
Tube inlet	Pad	138.00	138.00	1.000
Tube outlet	Nozzle	118.00	118.00	1.000
Tube outlet	Pad	138.00	138.00	1.000
Shell inlet	Nozzle	118.00	118.00	1.000
Shell inlet	Pad	138.00	138.00	1.000
Shell outlet	Nozzle	118.00	118.00	1.000
Shell outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Description	Ambient	Operating	Ratio
Ch. cover	187.00	187.00	1.000
LHS CH FL	138.00	138.00	1.000
Ch.shell	138.00	138.00	1.000
RHS CH FL	138.00	138.00	1.000
Shell flg-1	138.00	136.80	1.009
Shell	138.00	138.00	1.000
shell flg-3	138.00	136.80	1.009
shell flg-2	138.00	136.80	1.009
Shell -2	138.00	138.00	1.000
head- front	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Floating Head Matl.	187.00	171.60	1.090
Backing Ring Matl.	137.90	137.90	1.000
Flange Material	187.00	171.60	1.090
Tube Material	187.00	187.00	1.000
Tubesheet Material	187.00	187.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
Tube inlet	11.85	216.00	0.055
Tube outlet	11.85	216.00	0.055
Shell inlet	59.16	216.00	0.274
Shell outlet	67.81	153.40	0.442

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		3250.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2720.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2720.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5600
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5350.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5000.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	479.99 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1262.920 mm
Largest Center-to-Center Tube Distance	U1	15.8750 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	48355.9 cm ²
Perimeter of Tube Layout	Cp	8128.3 mm



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Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	175.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2806.000 mm
Dimension G for the Channel Side	Gc	2777.669 mm
Area of the Untubed Lanes	AL	435.8 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	3144.000 mm
Flange Inside Diameter	B	2724.000 mm
Flange Face Outside Diameter	Fod	2809.000 mm
Flange Face Inside Diameter	Fid	2724.000 mm
Gasket Outside Diameter	Go	2798.000 mm
Gasket Inside Diameter	Gi	2756.000 mm
Small end Hub thk.	g0	28.0000 mm
Large end Hub thk.	g1	42.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	2725.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	3004.000 mm
Nominal Bolt Diameter	dB	69.8500 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	32
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm
Alternate Flange Operating Bolt Load, Wm1		2183044.50 KgF
Alternate Flange Seating Bolt Load, Wm2		597492.62 KgF
Alternate Flange Design Bolt Load, W		1760277.62 KgF



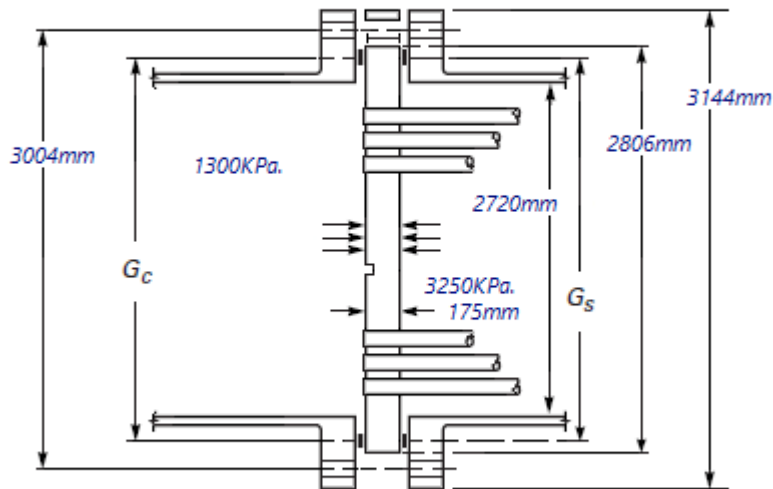
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Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No

Tubesheet Gasketed w/Shell and Channel



Configuration d:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.



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Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
 TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1300.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 1300.0)$
 $= 0.0881 + 0.0000 = 0.0881 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-5 at 21.11 °C
 Elastic Modulus for Material 195117184.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5000.00	12.04	50.0000	0.0075908	200.17

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 200.1697) / (3 \cdot 12.0379) = 22169.7383 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7193	25.40	5000.00	35.31	50.0000	0.0008822	86.08

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 86.0753) / (3 \cdot 35.3104) = 3250.0471 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7193 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	22169.74 KPa.
Required External Design Pressure	3250.00 KPa.
Required Thickness due to Shell Side pressure	0.7193 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

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Gasket Contact Width,	$N = (Goc - Gic) / 2$	21.000 mm
Basic Gasket Width,	$b0 = N / 2$	10.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	8.162 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2781.675 mm

Warning: Available bolt area, 1085.727 cm² for then shell flange is < required 1346.487 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME Stationary Tubesheet Calculations for Configuration d:



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Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \quad (\text{modified for corrosion if present})$$

$$= 0/175.0$$

$$= 0.0000 \quad (\text{must be } 0 \leq \rho \leq 1)$$

Effective tube hole diameter [d^*]:

$$= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.19994E+09/0.19994E+09) \cdot (186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 1262.92 + 25.4$$

$$= 2551.240 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2551.2402/2$$

$$= 1275.6201 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 31.75 / \sqrt{ 1 - 4 \cdot \min(435.77 \cdot 100.0; 4 \cdot 2551.24 \cdot 31.75) / (\pi \cdot 2551.24^2) }$$

$$= 31.8862 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8862 - 25.4) / 31.8862$$

$$= 0.2034$$

Ratio [ρ_s]:

$$= a_s / a_o$$

$$= 1390.8345/1275.6201$$

$$= 1.0903$$

Ratio [ρ_c]:

$$= a_c / a_o$$

$$= 1388.8345/1275.6201$$

$$= 1.0888$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 5600 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1275.6201))^2$$

$$= 0.6140$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot a_o))^2$$

$$= 1 - 5600 (25.4 / (2 \cdot 1275.6201))^2$$

$$= 0.4449$$

Determine the value [$h'g$]:

$$= \max((h_g - CATC); 0) \quad (\text{For pressure only cases})$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$



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ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

$\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:

$\beta_C, k_C, \lambda_C, \delta_C$ are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 5.511811$; $\mu^* = 0.203417$; $E^* = 31868330$ KPa.

$E^*/E = 0.159387$; $\nu^* = 0.453616$

Note: As h/p (5.512) is > 2, data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:

$$= \left(\left(24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.454^2)5600 \cdot 0.19994E+09 \cdot 2.11 \right. \right. \\ \left. \left. (25.4 - 2.11)1275.6201^2 \right) / (31868330 \cdot 5000.0 \cdot 175.0^3) \right)^{1/4}$$

$$= 6.6862$$

Values from Table 4.18.3

$$Z_d = 0.004955 ; Z_v = 0.022169 ; Z_m = 0.211858$$

$$Z_a = 0.198592E+03 ; Z_w = 0.022169$$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:

$$= A / Do$$

$$= 2806.0/2551.2402$$

$$= 1.0999$$

Compute coefficient $[F]$:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_C + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31868330 \cdot (0 + 0 +$$

$$0.19994E+09 \cdot \ln(1.1))$$

$$= 0.3263$$

Compute parameter $[\phi]$:

$$= (1 + \nu^*)F$$

$$= (1 + 0.4536)0.3263$$

$$= 0.4743$$

Compute parameter $[Q1]$:

$$= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0903 - 1 - 0.4743 \cdot 0.0222) / (1 + 0.4743 \cdot 0.2119)$$

$$= 0.072519$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:

$$= (G_c - G_s) / Do \text{ (config d)}$$

$$= (2777.6689 - 2781.6689) / 2551.2402 = -0.00157$$

Calculate parameter $[\omega_s]$:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.0903 \cdot 0 \cdot 0 \cdot 0 (1 + 175.0 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter $[\omega^*s]$:

$$= ao^2(\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$$

$$= 1275.62^2(1.09^2 - 1) \cdot (1.09 - 1) / 4 - 0$$

$$= 6936.9199 \text{ mm}^2$$



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Parameter [wc]:

$$= pc \cdot kc [\delta C (CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) - \zeta c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2)]$$

$$= 1.089 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 175.0) - 0 (0/175.0 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [w*c]:

$$= ao^2 [(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - wc$$

$$= 1275.62012^2 [(1.08875^2 + 1) \cdot (1.08875 - 1) / 4 - (1.09032 - 1) / 2] - 0$$

$$= 5417.4453 \text{ mm}^2$$

Pressure [P*S]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [P*C]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$Pe = Ps - Pt$$

$$= 3250.0 - 1300.0 = 1950.0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((w*s \cdot Ps - w*c \cdot Pt) - (ws \cdot P*s - wc \cdot P*c)) + W* \cdot yb / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((6936.92 \cdot 3250.0 - 5417.445 \cdot 1300.0) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0. + 2183044 \cdot -0.002 / (2 \cdot \pi)] / (1 + 0.47429 \cdot 0.21186)$$

$$= 941.5175 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= 0.073 + 2 \cdot 941.517 / (1950.0 \cdot 1275.62^2)$$

$$= 0.07834$$

Fm Value from Table 4.18.3 = 0.065883

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0659 / 0.2034) \cdot (2 \cdot 1275.6201 / (175.0 - 0))^2 \cdot 1950.0$$

$$= 201.3554 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot Sy = 1.5 \cdot 0.95 \cdot 479.99 = 683.99 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σf]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (h - h'g))^2 \cdot Pe$$

$$= (1.5 \cdot 0.0563 / 0.2034) \cdot (2 \cdot 1275.6201 / (87.801 - 0))^2 \cdot 1950.0$$

$$= 683.9876 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + Cats + Catc$$

$$= 87.8011 + 0 + 0$$

$$= 87.8011 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τlimit]:

$$= (2 \cdot \mu \cdot h / ao) \cdot (0.8 \cdot \beta T \cdot Sy)$$

$$= (2 \cdot 0.2 \cdot 175.0 / 1275.62) \cdot 364.793$$

$$= 20.0181 \text{ N./mm}^2$$



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Shear Stress is not required to be computed; [Pe] <= TauIlimit

Tube Sheet Shear Stress - final thickness [tf]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (48355.9 \cdot 1950.0) / (0.2 \cdot 15.901 \cdot 8128.3)$$

$$= 364.7934 \text{ N./mm}^2$$

Reqd Tube Sheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{atS} + C_{atC}$$

$$= 15.9014 + 0 + 0$$

$$= 15.9014 \text{ mm}$$

Reqd Tube Sheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(87.8011; 15.9014)$$

$$= 87.8011 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -0.8838, F_{tmax} = 6.6866$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((3250.0 \cdot 0.4449 - 1300.0 \cdot 0.614) - (1950.0) \cdot -0.884) / (0.614 - 0.4449)$$

$$= 14.0208 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((3250.0 \cdot 0.4449 - 1300.0 \cdot 0.614) - (1950.0) \cdot 6.687) / (0.614 - 0.4449)$$

$$= -73.2725 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 73.272 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= \beta_T \cdot S_{y,t} = 455.9918 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_a^4); 1.25)$$

$$= \max((3.25 - 0.25(0.005 + 0.078 \cdot 0.022)6.686^4); 1.25)$$

$$= 1.2500 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{y,t})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994E+09 / 479)}$$

$$= 93.8884$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 8.268$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (479 / 1.25) \cdot (1 - 0 / (2 \cdot 93.888))$$



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= 455.992 N./mm² (Never greater than Sot)

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

= $\sigma_{t,max} \cdot \text{Tube Area}$
 = 73.27 • 1.5438
 = 1153.49 Kgf

Tube Metal Area [At]:

= $\pi (d_o - t) t$
 = $\pi (25.4 - 2.11) 2.11$
 = 1.544 cm²

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

= $\pi \cdot t (d_o - t) S$
 = $\pi \cdot 2.11 (25.4 - 2.11) 455.992$
 = 7178.441 Kgf

Groove Weld Strength [Fg]:

= $0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w$ (but not > Ft)
 = $0.85 \cdot \pi \cdot 1.651 (25.4 + 0.67 \cdot 1.651) 455.99$
 = 5433.6445 Kgf

Weld Strength Factor [fw]:

= S / S_w where $S_w = \min(S \text{ and } S_t)$
 = 455.992/455.992
 = 1.000

Design Strength Ratio [fd]:

= 1 for full strength welds

Fillet Weld Strength [Ff]:

= $0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w$ (but not > Ft)
 = $0.55 \cdot \pi \cdot 1.651 (25.4 + 0.67 \cdot 1.651) 455.99$
 = 3515.8875 Kgf

Minimum Length of Weld [ar]:

= $2 \left(\sqrt{(0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d} - 0.75 \cdot d_o \right)$
 = $2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) 1.0 \cdot 1.0} - 0.75 \cdot 25.4 \right)$
 = 2.667 mm

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 7178.44 is >= tube strength 7178.44 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	201.4 <=	684.0 N./mm ²	Pass
Tubesheet Shear Stress	33.1 <=	364.8 N./mm ²	Pass
Maximum Stress in the Tubes	73.3 <=	456.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-73.3 <=	-456.0 N./mm ²	Pass
Maximum Force on any one Tube	1153.5 <=	7178.4 Kgf	Pass



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Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	87.801	175.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	1.651	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	71.735	5.988	138	684	22	365	Ps+Pt-Th	D1 Ok
D2uc	116.768	14.969	337	684	55	365	Ps+Pt-Th	D2 Ok
D3uc	87.801	8.982	201	684	33	365	Ps+Pt-Th	D3 Ok
Max:	116.7684	14.969	mm	0.492		0.151	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses				Tube Loads		Shell Stress		Channel Stress		Pass
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	47	455	-12	-455	745	7178	...	...	...	...	Ok
D2uc	120	455	-120	-455	1891	7178	...	...	...	...	Ok
D3uc	73	455	-73	-455	1153	7178	...	...	...	...	Ok
Max RATIO	0.263			0.263		0.263	...			...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	116.768	175.000	Ok
Tube thickness :	0.719	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	1.651	Ok
Tube-tubesheet groove weld leg :	1.333	1.651	Ok



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Note: This is a full strength Tube to Tubesheet Joint.

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Input Echo, Tubesheet Number 1, Description: Floating tubesh

Shell Data:

Main Shell Description: Shell

Shell Test Pressure		3250.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2720.000 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Test Pressure		1300.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2720.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5600
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5350.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5000.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	479.99 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	1262.920 mm
Largest Center-to-Center Tube Distance	U1	15.8750 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	48355.9 cm ²
Perimeter of Tube Layout	Cp	8128.3 mm



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Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	21.11 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	175.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2710.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	435.8 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2710.000 mm
Flange Inside Diameter	B	2568.000 mm
Flange Face Outside Diameter	Fod	2600.000 mm
Flange Face Inside Diameter	Fid	2568.000 mm
Gasket Outside Diameter	Go	2597.000 mm
Gasket Inside Diameter	Gi	2571.000 mm
Gasket Factor,	m	3.75
Gasket Design Seating Stress	y	52.40 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2650.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series		UNC Thread Series
Number of Bolts	n	52
Bolt Material		SA-193 B7M
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	138.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

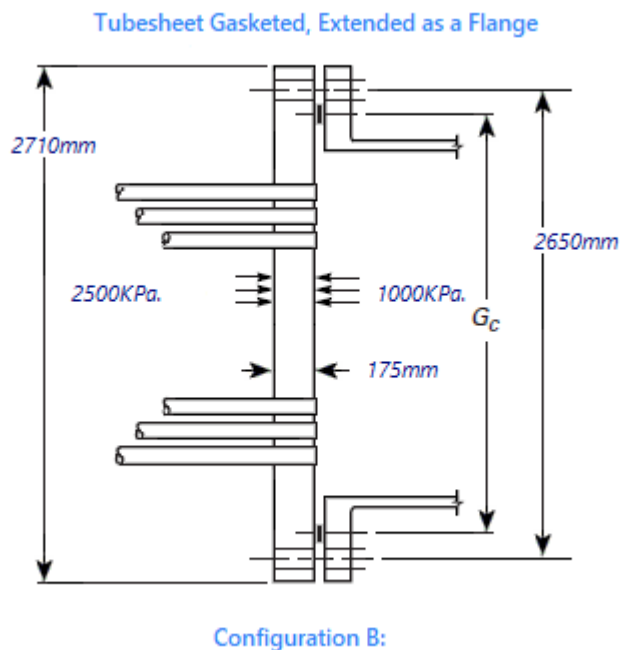
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	159.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No
---	--	----



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Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
TubeSheet - TM-1 Material Group H



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Elastic Mod. at Design Temperature	21.1 °C	0.19994E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$$
 per Appendix 1-1 (a)(1)

$$= (1300.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 1300.0)$$

$$= 0.0881 + 0.0000 = 0.0881 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	HA-5	at	21.11 °C
Elastic Modulus for Material			195117184.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5000.00	12.04	50.0000	0.0075908	200.17

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 200.1697) / (3 \cdot 12.0379) = 22169.7383 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.7193	25.40	5000.00	35.31	50.0000	0.0008822	86.08

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 86.0753) / (3 \cdot 35.3104) = 3250.0471 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.7193 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	1300.00 KPa.
Allowable External Pressure at Corroded thickness	22169.74 KPa.
Required External Design Pressure	3250.00 KPa.
Required Thickness due to Shell Side pressure	0.7193 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2584.156 mm

Warning: Available bolt area, 434.115 cm² for then shell flange is < required 530.95 cm²

Note that for the pressure test case, ASME VIII-2 paragraph 4.18.13 (c)(9) does not require that the bolt area requirement be met.

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$



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$$= 0/175.0$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/S) \cdot (p); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.19994E+09/0.19994E+09) \cdot (186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot ro + dt$$

$$= 2 \cdot 1262.92 + 25.4$$

$$= 2551.240 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [ao]:

$$= Do / 2$$

$$= 2551.2402/2$$

$$= 1275.6201 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{ 1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2) }$$

$$= 31.75 / \sqrt{ 1 - 4 \cdot \min(435.77 \cdot 100.0 ; 4 \cdot 2551.24 \cdot 31.75) / (\pi \cdot 2551.24^2) }$$

$$= 31.8862 \text{ mm}$$

Effective ligament efficiency for bending [μ*]:

$$= (p* - d*) / p*$$

$$= (31.8862 - 25.4) / 31.8862$$

$$= 0.2034$$

Ratio [ps]:

$$= as / ao$$

$$= 1292.0778/1275.6201$$

$$= 1.0129$$

Ratio [pc]:

$$= ac / ao$$

$$= 1292.0778/1275.6201$$

$$= 1.0129$$

Compute parameter [xt]:

$$= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$$

$$= 1 - 5600 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1275.6201))^2$$

$$= 0.6140$$

Determine parameter [xs]:

$$= 1 - Nt(dt / (2 \cdot ao))^2$$

$$= 1 - 5600 (25.4 / (2 \cdot 1275.6201))^2$$

$$= 0.4449$$

Determine the value [h'g]:

$$= \max((hg - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

βs, ks, λs, δs are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:



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β_C , k_c , λ_C , δ_C are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 5.511811$; $\mu^* = 0.203417$; $E^* = 31868330$ KPa.
 $E^*/E = 0.159387$; $\nu^* = 0.453616$

Note: As h/p (5.512) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:
 $= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.454^2)5600 \cdot 0.19994E+09 \cdot 2.11$
 $(25.4 - 2.11)1275.6201^2) / (31868330 \cdot 5000.0 \cdot 175.0^3))^{1/4}$
 $= 6.6862$

Values from Table 4.18.3

$Z_d = 0.004955$; $Z_v = 0.022169$; $Z_m = 0.211858$
 $Z_a = 0.198592E+03$; $Z_w = 0.022169$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:
 $= A / Do$
 $= 2710.0/2551.2402$
 $= 1.0622$

Compute coefficient $[F]$:
 $= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$
 $= (1 - 0.45) / 31868330 \cdot (0 + 0 +$
 $0.19994E+09 \cdot \ln(1.06))$
 $= 0.2069$

Compute parameter $[\phi]$:
 $= (1 + \nu^*) F$
 $= (1 + 0.4536) 0.2069$
 $= 0.3008$

Compute parameter $[Q_1]$:
 $= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.0129 - 1 - 0.3008 \cdot 0.0222) / (1 + 0.3008 \cdot 0.2119)$
 $= 0.005859$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:
 $= (G_c - C) / Do$ (config b)
 $= (2584.1555 - 2650.0) / 2551.2402 = -0.02581$

Calculate parameter $[\omega_s]$:
 $= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$
 $= 1.0129 \cdot 0 \cdot 0 \cdot 0 (1 + 175.0 \cdot 0)$
 $= 0.0000 \text{ mm}^2$

Calculate parameter $[\omega_{*s}]$:
 $= ao^2(p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$
 $= 1275.62^2(1.013^2 - 1) \cdot (1.013 - 1) / 4 - 0$
 $= 136.3001 \text{ mm}^2$

Parameter $[\omega_c]$:
 $= p_c \cdot k_c [\delta_C (Cm_{\theta} \cdot \beta_C + Cq_{\Delta} \cdot h \cdot \beta_C^2) -$
 $\zeta_c (Cm_{\theta} + h \cdot Cq_{\theta} \cdot h \cdot \beta_C / 2))$
 $= 1.013 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 175.0) -$
 $0 (0/175.0 + 0 \cdot 0/2))$
 $= 0.0000 \text{ mm}^2$



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Parameter [$\omega \cdot c$]:
 $= a_o^2 [(p_c^2 + 1) \cdot (p_c - 1)/4 - (p_s - 1)/2] - \omega c$
 $= 1275.62012^2 [(1.0129^2 + 1) \cdot (1.0129 - 1)/4 - (1.0129 - 1)/2] - 0$
 $= 136.3000 \text{ mm}^2$

Pressure [$P \cdot S$]:
 $= 0$ For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [$P \cdot C$]:
 $= 0$ For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$P_e = P_s - P_t$
 $= 3250.0 - 1300.0 = 1950.0 \text{ KPa.}$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q_2]:
 $= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot \gamma_b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$
 $= [((136.3 \cdot 3250.0 - 136.3 \cdot 1300.0) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0. + 610868 \cdot -0.026 / (2 \cdot \pi)] / (1 + 0.30082 \cdot 0.21186)$
 $= -2333.3936 \text{ Kgf}$

Calculate factor [Q_3]:
 $= Q_1 + 2 \cdot Q_2 / (P_e \cdot a_o^2)$
 $= 0.006 + 2 \cdot -2333.394 / (1950.0 \cdot 1275.62^2)$
 $= -0.00856$

Fm Value from Table 4.18.3 = 0.033708

Tubesheet Bending Stress - original thickness [σ]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (H - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.0337 / 0.2034) \cdot (2 \cdot 1275.6201 / (175.0 - 0))^2 \cdot 1950.0$
 $= 103.0190 \text{ N./mm}^2$

Allowable Tubesheet Bending Stress [σ allowed]:
 $= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 479.99 = 683.99 \text{ N./mm}^2, 4.18.233$

Tubesheet Bending Stress - final thickness [σ_f]:
 $= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$
 $= (1.5 \cdot 0.0106 / 0.2034) \cdot (2 \cdot 1275.6201 / (38.13 - 0))^2 \cdot 1950.0$
 $= 683.9684 \text{ N./mm}^2$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [H_{ReqB}]:
 $= h + C_{ats} + C_{atc}$
 $= 38.1299 + 0 + 0$
 $= 38.1299 \text{ mm}$

H reqd. = H_{ReqB} = 38.1299 mm

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	103.0 <=	684.0 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 3 Load Cases:



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Case#	--Reqd. Thk. + CA	Bend	---- Tubesheet		Stresses		Case Type	Pass/Fail	
	Tbsht Extnsn		Allwd	Shear	Allwd				
D1uc	49.646	34.929	94	684	...	...	Ps+Pt-Th	D1	Ok
D2uc	70.587	...	196	684	...	...	Ps+Pt-Th	D2	Ok
D3uc	38.130	...	103	684	...	...	Ps+Pt-Th	D3	Ok
Max:	70.5874	34.929 mm	0.286		...		(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	70.587	175.000	Ok
Tubesheet thickness flanged extension :	34.929	159.000	Ok
Tube thickness :	0.719	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	1.651	Ok
Tube-tubesheet groove weld leg :	0.000	1.651	Ok



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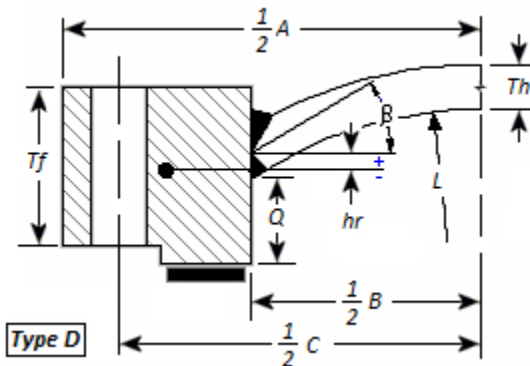
Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	2500.00	KPa.
Design Temperature for Spherical Head	Temp	230.00	°C
Head Material		SA-240	
Head Material UNS Number		S32205	
Head Allowable Stress at Temperature	Sho	187.00	N./mm ²
Head Allowable Stress at Ambient	Sha	187.00	N./mm ²
Crown Radius for Spherical Head	L	2316.000	mm
Head Thickness	Th	50.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS Number		S32205	
Flange Allowable Stress at Temperature	Sfo	187.00	N./mm ²
Flange Allowable Stress at Ambient	Sfa	187.00	N./mm ²
Flange Outside Diameter	Fod	2710.000	mm
Flange Inside Diameter	Fid	2568.000	mm
Flange Thickness	Tf	280.0000	mm
Bolt Material		SA-193 B7M	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	138.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00	N./mm ²
Diameter of Bolt Circle	C	2650.000	mm
Nominal Bolt Diameter	dB	38.1000	mm
Type of Threads		UNC Thread Series	
Number of Bolts		52	
Full face Gasket Flange Option		Program Selects	
Flange Face Outside Diameter	Ffod	2600.000	mm
Flange Face Inside Diameter	Ffid	2568.000	mm
Gasket Outside Diameter	Go	2597.000	mm
Gasket Inside Diameter	Gi	2571.000	mm
Gasket Factor,	m	3.7500	
Gasket Design Seating Stress	y	52.40	N./mm ²
Flange Facing Sketch		Code Sketch 1a	
Column for Gasket Seating		Code Column I	
Gasket Thickness		4.5000	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		2.1689	mm
The Flange is not Slotted.			
This unit does not have a Backing Ring.			
Use Full Bolt Load in Flange Design		Yes	



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Internal Pressure results for Spherical Heads:

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Thickness Due to Internal Pressure[thr]:
 $= 5PL / 6Sho$
 $= (5 \cdot 1000.0 \cdot 2316.0) / (6 \cdot 186)$
 $= 10.3216 \text{ mm}$

Maximum Allowable Working Pressure at Given Thickness[Pa]:
 $= 6Sho \cdot (Th - Cass - Cats) / 5L$
 $= (6 \cdot 186 \cdot 50.0) / (5 \cdot 2316.0)$
 $= 4844.19 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold[Pnc]:
 $= 6Sha / 5L$
 $= (6 \cdot 186 \cdot 50.0) / (5 \cdot 2316.0)$
 $= 4844.19 \text{ KPa.}$

Actual Stress at given Pressure and Thickness[Sact]:
 $= 5PL / 6(Th - Cass - Cats)$
 $= (5 \cdot 1000.0 \cdot 2316.0) / (6 \cdot 50.0)$
 $= 39. \text{ N./mm}^2$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 1.602 %

External Pressure results, Spherical Head ASME Code, Section VIII Division 1, 2025

External Pressure Chart HA-5 at 230.00 °C
 Elastic Modulus for Material 180805456.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	50.0000	mm
Outside Crown Radius of Shell	OD/2	2366.0000	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	47.3200	
Geometry Factor, A	f(D/T, LD)	0.0026416	
Materials Factor, B	f(A, Chart)	123.6042	N./mm ²
Maximum Allowable Working Pressure		2611.94	KPa.
$EMAWP = B / ((OD/T)/2) = 123.6042 / (94.64/2.0) = 2611.9395$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	48.2109	mm
Outside Diameter of Shell	ODCA	4732.000	mm
Diameter / Thickness Ratio	(D/T)	98.1521	
Geometry Factor, A	f(DT, LD)	0.0025471	
Materials Factor, B	f(A, Chart)	122.7097	N./mm ²
Maximum Allowable Working Pressure		2500.25	KPa.
$EMAWP = B / ((D/T)/2) = 122.7097 / (98.1521/2) = 2500.2527$			



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Intermediate Calculations for Flanged portion:

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Gasket Contact Width, $N = (Goc - Gic) / 2$ 13.000 mm
 Basic Gasket Width, $b0 = N / 2$ 6.500 mm
 Effective Gasket Width, $b = \text{sqrt}(b0) * 2.5$ 6.422 mm
 Gasket Reaction Diameter, $G = Go - 2.0 * b$ 2584.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:
 $= 2a + 6t / (m + 0.5)$
 $= 2 * 38.1 + 6 * 280.0 / (3.75 + 0.5)$
 $= 471.494 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C * \sin(\pi / n)$
 $= 2650.0 * \sin(\pi / 52)$
 $= 160.003 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:
 $= \max(\sqrt{Bs / (2a + t)}, 1)$
 $= \max(\sqrt{160.003 / (2 * 38.1 + 280.0)}, 1)$
 $= 1.0000$

Bolting Information for UNC Thread Series (Non Mandatory):

Total Area of Bolts 434.115 cm²

	Minimum	Actual	Maximum
<i>Radial Distance between Bolts and Edge:</i>	<i>38.100</i>	<i>30.000</i>	
Circ. Spacing between the Bolts:	82.550	160.003	471.494

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	534839.8	Kgf
Contact Load on Gasket Surfaces	Hp	39875.9	Kgf
Hydrostatic End Load at Flange ID	Hd	528173.3	Kgf
Pressure Force on Flange Face	Ht	6666.5	Kgf
Radial Component of Head Membrane Force	Hr	804758.4	Kgf
Operating Bolt Load:	Wm1	574715.8	Kgf
Gasket Seating Bolt Load	Wm2	278594.4	Kgf
Required Bolt Area	Am	408.423	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:
 $= Ab * Sb / (y * PI * (Go + Gi))$
 $= 434.115 * 138.0 / (52.4 * 3.14 * (2597.0 + 2571.0))$
 $= 7.041 \text{ mm}$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :
 = 38.100 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	610868.3	Kgf
Gasket Seating Force (Operating)	Hg	39875.9	Kgf
Distance to Gasket Load Reaction	hg	32.9223	mm
Distance to Face Pressure Reaction	ht	36.9612	mm
Distance to End Pressure Reaction	hd	41.0000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	528173.	41.0000	1.0000
				21656. Kg-m.



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Face Pressure,	Mt	6667.	36.9612	1.0000	246. Kg-m.
Gasket Load,	Mg	39876.	32.9223	1.0000	1313. Kg-m.
Floating Hd. Load,	Mh	804758.	2.1689	1.0000	-1745. Kg-m.
Gasket Seating,	Ma	610868.	32.9223	1.0000	20112. Kg-m.

Total Moment for Operation (Internal Pressure) 21469.322 Kg-m.
 Total Moment for Gasket Seating (Int. Pressure) 20111.596 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	1337099.5	Kgf
Hydrostatic End Load at Flange ID	Hd	1320433.2	Kgf
Pressure Force on Flange Face	Ht	16666.2	Kgf
Radial Component of Head Membrane Force	Hr	2011896.0	Kgf
Gasket Seating Bolt Load	Wm2	278594.4	Kgf
Required Bolt Area	Am	197.984	cm ²

Flange Design Bolt Load (Seating)	W	610868.3	Kgf
Distance to Gasket Load Reaction	hg	32.9223	mm
Distance to Face Pressure Reaction	ht	36.9612	mm
Distance to End Pressure Reaction	hd	41.0000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment	
End Pressure,	Md	1320433.	8.0777	1.0000	10666. Kg-m.
Face Pressure,	Mt	16666.	4.0389	1.0000	67. Kg-m.
Floating Hd. Load,	Mh	2011896.	2.1689	1.0000	-4364. Kg-m.
Gasket Seating,	Ma	610868.	32.9223	1.0000	20112. Kg-m.

Total Moment for Operation (External Pressure) 6369.947 Kg-m.
 Total Moment for Gasket Seating (Ext. Pressure) 20111.596 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000.0 \cdot 2568.0 \cdot \sqrt{(4 \cdot 2316.0 \cdot 2316.0 - 2568.0 \cdot 2568.0) / (8 \cdot 186.997(2710.0 - 2568.0))}$$

$$= 46.6046 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 21469(2710.0 + 2568.0) / (186.997 \cdot 2568.0)(2710.0 - 2568.0)$$

$$= 16296.3584 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J) \text{ per 1-6(g)}}$$

$$= 46.605 + \sqrt{(46.605 \cdot 46.605 + 16296.358)}$$

$$= 182.5029 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 20111(2710.0 + 2568.0) / (186.997 \cdot 2568.0)(2710.0 - 2568.0)$$

$$= 15265.7705 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J) \text{ per 1-6(g)}}$$

$$= 0 + \sqrt{(0 \cdot 0 + 15265.771)}$$

$$= 123.5547 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 2500.0 \cdot 2568.0 \cdot \sqrt{(4 \cdot 2316.0 \cdot 2316.0 - 2568.0 \cdot 2568.0) / (8 \cdot 186.997(2710.0 - 2568.0))}$$



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= 116.5116 mm

Factor [J]:

$$= Mo(A + B)/(S \cdot B)(A - B)$$

$$= 6369(2710.0+2568.0)/(186.997 \cdot 2568.0)(2710.0-2568.0)$$

$$= 4835.1279 \text{ mm}^2$$

Required thickness for Main Flange, External Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 116.512 + \sqrt{(116.512 \cdot 116.512 + 4835.128)}$$

$$= 252.1953 \text{ mm}$$

Factor [J]:

$$= Mo(A + B)/(S \cdot B)(A - B)$$

$$= 20111(2710.0+2568.0)/(186.997 \cdot 2568.0)(2710.0-2568.0)$$

$$= 15265.7705 \text{ mm}^2$$

Required thickness for Main Flange, External Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 15265.771)}$$

$$= 123.5547 \text{ mm}$$

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	10.3216	182.5029
Shellside (External) Pressure	48.2109	252.1953
Tubeside Gasket Seating Load		123.5547
Shellside Gasket Seating Load		123.5547
Maximum + Corrosion Allowance	48.2109	252.1953
Actual Thickness as Given	50.0000	280.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1849.11	1849.11	2796.51	2796.51
Floating head Stress	4844.09	4844.09	2611.84	3380.60
Bolt Stress	1062.81	1062.81	...	...
Minimum MAWP & MAPnc	1062.81	1062.81	2611.84	2796.51

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	127	No Calc	No Calc
20	30	No Calc	...	210	No Calc	No Calc
30	40	1500	2748	11	0.00063345	63.3309
40	50	No Calc	...	297	No Calc	No Calc
50	60	No Calc	...	279	No Calc	No Calc
60	70	4600	2773	25.5	0.00069686	69.6703
70	80	No Calc	...	279	No Calc	No Calc
80	90	No Calc	...	279	No Calc	No Calc
90	100	976.667	2773	25.5	0.0037549	115.577
100	110	No Calc	2781	29.5	0.0014733	96.3215

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	130	45.4914	1.013	No Calc
20	30	210	191.262	1.013	No Calc
30	40	14	4.08477	1.013	337.991
40	50	300	293.599	1.013	No Calc
50	60	280	264.033	...	No Calc
60	70	26.5	No Calc	...	854.185
70	80	280	257.988	...	No Calc
80	90	280	257.124	...	No Calc
90	100	26.5	No Calc	...	1417.02
100	110	30.5	No Calc	...	1135.21

Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	40	1500	107881	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc
60	70	4600	No Calc	No Calc	No Calc
70	80	No Calc	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	976.667	No Calc	No Calc	No Calc
100	110	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.



Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	7614.34	...	7614.34	...	761.434
20	30	2187.03	1557544	2187.03	1557544	218.703
30	40	1397.97	8717597	1099.61	8756099	139.797
40	50	4647.08	1960647	4580.25	1961366	464.708
50	60	4452.35	2007585	4429.53	2008065	445.235
60	70	8618.14	13678840	8313.37	13718170	861.814
70	80	4243.53	2007585	4220.71	2008065	424.353
80	90	4281.51	2001698	4258.72	2002176	428.151
90	100	1311.45	2821302	1265.08	2827287	131.145
100	110	2657.27	2925238	2583.46	2931481	265.727
Total		41410	37678040.00	40552	37770256.00	4141



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Liqd	...	65	-10	...	Liquid: 10
20	Liqd	1556.59	130	-15.5001	...	Liquid: 20
30	Liqd	8712.28	750	...	...	Liquid: 30
30	Nozl	168.632	750	1518.75	...	Tube inlet
30	Nozl	168.632	750	-1518.75	...	Tube outlet
30	Wght	330	750	...	...	Wght:[1 of 1]
40	Liqd	1959.45	164	-15.5001	...	Liquid: 40
50	Liqd	2006.36	168	-13.5002	...	Liquid: 50
50	Insl	89.5932	172.5	...	...	Insulation
50	Lini	146.085	178	...	...	ss clad
60	Sadl	2680.84	920	1704.75	...	Left Saddle fix
60	Sadl	2680.84	3680	1704.75	...	RH floating
60	Liqd	13670.5	2300	...	...	Liquid: 40
60	Insl	1463.17	2300	...	...	Ins: 40
60	Lini	943.44	2300	...	...	SS clad
60	Nozl	725.824	2300	1741	...	Shell inlet
60	Nozl	549.32	2300	-1647.35	...	Shell outlet
60	Wght	520	920	-2100	...	Fireproofing
60	Wght	520	3680	-2100	...	Fire proofing
70	Liqd	2006.36	168	-13.5002	...	Liquid: 70
70	Insl	85.3912	175	...	...	Insulation
70	Lini	162.499	198	...	...	ss Clad
80	Liqd	2000.48	168	-15.5001	...	Liquid: 80
80	Insl	86.1684	175	...	...	Insulation
80	Lini	147.508	180	...	...	ss clad
90	Liqd	2819.58	350	...	...	Liquid: 70
90	Insl	222.656	350	...	...	Ins: 70
90	Lini	143.567	350	...	...	Ss clad
100	Liqd	2923.45	276.667	...	...	Liquid: 80
100	Insl	414.943	310	...	...	Ins: 80
100	Lini	222.55	365	...	...	SS clad
40	FTsh	5028.36	421.85	...	...	Tubesheet
40	Tube	39715.3	3009.35	...	...	Tubes
40	FLTs	4403.43	5684.35	...	...	



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 Element and Detail Weights: Step: 17 10:19am Sep 9,2026

40 F1Hd	3827.22	5824.35	...	...
---------	---------	---------	-----	-----

Total Weight of Each Detail Type:

Saddles	5361.7
Liquid	37655.0
Insulation	2361.9
Lining	1765.6
Nozzles	1612.4
Weights	1370.0
Exchanger Components	52974.3
Liquid in Tubes	10551.0

Sum of the Detail Weights 113652.0 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	45551.8	45551.8	45551.8	45551.8	45551.8	45551.8
Saddles	5361.7	5361.7	5361.7	5361.7	5361.7	5361.7
Nozzles	1612.4	1612.4	1612.4	1612.4	1612.4	1612.4
Wld Weights	1370.0	1370.0	1370.0	1370.0	1370.0	1370.0
Exchanger	52974.3	52974.3	52974.3	52974.3	52974.3	52974.3
Insulation	...	...	...	...	2361.9	2361.9 *
Lining	1765.6	1765.6	1765.6	1765.6	1765.6	1765.6
Ope. Liquid	...	...	...	...	...	37655.0
Tube Ope Lqd	...	...	...	...	...	10551.0
Test Liquid	...	37434.7	...	...	...	...
Tube Tst Lqd	...	10551.1	...	...	...	...
Totals	108635.8	156621.5	108635.8	108635.8	110997.7	159203.8

Field Installation Options:

* Insulation installed after lifting.

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	108635.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	156621.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	108635.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	108635.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	110997.7 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	159203.8 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	158259.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	158131.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	10557508.0 cm ³
Weight of Ope Liq in tubes :	10551.0 kg.
Weight of Water in tubes :	10551.1 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area	Inside Surface Area
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Element and Detail Weights:

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		cm ²	cm ²
10	20	83692	...
20	30	37469.8	22901
30	40	129496	128177
40	50	52142.8	28711.6
50	60	52743	29438.5
60	70	401169	393076
70	80	51635.5	29438.5
80	90	51799.5	29395.3
90	100	61047.4	59815.9
100	110	88885.2	84470.1

Total 1010080.6 805424.1 cm²Total 1087.2 867.0 ft²**PV Elite is a trademark of Hexagon AB, 2026, All rights reserved.**



Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	- - - UG-44	Max Pressure 50%	DNV
Tube inlet	1820.4	1960.0	82	150	GR 1.1	2318.5	1686.4	1820.4	412.1
Tube outlet	1820.4	1960.0	82	150	GR 1.1	2318.5	1686.4	1820.4	412.1
Shell inlet	4272.0	5100.0	230	300	GR 1.1	1247.4	4272.0	4272.0	4272.0
Shell outlet	4272.0	5100.0	230	300	GR 1.1	1410.9	4272.0	4272.0	4272.0

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 4272.041 KPa.
 Lowest Flange Pressure Rating was (Amb)[ShellSide]: 5100.005 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 1686.430 KPa.
 Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1960.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
Tube inlet	389.001	1.20	2472	2677	1062.8	1686.4	Pass
Tube outlet	389.001	1.20	2472	2677	1062.8	1686.4	Pass
Shell inlet	793.750	0.39	11353	5736	2512.6	4272.0	Pass
Shell outlet	657.225	0.50	7266	4589	2512.6	4272.0	Pass

Evaluating Flange on Nozzle [Tube inlet]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(2472) + 4 \cdot \text{abs}(2677) 389.001 \leq \pi \cdot 389.001^3 [(1820.434 - 1062.805) + 1.200 \cdot 1820.434]$
 $= 0.43724E+05 \leq 0.55485E+05$ Check Passes

Maximum Pressure for Flange on Nozzle Tube inlet [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 43723.9 / (\pi \cdot 389.001^3))$
 $= \min(1820.434, 1686.430)$
 $= 1686.430 \text{ KPa.}$

Evaluating Flange on Nozzle [Tube outlet]:
 $= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$
 $= 16(2472) + 4 \cdot \text{abs}(2677) 389.001 \leq \pi \cdot 389.001^3 [(1820.434 - 1062.805) + 1.200 \cdot 1820.434]$
 $= 0.43724E+05 \leq 0.55485E+05$ Check Passes

Maximum Pressure for Flange on Nozzle Tube outlet [Pmax]:
 $= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$
 $= \min(1820.434, 1.200 \cdot 1820.434 + 1820.434 - 43723.9 / (\pi \cdot 389.001^3))$
 $= \min(1820.434, 1686.430)$
 $= 1686.430 \text{ KPa.}$

Evaluating Flange on Nozzle [Shell inlet]:



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Nozzle Flange MAWP: Step: 18 10:19am Sep 9, 2026

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(11353) + 4 \cdot \text{abs}(5736) 793.750 \leq \pi \cdot 793.750^3 [(4272.041 - 2512.621) + .386 \cdot 4272.041] \\
 &= 0.19986E+06 \leq 0.54588E+06 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell inlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(4272.041, 0.386 \cdot 4272.041 + 4272.041 - 199859.3 / (\pi \cdot 793.750^3)) \\
 &= \min(4272.041, 4672.331) \\
 &= 4272.041 \text{ KPa.}
 \end{aligned}$$

Evaluating Flange on Nozzle [Shell outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(7266) + 4 \cdot \text{abs}(4589) 657.225 \leq \pi \cdot 657.225^3 [(4272.041 - 2512.621) + .500 \cdot 4272.041] \\
 &= 0.12832E+06 \leq 0.35429E+06 \text{ Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle Shell outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(4272.041, 0.500 \cdot 4272.041 + 4272.041 - 128317.8 / (\pi \cdot 657.225^3)) \\
 &= \min(4272.041, 4997.182) \\
 &= 4272.041 \text{ KPa.}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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 Wind Load Calculation: Step: 19 10:19am Sep 9,2026

Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	9000 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.222 \cdot 0.938) / (1 + 1.7 \cdot 3.4 \cdot 0.222)))$
 $= \min(0.85, 0.893)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.540
Structure Height to Diameter ratio	3.417

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Zg]$:
 Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Zg = 274320.0$ mm

Effective Height $[z]$:
 $= \text{Centroid Height} + \text{Vessel Base Elevation}$
 $= 2100.0 + 9000.0 = 11100.0$ mm
 $= 36.417$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[Kz]$:
 Because z (36.417 ft.) > 15 ft.
 $= 2.01 (z / Zg)^{2/\alpha}$
 $= 2.01 (36.417 / 900.0)^{2/9.5}$
 $= 1.023$

Type of Hill: No Hill



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Wind Directionality Factor [Kd]:
 = 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:
 K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:
 = $(1 + K1 \cdot K2 \cdot K3)^2$
 = $(1 + 0.0 \cdot 0.0 \cdot 0.0)^2$
 = 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:
 = $\max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$
 = $\max(16, 0.00256 \cdot 1.023 \cdot 1.0 \cdot 1.0 \cdot 1.0 \cdot 60.399^2)$
 = 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:
 = $qz \cdot G \cdot Cf \cdot \text{WindArea}$
 = $16.0 \cdot 0.85 \cdot 0.54 \cdot 5.038$
 = 37.0 lbs. [16.8] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Ch. cover	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
LHS CH FL	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
Ch.shell	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
RHS CH FL	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
Shell flg-1	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
Shell	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
shell flg-3	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
shell flg-2	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
Shell -2	11100.0	0.000	0.000	0.000	1.023	1.000	78.120
head- front	11100.0	0.000	0.000	0.000	1.023	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Wght:[1 of 1]	0.00	11100.00	78.12	0.00
Fireproofing	0.00	9000.00	78.12	0.00
Fire proofing	0.00	9000.00	78.12	0.00

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	11100	3600	4680	78.12	18.469
20	30	11100	3264	8486.4	78.12	33.4905
30	40	11100	3297.6	49464	78.12	195.203
40	50	11100	3264	10705.9	78.12	42.2495
50	60	11100	3508.8	11789.6	78.12	46.526
60	70	11100	3567.6	164110	78.12	647.637
70	80	11100	3508.8	11789.6	78.12	46.526
80	90	11100	3504	11773.4	78.12	46.4624
90	100	11100	3567.6	24973.2	78.12	98.5535
100	110	11100	3577.2	20764.6	78.12	81.9448

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



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Wind Load Calculation:

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Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(11.8110.75)$
 = 0.127 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.127$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 153842 \cdot 1.25$
 = 9045.914 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 153842$
 = 4824.5 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1500	12820.2	824.154



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Earthquake Load Calculation:

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20	30	1360	12820.2	747.233
30	40	1360	12820.2	747.233
40	50	1360	12820.2	747.233
50	60	1362	12820.2	748.332
60	Sad1	1360	12820.2	747.233
Sad1	70	1360	12820.2	747.233
60	70	1360	12820.2	747.233
70	80	1362	12820.2	748.332
80	90	1360	12820.2	747.233
90	100	1360	12820.2	747.233
100	110	1360	12820.2	747.233

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation: Step: 21 10:19am Sep 9,2026

Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.
Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	5041.700 mm
Center of Gravity of the Liquid	4494.141 mm
Center of Gravity of the Insulation	6155.182 mm
Center of Gravity of the Lining	6084.877 mm
Center of Gravity of the Nozzles	4226.917 mm
Center of Gravity of the Added Weights (Operating)	4103.404 mm
Center of Gravity of the Added Weights (Empty)	4103.404 mm
Center of Gravity of the Tubesheet(s)	4775.107 mm
Center of Gravity of the Tubes	4905.700 mm
Center of Gravity of the Floating Head	7727.050 mm
Center of Gravity of Bare Shell New and Cold	4055.259 mm
Center of Gravity of Bare Shell Corroded	4055.148 mm
Vessel CG in the Operating Condition	4651.034 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	4642.846 mm
Vessel CG in the Test Condition	4626.880 mm

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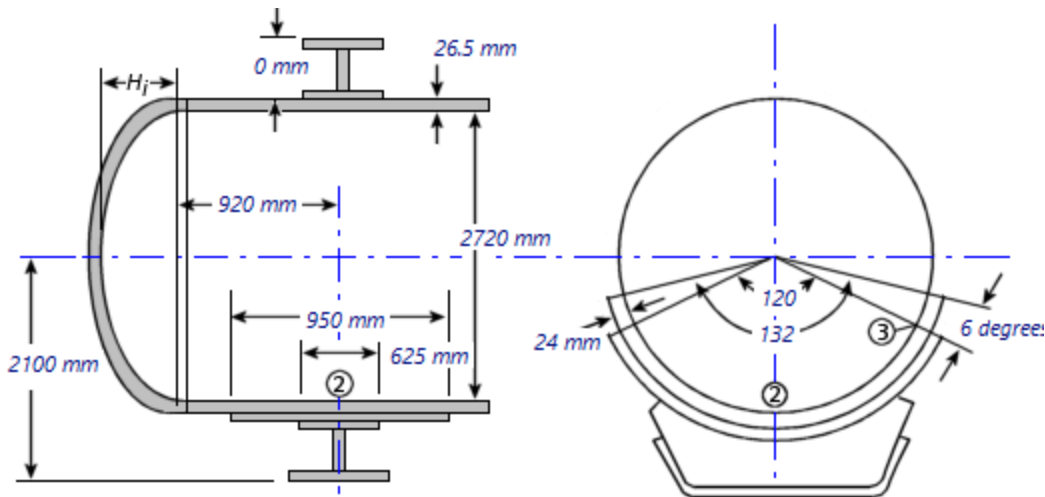
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 Saddle Calcs: Bundle Pull New: Step: 22 10:19am Sep 9,2026

ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, New Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.25	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		53000.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	67336.26	Kgf
Pressure used in Analysis	P	0.000 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-3.44	-234.00
Long. Stress at Top	of Midspan	3.44	138.00
Long. Stress at Bottom	of Midspan	3.44	138.00
Long. Stress at Top	of Saddles	-0.60	-234.00



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 Saddle Calcs: Bundle Pull New: Step: 22 10:19am Sep 9, 2026

Long. Stress at Top of Saddles	0.60	138.00
Long. Stress at Bottom of Saddles	0.33	138.00
Tangential Shear in Shell	15.10	110.40
Circ. Stress at Horn of Saddle	21.45	207.00
Circ. Compressive Stress in Shell	1.08	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 53000 • 1.5 • 2100.0/2760.0
 = 60489.2 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 6847 + max(60489, 0, 0, 0)
 = 67336.3 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	70017.10	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	79500.00	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \frac{(1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))}{(1 + (4hm)/(3L))} \right]$$

$$= -67336 \cdot 920.0 \left[1 - \frac{(1 - 920.0/6358.0 + (1373.25^2 - 0.0^2)/(2 \cdot 920.0 \cdot 6358.0))}{(1 + (4 \cdot 0.0)/(3 \cdot 6358.0))} \right]$$

$$= 1022.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 67336 \cdot 6358/4 \left(\frac{(1 + 2(1373^2 - 0^2)/(6358^2))}{(1 + (4 \cdot 0)/(3 \cdot 6358))} - 4 \cdot 920/6358 \right)$$

$$= 55068.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 1373.25/(2 \cdot 26.5) - 55068.9/(\pi \cdot 1373.3^2 \cdot 26.5)$$

$$= -3.44 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 1373.25/(2 \cdot 26.5) + 55068.9/(\pi \cdot 1373.3^2 \cdot 26.5)$$

$$= 3.44 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$



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 Saddle Calcs: Bundle Pull New: Step: 22 10:19am Sep 9,2026

$$= 0.0 \cdot 1373.25 / (2 \cdot 26.5) - 1022.1 / (0.1066 \cdot \pi \cdot 1373.3^2 \cdot 26.5)$$

$$= -0.60 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1373.25 / (2 \cdot 26.5) + 1022.1 / (0.1923 \cdot \pi \cdot 1373.3^2 \cdot 26.5)$$

$$= 0.33 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 67336(6358.0 - 2 \cdot 920.0) / (6358.0 + (4 \cdot 0.0 / 3))$$

$$= 47849.2 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 47849.2 / (1373.2501 \cdot 26.5)$$

$$= 15.10 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t}$$

$$= 0.78 \cdot \sqrt{1373.25 \cdot 26.5}$$

$$= 148.796 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$$

$$= \min(625.0 + 1.56 \cdot \sqrt{1373.25 \cdot 26.5}, 2 \cdot 920.0)$$

$$= 922.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 67336 \cdot 0.1 / (922.593(26.5 + 1.0 \cdot 24.0))$$

$$= -1.08 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r_*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -67336 / (4(26.5 + 1.0 \cdot 24.0)922.593) -$$

$$12 \cdot 0.027 \cdot 67336 \cdot 1373.25 / (6358.0(26.5 + 1.0 \cdot 24.0)^2)$$

$$= -21.45 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	700.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{1360.0 \cdot 26.5}) + 950.0)26.5^3 / 12$$

$$= 193.254 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 193.254 + 330.231(305.471 - 13.25)^2$$

$$= 282186.812 \text{ cm}^4$$



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Saddle Calcs: Bundle Pull New: Step: 22 10:19am Sep 9, 2026

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1246.2	26.5	13.2	330.2	437555.6	0.282E+06
Wearplate	950.0	24.0	38.5	228.0	877800.0	0.163E+06
Web	24.0	633.5	367.2	152.0	5583669.5	0.567E+05
BasePlate	650.0	56.0	712.0	364.0	25916802.0	0.603E+06
Totals	...	...	...	1074.3	32815826.0	0.110E+07

Distance to Centroid [C1]:

= AY / A

= 32815826/1074.271

= 305.471 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 120.0/2

= 120.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))

= 0.2035

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.204 • 67336.258

= 13704.3887 Kgf

Tension Stress, St = (Fh/As) = 1.8063 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 2100.0 - 1360.0 • sin(120.0/2)/1.0472

= 975.289 mm

Bending Moment, M = Fh * d = 13366.0049 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 3.6269 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(67336 + 2681)650.0/(4 • 2350.0 • 173.33))

= 28.667 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 120.0/2)(1360.0 + 26.5 + 24.0)

= 2443.058 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 2443.0576/(6 - 1)

= 488.612 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2

= 488.6115 • 650.0/2

= 1587.988 cm²



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 Saddle Calcs: Bundle Pull New: Step: 22 10:19am Sep 9,2026

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 70017.102 / (2350.0 \cdot 650.0)$
 $= 4.584 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1588.0 \cdot 4.58$
 $= 7278.971 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 144.24 + 58.633$
 $= 202.873 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 7279.0 / 202.8734$
 $= 3.519 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 625.0 - 24.0$
 $= 601.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 202.873$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 625.0 / 2$
 $= 312.500 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(48828.1 / 202.873)}$
 $= 155.140 \text{ mm}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 2100.0 - \cos(120.0 / 2) (1360.0 + 26.5 + 24.0) - 56.0$
 $= 1338.750 \text{ mm}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0 / 2) (1360.0 + 26.5 + 24.0)$
 $= 2443.058 \text{ mm}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 2350.0 - 50.8$
 $= 2299.200 \text{ mm}$

Length of Outer Rib, step 4, [Dd]:
 $= (L_w - B_d) / 2$



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$$= (2443.058 - 2299.2) / 2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{ (L_v^2 + D_d^2) }$$

$$= \sqrt{ (1338.75^2 + 71.929^2) }$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1340.681 / 155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{pln}) \cdot e \cdot L / 2$$

$$= 79500.0 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68 / 2$$

$$= 5540.348 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (8.6418 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (8.64) / (8 \cdot 123.21) - (8.64^3) / (8 \cdot 123.21^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.52 / 153.2 + (5540.35 \cdot 312.5 / 48828\text{E}+09) / 173.33$$

$$= 0.224$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 261.507$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (R_o + W_{pdthk})^2 - (\text{Pitch} / 2)^2 } - B_{pthk}$$

$$= 2100.0 - \sqrt{ (1410.5 + 24.0)^2 - (488.612 / 2)^2 } - 56.0$$

$$= 654.819 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (48881.6 / 261.507) }$$

$$= 136.720 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 654.819/136.72$$

$$= 4.789$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 79500.0 / (2 \cdot 2350.0)$$

$$= 16.915 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 16.915 \cdot 488.612 \cdot 654.819$$

$$= 5412.060 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5412.06/1564213$$

$$= 33.930 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (4.7895 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (4.79)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (4.79) / (8 \cdot 123.21) - (4.79^3) / (8 \cdot 123.21^3))$$

$$= 154.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 5.46/154.53 + (5412.06 \cdot 312.5/48881.609) / 173.33$$

$$= 0.231$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	10
Total Number of Bolts in Tension/Baseplate	Nbt	5
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0763 cm ²
Saddle Load Q0 (Weight)	Q0	70017.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	79500.0 Kg
Maximum Transverse Force	Ft	0.0 Kg
Maximum Longitudinal Force	F1	79500.0 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 79500.0 / (14.076 \cdot 10)$$

$$= 55.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (14.076 \cdot 10)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (79500.0 - 70017.1) / (172.0 \cdot 10.0)$$

$$= 0.5407 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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$$= 79500.0 / (103.2 \cdot 10.0)$$
$$= 7.5548 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$
$$= 2100.0 \cdot 0.0 + 0.0$$
$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$
$$= 0.0 / 70017.1$$
$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$
$$= \max[0.5407, 7.5548, 0.0]$$
$$= 7.5548 \text{ cm}^2$$

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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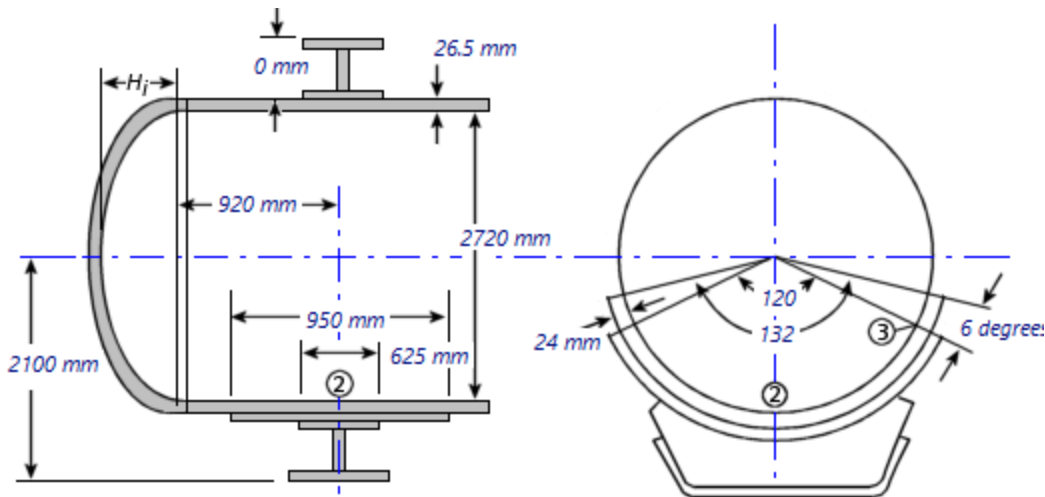
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Bundle Pull, Corroded Condition

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.75	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.00	
Tube Bundle Pull out Force		53000.00	Kgf
Tube Bundle Pull out Force Impact Factor		1.500	
Apply Bundle Pull out Force to Fixed Saddle only		Yes	



Results for Horizontal Vessel Analysis:

Saddle Force Q, + Bundle Pull, Force	66864.03	Kgf
Pressure used in Analysis	P	0.000 KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top	of Midspan	-3.55	-234.00
Long. Stress at Top	of Midspan	3.55	138.00
Long. Stress at Bottom	of Midspan	3.55	138.00
Long. Stress at Top	of Saddles	-0.62	-234.00



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Long. Stress at Top of Saddles	0.62	138.00
Long. Stress at Bottom of Saddles	0.34	138.00
Tangential Shear in Shell	15.57	110.40
Circ. Stress at Horn of Saddle	22.11	207.00
Circ. Compressive Stress in Shell	1.10	138.00

Intermediate Results: Saddle Reaction Q due to Bundle pull out Force:

Saddle Reaction Force due to Bundle Pull out Force [Fwl]:
 = Bundle Pull Out Force • Impact Factor • B / Ls
 = 53000 • 1.5 • 2100.0/2760.0
 = 60489.2 Kgf

Load Combination Results for Tube Bundle Pull Out [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 6375 + max(60489, 0, 0, 0)
 = 66864.0 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	69544.88	Kgf
Transverse Shear Load	0.00	Kgf
Longitudinal Shear Load	79500.00	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \frac{(1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))}{(1 + (4hm)/3L)} \right]$$

$$= -66864 \cdot 920.0 \left[1 - \frac{(1 - 920.0/6358.0 + (1373.75^2 - 0.0^2)/(2 \cdot 920.0 \cdot 6358.0))}{(1 + (4 \cdot 0.0)/(3 \cdot 6358.0))} \right]$$

$$= 1022.2 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 66864 \cdot 6358/4 \left(\frac{(1 + 2(1374^2 - 0^2)/(6358^2))}{(1 + (4 \cdot 0)/(3 \cdot 6358))} - 4 \cdot 920/6358 \right)$$

$$= 54689.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 1373.75/(2 \cdot 25.5) - 54689.9/(\pi \cdot 1373.8^2 \cdot 25.5)$$

$$= -3.55 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 1373.75/(2 \cdot 25.5) + 54689.9/(\pi \cdot 1373.8^2 \cdot 25.5)$$

$$= 3.55 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$



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$$= 0.0 \cdot 1373.75 / (2 \cdot 25.5) - 1022.2 / (0.1066 \cdot \pi \cdot 1373.8^2 \cdot 25.5)$$

$$= -0.62 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 1373.75 / (2 \cdot 25.5) + 1022.2 / (0.1923 \cdot \pi \cdot 1373.8^2 \cdot 25.5)$$

$$= 0.34 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 66864(6358.0 - 2 \cdot 920.0) / (6358.0 + (4 \cdot 0.0 / 3))$$

$$= 47513.6 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 47513.64 / (1373.75 \cdot 25.5)$$

$$= 15.57 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1373.75 \cdot 25.5)}$$

$$= 145.988 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(625.0 + 1.56 \cdot \sqrt{(1373.75 \cdot 25.5)}, 2 \cdot 920.0)$$

$$= 916.98 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000 \text{ Materials are the same, bundle pull case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 66864 \cdot 0.1 / (916.977(25.5 + 1.0 \cdot 24.0))$$

$$= -1.10 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -66864 / (4(25.5 + 1.0 \cdot 24.0)916.977) -$$

$$12 \cdot 0.027 \cdot 66864 \cdot 1373.75 / (6358.0(25.5 + 1.0 \cdot 24.0)^2)$$

$$= -22.11 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	700.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1361.0 \cdot 25.5)} + 950.0)25.5^3) / 12$$

$$= 171.426 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 171.426 + 316.358(308.443 - 12.75)^2$$

$$= 276776.406 \text{ cm}^4$$



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	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	1240.6	25.5	12.8	316.4	403356.2	0.277E+06
Wearplate	950.0	24.0	37.5	228.0	855000.0	0.167E+06
Web	24.0	633.5	366.2	152.0	5568464.0	0.559E+05
BasePlate	650.0	56.0	711.0	364.0	25880396.0	0.591E+06
Totals	...	...	...	1060.4	32707216.0	0.109E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 32707216/1060.398$$

$$= 308.443 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 66864.031$$

$$= 13608.2803 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 1.7936 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2100.0 - 1361.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 974.462 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 13261.0137 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.6766 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(66864 + 2681) \cdot 650.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 28.570 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2443.0576 / (6 - 1)$$

$$= 488.612 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 488.6115 \cdot 650.0/2$$

$$= 1587.988 \text{ cm}^2$$



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Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 69544.875 / (2350.0 \cdot 650.0)$
 $= 4.553 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1588.0 \cdot 4.55$
 $= 7229.878 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 144.24 + 58.633$
 $= 202.873 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 7229.9 / 202.8734$
 $= 3.495 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 625.0 - 24.0$
 $= 601.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 202.873$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$
 $= 625.0 / 2$
 $= 312.500 \text{ mm}$

Radius of Gyration [r]:
 $= \sqrt{ (\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{ (48828.1 / 202.873)}$
 $= 155.140 \text{ mm}$

Length of Outer Rib, step 1, [Lv]:
 $= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$
 $= 2100.0 - \cos(120.0 / 2) (1360.0 + 26.5 + 24.0) - 56.0$
 $= 1338.750 \text{ mm}$

Length of Outer Rib, step 2, [Lw]:
 $= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$
 $= 2 \cdot \sin(120.0 / 2) (1360.0 + 26.5 + 24.0)$
 $= 2443.058 \text{ mm}$

Length of Outer Rib, step 3, [Bd]:
 $= \text{Baseplate length} - \text{clearance}$
 $= 2350.0 - 50.8$
 $= 2299.200 \text{ mm}$

Length of Outer Rib, step 4, [Dd]:
 $= (L_w - B_d) / 2$



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$$= (2443.058 - 2299.2) / 2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{ (L_v^2 + D_d^2) }$$

$$= \sqrt{ (1338.75^2 + 71.929^2) }$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0) }$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1340.681 / 155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 79500.0 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68 / 2$$

$$= 5540.348 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ (8.6418 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (8.64) / (8 \cdot 123.21) - (8.64^3) / (8 \cdot 123.21^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.49 / 153.2 + (5540.35 \cdot 312.5 / 48828\text{E}+09) / 173.33$$

$$= 0.223$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 261.507$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (R_o + W_{pdthk})^2 - (\text{Pitch} / 2)^2 } - B_{pthk}$$

$$= 2100.0 - \sqrt{ (1410.5 + 24.0)^2 - (488.612 / 2)^2 } - 56.0$$

$$= 654.819 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (\text{Total Inertia} / \text{Total Area}) }$$

$$= \sqrt{ (48881.6 / 261.507) }$$

$$= 136.720 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL / r$$



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$$= 1 \cdot 654.819/136.72$$

$$= 4.789$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 79500.0 / (2 \cdot 2350.0)$$

$$= 16.915 \text{ Kgf/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 16.915 \cdot 488.612 \cdot 654.819$$

$$= 5412.060 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 5412.06/1564213$$

$$= 33.930 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ ($4.7895 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (4.79)^2 / (2 \cdot 123.21^2)) 260 / (5/3 + 3 \cdot (4.79) / (8 \cdot 123.21) - (4.79)^3 / (8 \cdot 123.21^3))$$

$$= 154.5 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase,c \cdot C1/I) / Sba$$

$$= 5.42/154.53 + (5412.06 \cdot 312.5/48881.609) / 173.33$$

$$= 0.231$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	10
Total Number of Bolts in Tension/Baseplate	Nbt	5
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0763 cm ²
Saddle Load Q0 (Weight)	Q0	69544.9 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	79500.0 Kgf
Maximum Transverse Force	Ft	0.0 Kgf
Maximum Longitudinal Force	F1	79500.0 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 79500.0 / (14.076 \cdot 10)$$

$$= 55.4 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 0.0 / (14.076 \cdot 10)$$

$$= 0.0 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= (QL - Q0) / (Stba \cdot Rnbolts)$$

$$= (79500.0 - 69544.88) / (172.0 \cdot 10.0)$$

$$= 0.5676 \text{ cm}^2$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$



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$$= 79500.0 / (103.2 \cdot 10.0)$$
$$= 7.5548 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot Ft + \text{Sum of X Moments}$$
$$= 2100.0 \cdot 0.0 + 0.0$$
$$= 0.00 \text{ Kg-m.}$$

Eccentricity (e):

$$= Rmom / QO$$
$$= 0.0 / 69544.88$$
$$= 0.00 \text{ mm} < Bplen/6 \text{ --> No Uplift in Transverse direction}$$

Bolt Area due to Transverse Load [Bltareart]:

$$= 0 \text{ (No Uplift)}$$

Required Area of a Single Bolt [Bltarear]:

$$= \max[Bltarearl, Bltarears, Bltareart]$$
$$= \max[0.5676, 7.5548, 0.0]$$
$$= 7.5548 \text{ cm}^2$$

ASME Horizontal Vessel Analysis: Bundle pullout, Right Saddle

This saddle was not processed as the coefficient of friction was specified and the directive was set to only analyze the fixed saddle for this case.

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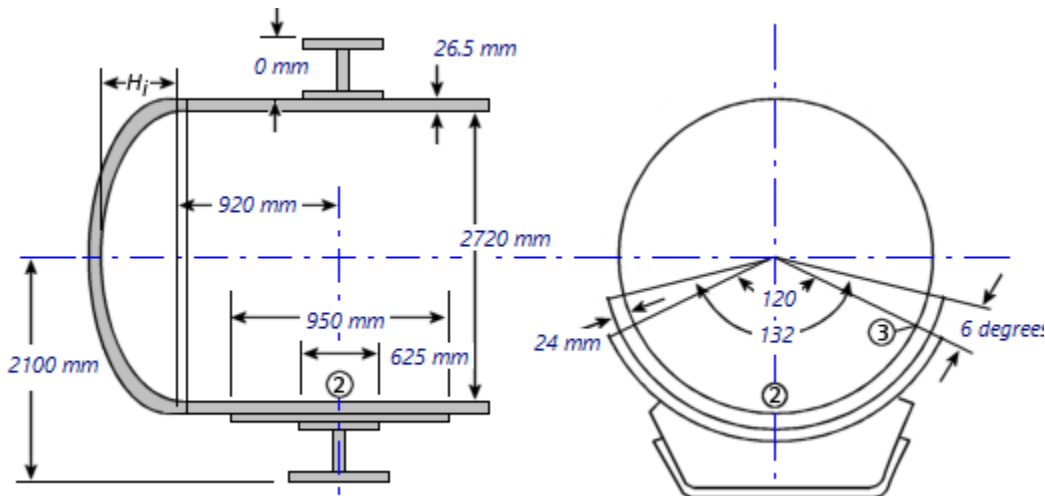
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.25	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.00	

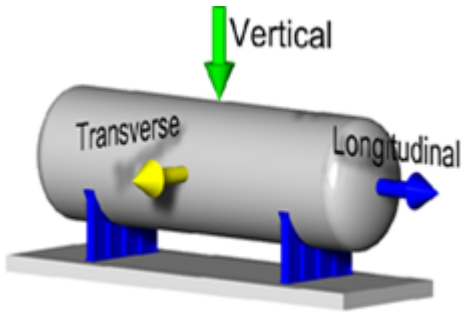


Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		78655.23	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-4.02	-234.00
Long. Stress at Top of Midspan	4.02	138.00
Long. Stress at Bottom of Midspan	4.02	138.00
Long. Stress at Top of Saddles	-0.70	-234.00
Long. Stress at Top of Saddles	0.70	138.00
Long. Stress at Bottom of Saddles	0.39	138.00
Tangential Shear in Shell	17.63	110.40
Circ. Stress at Horn of Saddle	25.06	207.00
Circ. Compressive Stress in Shell	1.26	138.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 342.87 • 0.00256 • 57.6² • 1.0
 = 2912.4 lbf
 = 1321.075 Kgf

Longitudinal Force due to Wind during Transport [F_{lw}]:
 = π • D²/4 • 0.00256 • wv² • wlsf • Wind Dia. Mult
 = π • 9.098²/4 • 0.00256 • 57.6² • 1.0 • 1.2
 = 662.6 lbf
 = 300.6 Kgf

Transverse Force due to Lateral Acceleration [F_t]:
 = g_z • Shipping Weight • g_w + F_{tw}
 = 0.1 • 101508.43 • 1.0 + 1321.07
 = 11471.9 Kgf

Longitudinal Force due to Axial Acceleration [F_l]:
 = g_x • Shipping Weight • g_w + F_{lw}
 = 0.1 • 101508.43 • 1.0 + 300.57
 = 10451.4 Kgf

Saddle Reaction Force due to Transverse Load [F_{wt}]:
 = F_t/Num of Saddles • B / E
 = 11471.9/2 • 2100.0/2350.0
 = 5125.8 Kgf

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:
 = F_l • B / L_s
 = 10451 • 2100.0/2760.0
 = 7952.2 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:
 = Saddle Load • g_y
 = 67336.3 • 0.05



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= 3366.813 Kgf

Load Combination Results due to Transport Loading [Q]:

= Saddle Load + Frv + max(Fwl, Fwt)
 = 67336 + 3367 + max(7952, 5126)
 = 78655.2 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	81336.07	Kgf
Transverse Shear Load	5735.96	Kgf
Longitudinal Shear Load	10451.41	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= Baseplate Length
 = 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q*a [1 - (1 - a/L + (Rm²-hm²)/(2a*L))/(1+(4hm)/3L)]
 = -78655*920.0[1-(1-920.0/6358.0+(1373.25²-0.0²)/
 (2*920.0*6358.0))/(1+(4*0.0)/(3*6358.0))]
 = 1193.9 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q*L/4*((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 78655*6358/4*((1+2(1373²-0²)/(6358²))/(1+(4*0)/
 (3*6358))-4*920/6358)
 = 64325.7 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ1]:

= P * Rm/(2t) - M2/(π*Rm²t)
 = 0.0 * 1373.25/(2*26.5) - 64325.7/(π*1373.3²*26.5)
 = -4.02 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ2]:

= P * Rm/(2t) + M2/(π * Rm² * t)
 = 0.0 * 1373.25/(2 * 26.5) + 64325.7/(π * 1373.3² * 26.5)
 = 4.02 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ*3]:

= P * Rm/(2t) - M1/(K1*π*Rm²t)
 = 0.0*1373.25/(2*26.5)-1193.9/(0.1066*π*1373.3²*26.5)
 = -0.70 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ*4]:

= P * Rm/(2t) + M1/(K1*π * Rm² * t)
 = 0.0*1373.25/(2*26.5)+1193.9/(0.1923*π*1373.3²*26.5)
 = 0.39 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4*hm/3))
 = 78655(6358.0 - 2 * 920.0)/(6358.0 + (4 * 0.0/3))
 = 55892.5 Kgf



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Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.1707 \cdot 55892.47 / (1373.2501 \cdot 26.5)$$

$$= 17.63 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1373.25 \cdot 26.5)}$$

$$= 148.796 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(625.0 + 1.56 \cdot \sqrt{(1373.25 \cdot 26.5)}, 2 \cdot 920.0)$$

$$= 922.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, transport case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 78655 \cdot 0.1 / (922.593(26.5 + 1.0 \cdot 24.0))$$

$$= -1.26 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 12 \cdot K_7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2)$$

$$= -78655 / (4(26.5 + 1.0 \cdot 24.0)922.593) -$$

$$12 \cdot 0.027 \cdot 78655 \cdot 1373.25 / (6358.0(26.5 + 1.0 \cdot 24.0)^2)$$

$$= -25.06 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	700.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1360.0 \cdot 26.5)} + 950.0)^3) / 12$$

$$= 193.254 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 193.254 + 330.231(305.471 - 13.25)^2$$

$$= 282186.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1246.2	26.5	13.2	330.2	437555.6	0.282E+06
Wearplate	950.0	24.0	38.5	228.0	877800.0	0.163E+06
Web	24.0	633.5	367.2	152.0	5583669.5	0.567E+05
BasePlate	650.0	56.0	712.0	364.0	25916802.0	0.603E+06
Totals	...	...	...	1074.3	32815826.0	0.110E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 32815826 / 1074.271$$



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$$= 305.471 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120.0/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 78655.227$$

$$= 16008.0449 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	2.1099	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	155.9972	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2100.0 - 1360.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 975.289 \text{ mm}$$

Bending Moment, M = Fh * d	=	15612.7812	Kg-m.
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Bending Stress, Sb = (M * C1 / I)	=	4.2366	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	173.3302	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(78655 + 2681)650.0 / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 30.897 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2443.0576 / (6 - 1)$$

$$= 488.612 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 488.6115 \cdot 650.0/2$$

$$= 1587.988 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 81336.07 / (2350.0 \cdot 650.0)$$

$$= 5.325 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1588.0 \cdot 5.32$$

$$= 8455.690 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 144.24 + 58.633$$



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$$= 202.873 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 8455.7/202.8734$$

$$= 4.087 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 625.0 - 24.0$$

$$= 601.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/202.873$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0/2$$

$$= 312.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(48828.1/202.873)}$$

$$= 155.140 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2100.0 - \cos(120.0/2) (1360.0 + 26.5 + 24.0) - 56.0$$

$$= 1338.750 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350.0 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2443.058 - 2299.2) / 2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(Lv^2 + Dd^2)}$$

$$= \sqrt{(1338.75^2 + 71.929^2)}$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$$

$$= 123.211$$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1340.681/155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 10451.4 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68/2$$

$$= 728.358 \text{ Kg-m.}$$

Compressive Allowable, KL/r < Cc (8.6418 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (Klr) / (8 \cdot Cc) - (Klr^3) / (8 \cdot Cc^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (8.64) / (8 \cdot 123.21) - (8.64^3) / (8 \cdot 123.21^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.09/153.2 + (728.36 \cdot 312.5 / 48828E+09) / 173.33$$

$$= 0.053$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/261.507$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0/2$$

$$= 312.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpdthk$$

$$= 2100.0 - \sqrt{ (1410.5 + 24.0)^2 - (488.612/2)^2 } - 56.0$$

$$= 654.819 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 48881.6/261.507 }$$

$$= 136.720 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 654.819/136.72$$

$$= 4.789$$

Unit Force [Force,u]:

$$= Fl / (2 \cdot \text{Baseplate Length})$$

$$= 10451.409 / (2 \cdot 2350.0)$$

$$= 2.224 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 2.224 \cdot 488.612 \cdot 654.819$$

$$= 711.492 \text{ Kg-m.}$$



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Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus
 = 711.492/1564213
 = 4.461 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.7895 < 123.2105) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2• C_c ²)) F_y /(5/3+3•(KL/r)/(8• C_c)-(KL/r)³)/(8• C_c ³)
 = (1-(4.79)²/(2 • 123.21²))260/
 (5/3+3•(4.79)/(8• 123.21)-(4.79³)/(8•123.21³)
 = 154.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} \cdot C_{1/I})/S_{ba}$
 = 6.34/154.53 + (711.49 • 312.5/48881.609)/173.33
 = 0.067

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.25	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		45686.88	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual	Allowable
		N./mm ²	N./mm ²
Long. Stress at Top	of Midspan	-2.33	-234.00
Long. Stress at Top	of Midspan	2.33	138.00
Long. Stress at Bottom	of Midspan	2.33	138.00
Long. Stress at Top	of Saddles	-0.41	-234.00
Long. Stress at Top	of Saddles	0.41	138.00
Long. Stress at Bottom	of Saddles	0.23	138.00
Tangential Shear in Shell		10.24	110.40
Circ. Stress at Horn of Saddle		14.56	207.00
Circ. Compressive Stress in Shell		0.73	138.00



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Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:
 = Transverse Area • 0.00256 • wv² • wlsf
 = 342.87 • 0.00256 • 57.6² • 1.0
 = 2912.4 lbf
 = 1321.075 Kgf

Longitudinal Force due to Wind during Transport [Flw]:
 = $\pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$
 = $\pi \cdot 9.098^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$
 = 662.6 lbf
 = 300.6 Kgf

Transverse Force due to Lateral Acceleration [Ft]:
 = gz • Shipping Weight • gw + Ftw
 = 0.1 • 101508.43 • 1.0 + 1321.07
 = 11471.9 Kgf

Longitudinal Force due to Axial Acceleration [Fl]:
 = gx • Shipping Weight • gw + Flw
 = 0.1 • 101508.43 • 1.0 + 300.57
 = 10451.4 Kgf

Saddle Reaction Force due to Transverse Load [Fwt]:
 = Ft/Num of Saddles • B / E
 = 11471.9/2 • 2100.0/2350.0
 = 5125.8 Kgf

Saddle Reaction Force due to Longitudinal Load [Fwl]:
 = Fl • B / Ls
 = 10451 • 2100.0/2760.0
 = 7952.2 Kgf

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:
 = Saddle Load • gy
 = 35937.8 • 0.05
 = 1796.891 Kgf

Load Combination Results due to Transport Loading [Q]:
 = Saddle Load + Frv + max(Fwl, Fwt)
 = 35938 + 1797 + max(7952, 5126)
 = 45686.9 Kgf

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	48367.72	Kgf
Transverse Shear Load	5735.96	Kgf
Longitudinal Shear Load	10451.41	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.



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Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -45687 \cdot 920.0 [1 - (1 - 920.0/6358.0 + (1373.25^2 - 0.0^2) / (2 \cdot 920.0 \cdot 6358.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 6358.0))]$$

$$= 693.5 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 45687 \cdot 6358 / 4 \cdot ((1 + 2(1373^2 - 0^2)/(6358^2)) / (1 + (4 \cdot 0) / (3 \cdot 6358)) - 4 \cdot 920 / 6358)$$

$$= 37363.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 0.0 \cdot 1373.25 / (2 \cdot 26.5) - 37363.6 / (\pi \cdot 1373.3^2 \cdot 26.5)$$

$$= -2.33 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 1373.25 / (2 \cdot 26.5) + 37363.6 / (\pi \cdot 1373.3^2 \cdot 26.5)$$

$$= 2.33 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$$

$$= 0.0 \cdot 1373.25 / (2 \cdot 26.5) - 693.5 / (0.1066 \cdot \pi \cdot 1373.3^2 \cdot 26.5)$$

$$= -0.41 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 1373.25 / (2 \cdot 26.5) + 693.5 / (0.1923 \cdot \pi \cdot 1373.3^2 \cdot 26.5)$$

$$= 0.23 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 45687(6358.0 - 2 \cdot 920.0) / (6358.0 + (4 \cdot 0.0 / 3))$$

$$= 32465.1 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 32465.13 / (1373.2501 \cdot 26.5)$$

$$= 10.24 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1, x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1373.25 \cdot 26.5)}$$

$$= 148.796 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(625.0 + 1.56 \cdot \sqrt{(1373.25 \cdot 26.5)}, 2 \cdot 920.0)$$

$$= 922.59 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= 1.0000$$
 Materials are the same, transport case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 45687 \cdot 0.1 / (922.593(26.5 + 1.0 \cdot 24.0))$$

$$= -0.73 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L < 8Rm (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -45687 / (4(26.5 + 1.0 \cdot 24.0)922.593) -$$



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$$12 \cdot 0.027 \cdot 45687 \cdot 1373.25 / (6358.0(26.5 + 1.0 \cdot 24.0)^2) \\ = -14.56 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	700.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3)/12 \\ = ((1.56 \cdot \sqrt{(1360.0 \cdot 26.5)} + 950.0)26.5^3)/12 \\ = 193.254 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2 \\ = 193.254 + 330.231(305.471 - 13.25)^2 \\ = 282186.812 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1246.2	26.5	13.2	330.2	437555.6	0.282E+06
Wearplate	950.0	24.0	38.5	228.0	877800.0	0.163E+06
Web	24.0	633.5	367.2	152.0	5583669.5	0.567E+05
BasePlate	650.0	56.0	712.0	364.0	25916802.0	0.603E+06
Totals	...	...	...	1074.3	32815826.0	0.110E+07

Distance to Centroid [C1]:

$$= AY / A \\ = 32815826 / 1074.271 \\ = 305.471 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2 \\ = 180 - 120.0/2 \\ = 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\ = (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0)\cos(120.0)) \\ = 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q \\ = 0.204 \cdot 45686.879 \\ = 9298.2695 \text{ Kgf}$$

Tension Stress, St = (Fh/As) = 1.2256 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta}/2) / (\text{theta}/2 \text{ in radians}) \\ = 2100.0 - 1360.0 \cdot \sin(120.0/2) / 1.0472 \\ = 975.289 \text{ mm}$$



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Bending Moment, $M = F_h \cdot d = 9068.6797 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 2.4608 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(45687 + 2681) \cdot 650.0) / (4 \cdot 2350.0 \cdot 173.33)}$
 $= 23.826 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 120.0/2) (1360.0 + 26.5 + 24.0)$
 $= 2443.058 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 2443.0576 / (6 - 1)$
 $= 488.612 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 488.6115 \cdot 650.0/2$
 $= 1587.988 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 48367.723 / (2350.0 \cdot 650.0)$
 $= 3.166 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1588.0 \cdot 3.17$
 $= 5028.304 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 144.24 + 58.633$
 $= 202.873 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 5028.3 / 202.8734$
 $= 2.431 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 625.0 - 24.0$
 $= 601.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0/202.873$
 $= 0.000 \text{ mm}$



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Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(48828.1 / 202.873)}$$

$$= 155.140 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2100.0 - \cos(120.0 / 2) (1360.0 + 26.5 + 24.0) - 56.0$$

$$= 1338.750 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350.0 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2443.058 - 2299.2) / 2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1338.75^2 + 71.929^2}$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994 \text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1340.681 / 155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 10451.4 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68 / 2$$

$$= 728.358 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($8.6418 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (8.64) / (8 \cdot 123.21) - (8.64^3) / (8 \cdot 123.21^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.43 / 153.2 + (728.36 \cdot 312.5 / 48828 \text{E}+09) / 173.33$$

$$= 0.042$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

B	D	Y	A	AY	Io
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Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= $0.0/261.507$

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= $625.0/2$

= 312.500 mm

Length of Inner Rib [L]:

= $\text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$

= $2100.0 - \sqrt{(1410.5 + 24.0)^2 - (488.612/2)^2} - 56.0$

= 654.819 mm

Radius of Gyration [r]:

= $\sqrt{(\text{Total Inertia} / \text{Total Area})}$

= $\sqrt{48881.6/261.507}$

= 136.720 mm

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 654.819/136.72$

= 4.789

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$

= $10451.409 / (2 \cdot 2350.0)$

= 2.224 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= $\text{Unit Force} \cdot e \cdot L$

= $2.224 \cdot 488.612 \cdot 654.819$

= 711.492 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= $711.492/1564213$

= 4.461 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.7895 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (4.79)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (4.79) / (8 \cdot 123.21) - (4.79^3) / (8 \cdot 123.21^3))$

= 154.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$

= $3.77/154.53 + (711.49 \cdot 312.5/48881.609) / 173.33$

= 0.050

Review notes about nozzle loadings and supports in the Warnings report.

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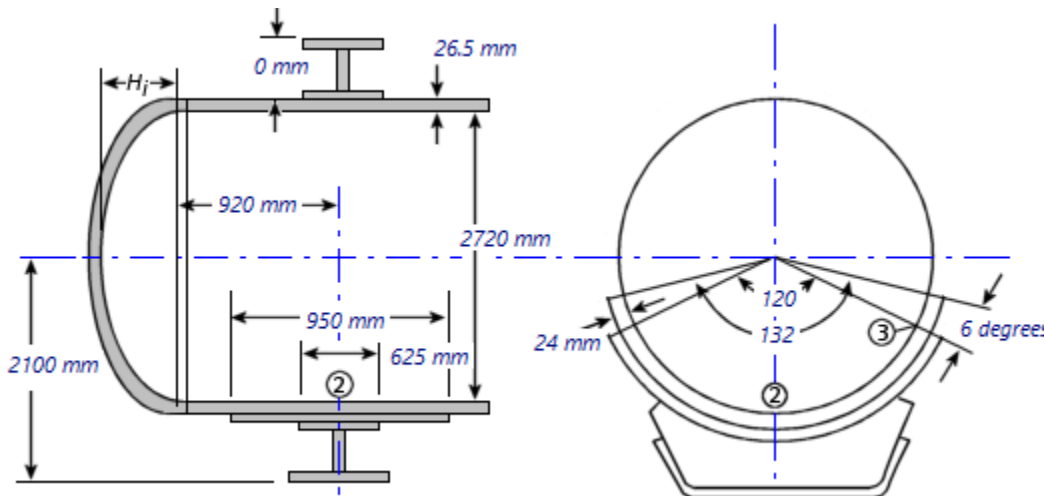
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.75	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	114195.34	Kgf
Pressure used in Analysis	P 2513.331	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	61.65	138.00
Long. Stress at Bottom of Midspan	73.76	138.00
Long. Stress at Top of Saddles	66.64	138.00
Long. Stress at Bottom of Saddles	68.29	138.00
Tangential Shear in Shell	26.59	110.40
Circ. Stress at Horn of Saddle	37.77	172.50
Circ. Compressive Stress in Shell	1.88	138.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(1257.1/2 + 0) \cdot 2100.0/2350.0$
 $= 1685.0 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(747.32, 0) \cdot 2100.0/2760.0$
 $= 568.6 \text{ Kg}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(9045.91, 0) \cdot 2100.0/2760.0$
 $= 6882.8 \text{ Kg}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(9046/2 + 0) \cdot 2100.0/2350.0$
 $= 12125.4 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 102070 + \max(569, 1685, 6883, 12125)$
 $= 114195.3 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(2.773 \cdot 1.2)^2)$
 $= 7328.618 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	116876.18	Kgf
Transverse Shear Load	4522.96	Kgf
Longitudinal Shear Load	9045.91	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 2350.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -114195 \cdot 920.0 [1 - (1 - 920.0/6358.0 + (1373.75^2 - 0.0^2)/(2 \cdot 920.0 \cdot 6358.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6358.0))]$
 $= 1745.7 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 $= 114195 \cdot 6358/4((1 + 2(1374^2 - 0^2)/(6358^2))/(1 + (4 \cdot 0)/(3 \cdot 6358)) - 4 \cdot 920/6358)$



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= 93403.4 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 2513.331 \cdot 1373.75 / (2 \cdot 25.5) - 93403.4 / (\pi \cdot 1373.8^2 \cdot 25.5)$$

$$= 61.65 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 2513.331 \cdot 1373.75 / (2 \cdot 25.5) + 93403.4 / (\pi \cdot 1373.8^2 \cdot 25.5)$$

$$= 73.76 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2513.331 \cdot 1373.75 / (2 \cdot 25.5) - 1745.7 / (0.1066 \cdot \pi \cdot 1373.8^2 \cdot 25.5)$$

$$= 66.64 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 2513.331 \cdot 1373.75 / (2 \cdot 25.5) + 1745.7 / (0.1923 \cdot \pi \cdot 1373.8^2 \cdot 25.5)$$

$$= 68.29 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot hm/3))$$

$$= 114195(6358.0 - 2 \cdot 920.0) / (6358.0 + (4 \cdot 0.0/3))$$

$$= 81147.3 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 81147.31 / (1373.75 \cdot 25.5)$$

$$= 26.59 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(1373.75 \cdot 25.5)}$$

$$= 145.988 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(625.0 + 1.56 \cdot \sqrt{(1373.75 \cdot 25.5)}, 2 \cdot 920.0)$$

$$= 916.98 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 114195 \cdot 0.1 / (916.977(25.5 + 1.0 \cdot 24.0))$$

$$= -1.88 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r_{*}]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -114195 / (4(25.5 + 1.0 \cdot 24.0)916.977) -$$

$$12 \cdot 0.027 \cdot 114195 \cdot 1373.75 / (6358.0(25.5 + 1.0 \cdot 24.0)^2)$$

$$= -37.77 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 2760.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000013 \cdot 2760.0(230.0 - 21.1)$$

$$= 7.463 \text{ mm}$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	700.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(1361.0 \cdot 25.5)} + 950.0)^3) / 12$$

$$= 171.426 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 171.426 + 316.358(308.443 - 12.75)^2$$

$$= 276776.406 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1240.6	25.5	12.8	316.4	403356.2	0.277E+06
Wearplate	950.0	24.0	37.5	228.0	855000.0	0.167E+06
Web	24.0	633.5	366.2	152.0	5568464.0	0.559E+05
BasePlate	650.0	56.0	711.0	364.0	25880396.0	0.591E+06
Totals	...	...	...	1060.4	32707216.0	0.109E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 32707216 / 1060.398$$

$$= 308.443 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 114195.344$$

$$= 23241.2285 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 3.0633 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 138.2389 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 2100.0 - 1361.0 \cdot \sin(120.0 / 2) / 1.0472$$

$$= 974.462 \text{ mm}$$

Bending Moment, M = Fh * d = 22648.1406 Kg-m.



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Bending Stress, $S_b = (M * C_1 / I) = 6.2791 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 * \text{Yield Str} = 153.5988 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress}))}$
 $= \sqrt{(3(114195 + 2681)650.0 / (4 * 2350.0 * 153.599))}$
 $= 39.345 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120.0/2) (1360.0 + 26.5 + 24.0)$
 $= 2443.058 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 2443.0576 / (6 - 1)$
 $= 488.612 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 488.6115 * 650.0/2$
 $= 1587.988 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 116876.18 / (2350.0 * 650.0)$
 $= 7.651 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p * B_p$
 $= 1588.0 * 7.65$
 $= 12150.437 \text{ Kgf}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 144.24 + 58.633$
 $= 202.873 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 12150.4 / 202.8734$
 $= 5.873 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 625.0 - 24.0$
 $= 601.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 202.873$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$



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$$= 625.0/2$$

$$= 312.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(48828.1/202.873)}$$

$$= 155.140 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2100.0 - \cos(120.0/2) (1360.0 + 26.5 + 24.0) - 56.0$$

$$= 1338.750 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0/2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350.0 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd})/2$$

$$= (2443.058 - 2299.2)/2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(\text{Lv}^2 + \text{Dd}^2)}$$

$$= \sqrt{(1338.75^2 + 71.929^2)}$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/230.4)}$$

$$= 130.885$$

Slenderness ratio [KL/r]:

$$= \text{KL}/r$$

$$= 1 \cdot 1340.681/155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 9045.9 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68/2$$

$$= 630.409 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($8.6418 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (8.64) / (8 \cdot 130.89) - (8.64^3) / (8 \cdot 130.89^3))$$

$$= 135.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 5.87/135.92 + (630.41 \cdot 312.5 / 48828\text{E}+09) / 153.6$$

$$= 0.069$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05



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Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/261.507

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 625.0/2

= 312.500 mm

Length of Inner Rib [L]:

= $\text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$

= 2100.0 - $\sqrt{(1410.5 + 24.0)^2 - (488.612/2)^2} - 56.0$

= 654.819 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{48881.6/261.507}$

= 136.720 mm

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 654.819/136.72$

= 4.789

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$

= $9045.914 / (2 \cdot 2350.0)$

= 1.925 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force $\cdot e \cdot L$

= $1.925 \cdot 488.612 \cdot 654.819$

= 615.812 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= $615.812/1564213$

= 3.861 N./mm²

Compressive Allowable, $KL/r < C_c$ (4.7895 < 130.8854) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (4.79)^2 / (2 \cdot 130.89^2)) 230 /$

$(5/3 + 3 \cdot (4.79) / (8 \cdot 130.89) - (4.79^3) / (8 \cdot 130.89^3))$

= 137 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$

= $9.12/137.02 + (615.81 \cdot 312.5/48881.609) / 153.6$

= 0.092

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	10
Total Number of Bolts in Tension/Baseplate	Nbt	5
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²



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Area Available in a Single Bolt	BltArea	14.0763 cm ²
Saddle Load QO (Weight)	QO	104750.8 Kgf
Saddle Load QL (Wind/Seismic contribution)	QL	6882.8 Kgf
Maximum Transverse Force	Ft	4523.0 Kgf
Maximum Longitudinal Force	Fl	9045.9 Kgf
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:
 = 17783.2 Kgf

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
 = max(Ff, Fl) / (Bolt Area • Number of Bolts)
 = 17783.2 / (14.076 • 10)
 = 12.4 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
 = Ft / (Bolt Area • Number of Bolts)
 = 4523.0 / (14.076 • 10)
 = 3.2 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
 = 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
 = Fl / (BoltShearAllowable • Nbolts)
 = 9045.91 / (82.8 • 10.0)
 = 1.0714 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = B • Ft + Sum of X Moments
 = 2100.0 • 4522.96 + 0.0
 = 9498.40 Kg-m.

Eccentricity (e):
 = Rmom / QO
 = 9498.4 / 104750.81
 = 90.67 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = max[Bltarearl, Bltarears, Bltareart]
 = max[0.0, 1.0714, 0.0]
 = 1.0714 cm²

MDMT Calculations for: Left Saddle fix, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 24.0, c = 1.0 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.983, Temp. Reduction = 1 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-3 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:



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Vessel Mean Radius	Rm	1373.75	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		136.80	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case 68721.98 Kgf
 Pressure used in Analysis P 2513.331 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	64.06	138.00
Long. Stress at Bottom of Midspan	71.35	138.00
Long. Stress at Top of Saddles	67.06	138.00
Long. Stress at Bottom of Saddles	68.06	138.00
Tangential Shear in Shell	16.00	110.40
Circ. Stress at Horn of Saddle	22.73	172.50
Circ. Compressive Stress in Shell	1.13	138.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (1257.1/2 + 0) \cdot 2100.0/2350.0$
 $= 1685.0 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(747.32, 0) \cdot 2100.0/2760.0$
 $= 568.6 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(9045.91, 0) \cdot 2100.0/2760.0$
 $= 6882.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (9046/2 + 0) \cdot 2100.0/2350.0$
 $= 12125.4 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 56597 + \max(569, 1685, 6883, 12125)$
 $= 68722.0 \text{ Kgf}$



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Longitudinal Wind Force [F_L]:
 = WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 766.08(0.0 + $\pi/4$ (2.773 • 1.2)²)
 = 7328.618 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	71402.83	Kgf
Transverse Shear Load	4522.96	Kgf
Longitudinal Shear Load	9045.91	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 = Baseplate Length
 = 2350.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:
 = -Q•a [1 - (1- a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -68722•920.0[1-(1-920.0/6358.0+(1373.75²-0.0²)/
 (2•920.0•6358.0))/(1+(4•0.0)/(3•6358.0))]
 = 1050.6 Kg-m.

Moment per Equation 4.15.2 [M2]:
 = Q•L/4((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L))-4a/L)
 = 68722•6358/4((1+2(1374²-0²)/(6358²))/(1+(4•0)/
 (3•6358))-4•920/6358)
 = 56209.6 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 = P • Rm/(2t) - M2/(π •Rm²t)
 = 2513.331 • 1373.75/(2•25.5) - 56209.6/(π •1373.8²•25.5)
 = 64.06 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 = P • Rm/(2t) + M2/(π • Rm² • t)
 = 2513.331 • 1373.75/(2 • 25.5) + 56209.6/(π • 1373.8² • 25.5)
 = 71.35 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:
 = P • Rm/(2t) - M1/(K1• π •Rm²t)
 = 2513.331•1373.75/(2•25.5)-1050.6/(0.1066• π •1373.8²•25.5)
 = 67.06 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:
 = P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = 2513.331•1373.75/(2•25.5)+1050.6/(0.1923• π •1373.8²•25.5)
 = 68.06 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:
 = Q(L-2a)/(L+(4•hm/3))
 = 68722(6358.0 - 2 • 920.0)/(6358.0 + (4 • 0.0/3))
 = 48833.9 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:



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$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.1707 \cdot 48833.9 / (1373.75 \cdot 25.5)$$

$$= 16.00 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{ (Rm \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (1373.75 \cdot 25.5) }$$

$$= 145.988 \text{ mm}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{ (Rm \cdot t) }, 2a)$$

$$= \min(625.0 + 1.56 \cdot \sqrt{ (1373.75 \cdot 25.5) }, 2 \cdot 920.0)$$

$$= 916.98 \text{ mm}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(137.997/137.997, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.7603 \cdot 68722 \cdot 0.1 / (916.977(25.5 + 1.0 \cdot 24.0))$$

$$= -1.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.29) [σ7,r*]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$$

$$= -68722 / (4(25.5 + 1.0 \cdot 24.0) 916.977) -$$

$$12 \cdot 0.027 \cdot 68722 \cdot 1373.75 / (6358.0(25.5 + 1.0 \cdot 24.0)^2)$$

$$= -22.73 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	700.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (1361.0 \cdot 25.5) } + 950.0) 25.5^3) / 12$$

$$= 171.426 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 171.426 + 316.358(308.443 - 12.75)^2$$

$$= 276776.406 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1240.6	25.5	12.8	316.4	403356.2	0.277E+06
Wearplate	950.0	24.0	37.5	228.0	855000.0	0.167E+06
Web	24.0	633.5	366.2	152.0	5568464.0	0.559E+05
BasePlate	650.0	56.0	711.0	364.0	25880396.0	0.591E+06
Totals	...	...	...	1060.4	32707216.0	0.109E+07



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Distance to Centroid [C1]:

$$= AY / A$$

$$= 32707216 / 1060.398$$

$$= 308.443 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 68721.984$$

$$= 13986.4150 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 1.8435 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2100.0 - 1361.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 974.462 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 13629.4990 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 3.7787 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3 \cdot (Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3 \cdot (68722 + 2681) \cdot 650.0) / (4 \cdot 2350.0 \cdot 153.599)}$$

$$= 30.753 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) \cdot (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 2443.0576 / (6 - 1)$$

$$= 488.612 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 488.6115 \cdot 650.0 / 2$$

$$= 1587.988 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 71402.836 / (2350.0 \cdot 650.0)$$

$$= 4.674 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1588.0 \cdot 4.67$$

$$= 7423.032 \text{ Kgf}$$



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Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 144.24 + 58.633$$

$$= 202.873 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 7423.0 / 202.8734$$

$$= 3.588 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 625.0 - 24.0$$

$$= 601.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 202.873$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(48828.1 / 202.873)}$$

$$= 155.140 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpthk}) - \text{bpthk}$$

$$= 2100.0 - \cos(120.0 / 2) (1360.0 + 26.5 + 24.0) - 56.0$$

$$= 1338.750 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpthk})$$

$$= 2 \cdot \sin(120.0 / 2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350.0 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (2443.058 - 2299.2) / 2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{L_v^2 + D_d^2}$$

$$= \sqrt{1338.75^2 + 71.929^2}$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$



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$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994E+09 / 230.4)}$$

$$= 130.885$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 1340.681 / 155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 9045.9 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68 / 2$$

$$= 630.409 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($8.6418 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (8.64) / (8 \cdot 130.89) - (8.64^3) / (8 \cdot 130.89^3))$$

$$= 135.9 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 3.59 / 135.92 + (630.41 \cdot 312.5 / 48828E+09) / 153.6$$

$$= 0.052$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 261.507$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + W_{pdthk})^2 - (Pitch/2)^2} - B_{pthk}$$

$$= 2100.0 - \sqrt{(1410.5 + 24.0)^2 - (488.612/2)^2} - 56.0$$

$$= 654.819 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{48881.6 / 261.507}$$

$$= 136.720 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 654.819 / 136.72$$

$$= 4.789$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 9045.914 / (2 \cdot 2350.0)$$

$$= 1.925 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$



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$$= 1.925 \cdot 488.612 \cdot 654.819$$

$$= 615.812 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 615.812 / 1564213$$

$$= 3.861 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($4.7895 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (4.79)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (4.79) / (8 \cdot 130.89) - (4.79^3) / (8 \cdot 130.89^3))$$

$$= 137 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 5.57 / 137.02 + (615.81 \cdot 312.5 / 48881.609) / 153.6$$

$$= 0.066$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	10
Total Number of Bolts in Tension/Baseplate	Nbt	5
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0763 cm ²
Saddle Load Q0 (Weight)	Q0	59277.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	6882.8 Kg
Maximum Transverse Force	Ft	4523.0 Kg
Maximum Longitudinal Force	F1	9045.9 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 9045.9 / (14.076 \cdot 10)$$

$$= 6.3 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 4523.0 / (14.076 \cdot 10)$$

$$= 3.2 \text{ N./mm}^2. \text{ Must be less than } 82.8 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 9045.91 / (82.8 \cdot 10.0)$$

$$= 1.0714 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$= B \cdot F_t + \text{Sum of X Moments}$$

$$= 2100.0 \cdot 4522.96 + 0.0$$

$$= 9498.40 \text{ Kg-m.}$$

Eccentricity (e):



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= Rmom / QO
= 9498.4/59277.45
= 160.23 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= max[Bltarearl, Bltarears, Bltareart]
= max[0.0, 1.0714, 0.0]
= 1.0714 cm²

MDMT Calculations for: RH floating, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 24.0, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr * E^* / (tg - c) = 0.983$, Temp. Reduction = 1 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-2 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-3 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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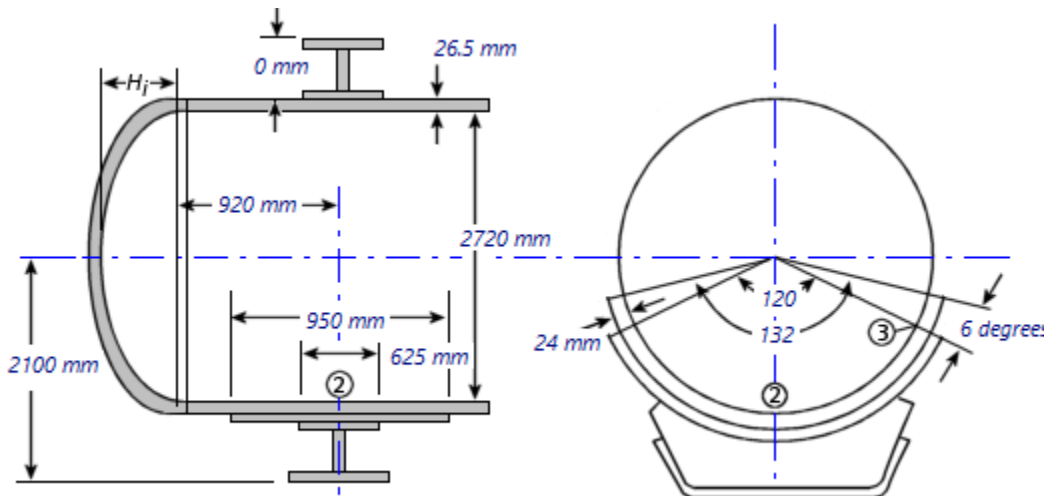


ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.75	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.00	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	100155.47	Kgf
Pressure used in Analysis	P 3263.465	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	82.60	234.00
Long. Stress at Bottom of Midspan	93.22	234.00
Long. Stress at Top of Saddles	86.98	234.00
Long. Stress at Bottom of Saddles	88.43	234.00
Tangential Shear in Shell	23.33	138.58
Circ. Stress at Horn of Saddle	33.13	350.99
Circ. Compressive Stress in Shell	1.65	234.00



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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3.0(628.5/2 + 0) \cdot 2100.0/2350.0$
 $= 842.5 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(373.66, 0) \cdot 2100.0/2760.0$
 $= 284.3 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 99313 + \max(284, 842, 0, 0)$
 $= 100155.5 \text{ Kg}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08(0.0 + \pi/4(2.773 \cdot 1.2)^2)$
 $= 7328.618 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	102836.30	Kgf
Transverse Shear Load	314.27	Kgf
Longitudinal Shear Load	373.66	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 2350.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$
 $= -100155 \cdot 920.0 [1 - (1 - 920.0/6358.0 + (1373.75^2 - 0.0^2)/(2 \cdot 920.0 \cdot 6358.0))/(1 + (4 \cdot 0.0)/(3 \cdot 6358.0))]$
 $= 1531.1 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 $= 100155 \cdot 6358/4((1 + 2(1374^2 - 0^2)/(6358^2))/(1 + (4 \cdot 0)/(3 \cdot 6358)) - 4 \cdot 920/6358)$
 $= 81919.8 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 $= 3263.465 \cdot 1373.75/(2 \cdot 25.5) - 81919.8/(\pi \cdot 1373.8^2 \cdot 25.5)$
 $= 82.60 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 $= 3263.465 \cdot 1373.75/(2 \cdot 25.5) + 81919.8/(\pi \cdot 1373.8^2 \cdot 25.5)$



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$$= 93.22 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 t)$
 $= 3263.465 \cdot 1373.75 / (2 \cdot 25.5) - 1531.1 / (0.1066 \cdot \pi \cdot 1373.8^2 \cdot 25.5)$
 $= 86.98 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:
 $= P \cdot Rm / (2t) + M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$
 $= 3263.465 \cdot 1373.75 / (2 \cdot 25.5) + 1531.1 / (0.1923 \cdot \pi \cdot 1373.8^2 \cdot 25.5)$
 $= 88.43 \text{ N./mm}^2$

Maximum Shear Force in the Saddle (4.15.3) [T]:
 $= Q(L-2a) / (L + (4 \cdot hm/3))$
 $= 100155(6358.0 - 2 \cdot 920.0) / (6358.0 + (4 \cdot 0.0/3))$
 $= 71170.6 \text{ Kg}$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:
 $= K2 \cdot T / (Rm \cdot t)$
 $= 1.1707 \cdot 71170.56 / (1373.75 \cdot 25.5)$
 $= 23.33 \text{ N./mm}^2$

Decay Length (4.15.20) [x_1, x_2]:
 $= 0.78 \cdot \sqrt{(Rm \cdot t)}$
 $= 0.78 \cdot \sqrt{(1373.75 \cdot 25.5)}$
 $= 145.988 \text{ mm}$

Effective reinforcing plate width (4.15.24) [B1]:
 $= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$
 $= \min(625.0 + 1.56 \cdot \sqrt{(1373.75 \cdot 25.5)}, 2 \cdot 920.0)$
 $= 916.98 \text{ mm}$

Wear Plate/Shell Stress ratio (4.15.26) [η]:
 $= 1.0000$ Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:
 $= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$
 $= -0.7603 \cdot 100155 \cdot 0.1 / (916.977(25.5 + 1.0 \cdot 24.0))$
 $= -1.65 \text{ N./mm}^2$

Circ. Comp. Stress at Horn of Saddle, $L < 8Rm$ (4.15.29) [σ_7, r^*]:
 $= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot Rm / (L(t + \eta \cdot tr)^2)$
 $= -100155 / (4(25.5 + 1.0 \cdot 24.0)916.977) -$
 $12 \cdot 0.027 \cdot 100155 \cdot 1373.75 / (6358.0(25.5 + 1.0 \cdot 24.0)^2)$
 $= -33.13 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	700.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$
 $= ((1.56 \cdot \sqrt{(1361.0 \cdot 25.5)} + 950.0)^3) / 12$
 $= 171.426 \text{ cm}^4$



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Resolved Inertia for the Shell [I,shell]:
 = ShellInertia + AShell(C1 - Y)²
 = 171.426 + 316.358(308.443 - 12.75)²
 = 276776.406 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	1240.6	25.5	12.8	316.4	403356.2	0.277E+06
Wearplate	950.0	24.0	37.5	228.0	855000.0	0.167E+06
Web	24.0	633.5	366.2	152.0	5568464.0	0.559E+05
BasePlate	650.0	56.0	711.0	364.0	25880396.0	0.591E+06
Totals	...	...	...	1060.4	32707216.0	0.109E+07

Distance to Centroid [C1]:
 = AY / A
 = 32707216/1060.398
 = 308.443 mm

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120.0/2
 = 120.0

Saddle Splitting Coefficient [K1]:
 = (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))
 = (1 + cos(120.0) - 0.5•sin(120.0)²)/(π - 2.094 + sin(120.0)cos(120.0))
 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.204 • 100155.469
 = 20383.8086 Kgf

Tension Stress, St = (Fh/As) = 2.6867 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 2100.0 - 1361.0 • sin(120.0/2)/1.0472
 = 974.462 mm

Bending Moment, M = Fh * d = 19863.6406 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 5.5071 N./mm²
 Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:
 = √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))
 = √(3(100155 + 2681)650.0/(4 • 2350.0 • 173.33))
 = 34.742 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2) (Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120.0/2) (1360.0 + 26.5 + 24.0)
 = 2443.058 mm

Distance between Ribs [e]:
 = Web Length / (Nribs - 1)
 = 2443.0576/(6 - 1)
 = 488.612 mm

Baseplate Pressure Area [Ap]:



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$$= e \cdot Bpwid / 2$$

$$= 488.6115 \cdot 650.0 / 2$$

$$= 1587.988 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 102836.305 / (2350.0 \cdot 650.0)$$

$$= 6.732 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1588.0 \cdot 6.73$$

$$= 10690.853 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 144.24 + 58.633$$

$$= 202.873 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 10690.9 / 202.8734$$

$$= 5.168 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 625.0 - 24.0$$

$$= 601.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 202.873$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 48828.1 / 202.873 }$$

$$= 155.140 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 2100.0 - \cos(120.0 / 2)(1360.0 + 26.5 + 24.0) - 56.0$$

$$= 1338.750 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(120.0 / 2)(1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350.0 - 50.8$$



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= 2299.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2
 = (2443.058 - 2299.2)/2
 = 71.929 mm

Length of Outer Rib [L]:

= $\sqrt{(Lv^2 + Dd^2)}$
 = $\sqrt{(1338.75^2 + 71.929^2)}$
 = 1340.681 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$
 = $\sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/260.0)}$
 = 123.211

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 1340.681/155.14
 = 8.642

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2
 = 373.7 / (2 • 2350.0) • 488.612 • 1340.68/2
 = 26.040 Kg-m.

Compressive Allowable, KL/r < Cc (8.6418 < 123.2105) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))
 = (1 - (8.64)² / (2 • 123.21²)) 260 /
 (5/3 + 3 • (8.64) / (8 • 123.21) - (8.64³) / (8 • 123.21³))
 = 153.2 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba
 = 5.17/153.2 + (26.04 • 312.5/48828E+09) / 173.33
 = 0.035

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/261.507
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 625.0/2
 = 312.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk
 = 2100.0 - $\sqrt{(1410.5 + 24.0)^2 - (488.612/2)^2}$ - 56.0
 = 654.819 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$
 = $\sqrt{(48881.6/261.507)}$



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= 136.720 mm

Slenderness ratio [KL/r]:

= KL/r
 = 1 • 654.819/136.72
 = 4.789

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)
 = 373.661/(2 • 2350.0)
 = 0.080 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L
 = 0.08 • 488.612 • 654.819
 = 25.437 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σB,base,c]:

= Bending Moment / Section Modulus
 = 25.437/1564213
 = 0.159 N./mm²

Compressive Allowable, KL/r < Cc (4.7895 < 123.2105) per AISC E2-1 [Sca]:

= (1-(KL/r)²/(2•Cc²))Fy/(5/3+3•(KL/r)/(8•Cc)-(KL/r)³/(8•Cc³))
 = (1-(4.79)²/(2 • 123.21²))260/
 (5/3+3•(4.79)/(8• 123.21)-(4.79³)/(8•123.21³))
 = 154.5 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I)/Sba
 = 8.02/154.53 + (25.44 • 312.5/48881.609)/173.33
 = 0.053

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	10
Total Number of Bolts in Tension/Baseplate	Nbt	5
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0763 cm ²
Saddle Load Q0 (Weight)	Q0	101993.8 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	284.3 Kg
Maximum Transverse Force	Ft	314.3 Kg
Maximum Longitudinal Force	Fl	373.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τb,l]:

= FI / (Bolt Area • Number of Bolts)
 = 373.7/(14.076 • 10)
 = 0.3 N./mm². Must be less than 103.2 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τb,t]:

= Ft / (Bolt Area • Number of Bolts)
 = 314.3/(14.076 • 10)
 = 0.2 N./mm². Must be less than 103.2 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)



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Bolt Area due to Shear Load [Bltarears]:
 $= F / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 $= 373.66 / (103.2 \cdot 10.0)$
 $= 0.0355 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 $= B \cdot Ft + \text{Sum of X Moments}$
 $= 2100.0 \cdot 314.27 + 0.0$
 $= 659.97 \text{ Kg-m.}$

Eccentricity (e):
 $= \text{Rmom} / \text{QO}$
 $= 659.97 / 101993.8$
 $= 6.47 \text{ mm} < \text{Bplen}/6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 $= 0 \text{ (No Uplift)}$

Required Area of a Single Bolt [Bltarear]:
 $= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
 $= \max[0.0, 0.0355, 0.0]$
 $= 0.0355 \text{ cm}^2$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	1373.75	mm
Shell Thickness used in this Case	t	26.500	mm
Stiffened Vessel Length per 4.15.6	L	6358.00	mm
Distance from Saddle to Vessel tangent	a1 or a	920.00	mm
Saddle Width	b1 or b	625.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	950.00	mm
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	24.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	2100.00	mm
Coefficient of Friction	mu	0.30	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		53009.71	Kgf
Pressure used in Analysis	P	3263.465	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		85.10	234.00
Long. Stress at Bottom of Midspan		90.72	234.00
Long. Stress at Top of Saddles		87.42	234.00
Long. Stress at Bottom of Saddles		88.18	234.00
Tangential Shear in Shell		12.35	138.58



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Circ. Stress at Horn of Saddle	17.53	350.99
Circ. Compressive Stress in Shell	0.87	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (628.5/2 + 0) \cdot 2100.0/2350.0$
 $= 842.5 \text{ Kg}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(373.66, 0) \cdot 2100.0/2760.0$
 $= 284.3 \text{ Kg}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 52167 + \max(284, 842, 0, 0)$
 $= 53009.7 \text{ Kg}$

Longitudinal Wind Force [F]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 766.08 (0.0 + \pi/4 (2.773 \cdot 1.2)^2)$
 $= 7328.618 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	55690.55	Kgf
Transverse Shear Load	314.27	Kgf
Longitudinal Shear Load	373.66	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \text{Baseplate Length}$
 $= 2350.000 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0267	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0219			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$
 $= -53010 \cdot 920.0 [1 - (1 - 920.0/6358.0 + (1373.75^2 - 0.0^2) / (2 \cdot 920.0 \cdot 6358.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 6358.0))]$
 $= 810.4 \text{ Kg-m.}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$
 $= 53010 \cdot 6358/4 ((1 + 2(1374^2 - 0^2)/(6358^2)) / (1 + (4 \cdot 0)/(3 \cdot 6358)) - 4 \cdot 920/6358)$
 $= 43358.1 \text{ Kg-m.}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$
 $= 3263.465 \cdot 1373.75 / (2 \cdot 25.5) - 43358.1 / (\pi \cdot 1373.8^2 \cdot 25.5)$
 $= 85.10 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$\begin{aligned}
 &= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 3263.465 \cdot 1373.75 / (2 \cdot 25.5) + 43358.1 / (\pi \cdot 1373.8^2 \cdot 25.5) \\
 &= 90.72 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 3263.465 \cdot 1373.75 / (2 \cdot 25.5) - 810.4 / (0.1066 \cdot \pi \cdot 1373.8^2 \cdot 25.5) \\
 &= 87.42 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t) \\
 &= 3263.465 \cdot 1373.75 / (2 \cdot 25.5) + 810.4 / (0.1923 \cdot \pi \cdot 1373.8^2 \cdot 25.5) \\
 &= 88.18 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 53010(6358.0 - 2 \cdot 920.0) / (6358.0 + (4 \cdot 0.0 / 3)) \\
 &= 37668.7 \text{ Kgf}
 \end{aligned}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$\begin{aligned}
 &= K2 \cdot T / (R_m \cdot t) \\
 &= 1.1707 \cdot 37668.75 / (1373.75 \cdot 25.5) \\
 &= 12.35 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{R_m \cdot t} \\
 &= 0.78 \cdot \sqrt{1373.75 \cdot 25.5} \\
 &= 145.988 \text{ mm}
 \end{aligned}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$\begin{aligned}
 &= \min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a) \\
 &= \min(625.0 + 1.56 \cdot \sqrt{1373.75 \cdot 25.5}, 2 \cdot 920.0) \\
 &= 916.98 \text{ mm}
 \end{aligned}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr)) \\
 &= -0.7603 \cdot 53010 \cdot 0.1 / (916.977(25.5 + 1.0 \cdot 24.0)) \\
 &= -0.87 \text{ N./mm}^2
 \end{aligned}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.29) [σ_7, r^*]:

$$\begin{aligned}
 &= -Q / (4(t + \eta \cdot tr)b1) - 12 \cdot K7 \cdot Q \cdot R_m / (L(t + \eta \cdot tr)^2) \\
 &= -53010 / (4(25.5 + 1.0 \cdot 24.0)916.977) - \\
 &\quad 12 \cdot 0.027 \cdot 53010 \cdot 1373.75 / (6358.0(25.5 + 1.0 \cdot 24.0)^2) \\
 &= -17.53 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	2350.0000	mm
Baseplate Thickness	Bpthk	56.0000	mm
Baseplate Width	Bpwid	650.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	6	
Rib Thickness	Ribtk	24.0000	mm
Web Thickness	Webtk	24.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	700.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{R \cdot t}) + \text{WearPlateWidth})^3 / 12$$



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$$= ((1.56 \cdot \sqrt{(1361.0 \cdot 25.5)} + 950.0) 25.5^3) / 12$$

$$= 171.426 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + \text{AShell}(C1 - Y)^2$$

$$= 171.426 + 316.358(308.443 - 12.75)^2$$

$$= 276776.406 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	1240.6	25.5	12.8	316.4	403356.2	0.277E+06
Wearplate	950.0	24.0	37.5	228.0	855000.0	0.167E+06
Web	24.0	633.5	366.2	152.0	5568464.0	0.559E+05
BasePlate	650.0	56.0	711.0	364.0	25880396.0	0.591E+06
Totals	...	...	...	1060.4	32707216.0	0.109E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 32707216 / 1060.398$$

$$= 308.443 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120.0 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120.0) - 0.5 \cdot \sin(120.0)^2) / (\pi - 2.094 + \sin(120.0) \cos(120.0))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.204 \cdot 53009.711$$

$$= 10788.6260 \text{ Kg}$$

Tension Stress, $St = (Fh / As) = 1.4220 \text{ N./mm}^2$
 Allowed Stress, $Sa = 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 2100.0 - 1361.0 \cdot \sin(120.0/2) / 1.0472$$

$$= 974.462 \text{ mm}$$

Bending Moment, $M = Fh \cdot d = 10513.3135 \text{ Kg-m.}$

Bending Stress, $Sb = (M \cdot C1 / I) = 2.9148 \text{ N./mm}^2$
 Allowed Stress, $Sa = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(53010 + 2681)650.0) / (4 \cdot 2350.0 \cdot 173.33)}$$

$$= 25.567 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120.0/2) (1360.0 + 26.5 + 24.0)$$

$$= 2443.058 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (N_{\text{ribs}} - 1)$$

$$= 2443.0576 / (6 - 1)$$



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= 488.612 mm

Baseplate Pressure Area [Ap]:

= e • Bpwid / 2
 = 488.6115 • 650.0/2
 = 1587.988 cm²

Bearing Pressure [Bp]:

= (Q + SaddleWt) / (BasePlateLength • BasePlateWidth)
 = 55690.555/(2350.0 • 650.0)
 = 3.646 Kg/cm²

Axial Load [P]:

= Ap • Bp
 = 1588.0 • 3.65
 = 5789.585 Kg

Area of the Rib and Web [Ar]:

= Rib Area + Web Area
 = 144.24 + 58.633
 = 202.873 cm²

Compressive Stress [Sc]:

= P/Ar
 = 5789.6/202.8734
 = 2.799 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	24.0	625.0	...	150.0	...	0.488E+05

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 625.0 - 24.0
 = 601.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 0.0/202.873
 = 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2
 = 625.0/2
 = 312.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{48828.1/202.873}$
 = 155.140 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpthk) - bpthk
 = 2100.0 - cos(120.0/2)(1360.0 + 26.5 + 24.0) - 56.0
 = 1338.750 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpthk)
 = 2 • sin(120.0/2)(1360.0 + 26.5 + 24.0)
 = 2443.058 mm



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Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 2350.0 - 50.8$$

$$= 2299.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (2443.058 - 2299.2) / 2$$

$$= 71.929 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1338.75^2 + 71.929^2}$$

$$= 1340.681 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1340.681 / 155.14$$

$$= 8.642$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bpln) \cdot e \cdot L / 2$$

$$= 373.7 / (2 \cdot 2350.0) \cdot 488.612 \cdot 1340.68 / 2$$

$$= 26.040 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($8.6418 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (8.64)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (8.64) / (8 \cdot 123.21) - (8.64^3) / (8 \cdot 123.21^3))$$

$$= 153.2 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 2.8 / 153.2 + (26.04 \cdot 312.5 / 48828\text{E}+09) / 173.33$$

$$= 0.019$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	24.0	601.0	0.0	144.2	0.0	0.488E+05
Web	488.6	24.0	0.0	117.3	0.0	56.3
Totals	...	...	...	261.5	...	0.489E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 261.507$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 625.0 / 2$$

$$= 312.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 2100.0 - \sqrt{(1410.5 + 24.0)^2 - (488.612/2)^2} - 56.0$$

$$= 654.819 \text{ mm}$$



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Radius of Gyration [r]:
 $= \sqrt{(\text{Total Inertia} / \text{Total Area})}$
 $= \sqrt{(48881.6/261.507)}$
 $= 136.720 \text{ mm}$

Slenderness ratio [KL/r]:
 $= KL/r$
 $= 1 \cdot 654.819/136.72$
 $= 4.789$

Unit Force [Force,u]:
 $= FI / (2 \cdot \text{Baseplate Length})$
 $= 373.661/(2 \cdot 2350.0)$
 $= 0.080 \text{ Kg/mm}$

Moment at base of inner Rib [Mbase,c]:
 $= \text{Unit Force} \cdot e \cdot L$
 $= 0.08 \cdot 488.612 \cdot 654.819$
 $= 25.437 \text{ Kg-m.}$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
 $= \text{Bending Moment} / \text{Section Modulus}$
 $= 25.437/1564213$
 $= 0.159 \text{ N./mm}^2$

Compressive Allowable, $KL/r < C_c$ ($4.7895 < 123.2105$) per AISC E2-1 [Sca]:
 $= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$
 $= (1 - (4.79)^2 / (2 \cdot 123.21^2)) 260 /$
 $(5/3 + 3 \cdot (4.79) / (8 \cdot 123.21) - (4.79^3) / (8 \cdot 123.21^3))$
 $= 154.5 \text{ N./mm}^2$

AISC Unity Check of Inside Ribs (must be ≤ 1)
 $= S_c/S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$
 $= 4.34/154.53 + (25.44 \cdot 312.5/48881.609) / 173.33$
 $= 0.029$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	10
Total Number of Bolts in Tension/Baseplate	Nbt	5
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	70.0 mm
User Defined Area in a Single Bolt	UserBoltArea	16.1400 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	14.0763 cm ²
Saddle Load Q0 (Weight)	Q0	54848.1 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	284.3 Kg
Maximum Transverse Force	Ft	314.3 Kg
Maximum Longitudinal Force	F1	373.7 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
 $= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 373.7 / (14.076 \cdot 10)$
 $= 0.3 \text{ N./mm}^2$. Must be less than 103.2 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
 $= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 314.3 / (14.076 \cdot 10)$
 $= 0.2 \text{ N./mm}^2$. Must be less than 103.2 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss



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Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot N_{\text{bolts}})$
= $373.66 / (103.2 \cdot 10.0)$
= 0.0355 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $2100.0 \cdot 314.27 + 0.0$
= 659.97 Kg-m.

Eccentricity (e):
= R_{mom} / QO
= $659.97 / 54848.05$
= 12.03 mm < $B_{\text{plen}}/6$ --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0.0, 0.0355, 0.0]$
= 0.0355 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: Tube inlet

From: 30

Pressure for Reinforcement Calculations	P	1000.000	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	1.01	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2720.00	mm
Design Length of Section	L	1500.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1146.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	755.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

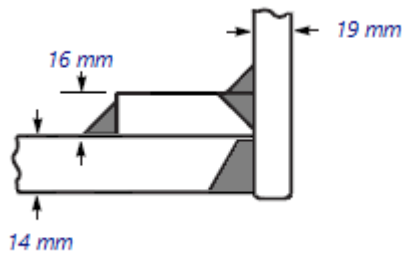
The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Tube inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 12.500 in.
 Actual Thickness Used in Calculation 0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 1363.0 / (138 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 9.9207 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1000.0 \cdot 158.75 / (118 \cdot 1.0 - 0.6 \cdot 1000.0)$
 $= 1.3523 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.1791 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	635.0000	mm
Parallel to Vessel Wall, opening length	d	317.5000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000	mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

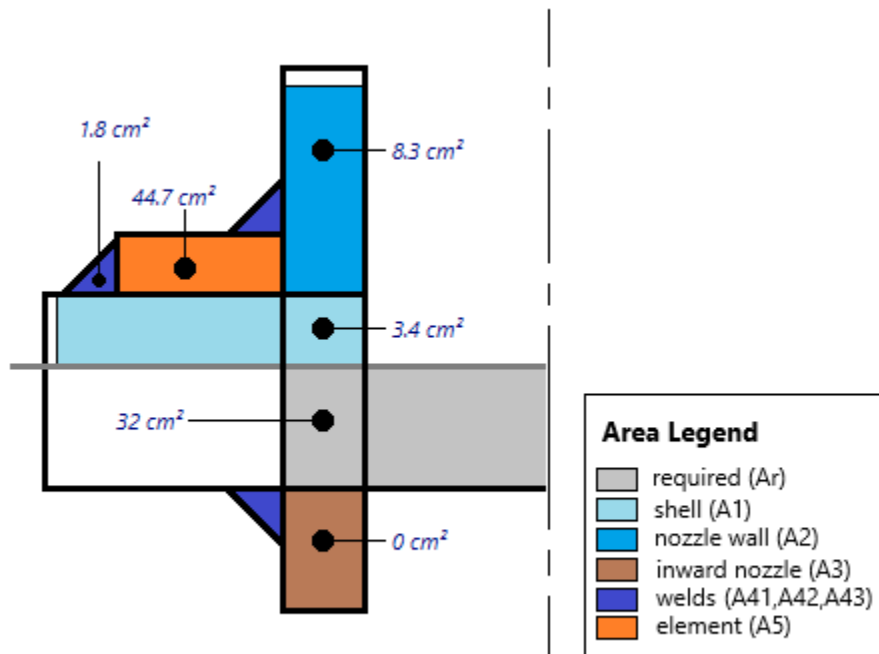


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	32.046	1.752	NA
Area in Shell	A1	3.367	30.933	NA
Area in Nozzle Wall	A2	8.323	8.875	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.819	1.819	NA
Area in Element	A5	44.704	44.704	NA
TOTAL AREA AVAILABLE	Atot	58.213	86.331	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 471.4540 16.0000 mm
 Based on given Pad Diameter: 755.6000 6.6344 mm
 Based on Shell or Nozzle Thickness: 488.0045 14.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (317.5 \cdot 9.9207 \cdot 1.0 + 2 \cdot 19.05 \cdot 9.9207 \cdot 1.0 \cdot (1 - 0.86))$
 $= 32.046 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 317.5(1.0 \cdot 11.0 - 1.0 \cdot 9.921) - 2 \cdot 19.05$
 $(1.0 \cdot 11.0 - 1.0 \cdot 9.9207) \cdot (1 - 0.855)$



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$$= 3.367 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (19.05 - 1.35) \cdot 0.8551 \\ &= 8.323 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\ &= (2.1275) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0 \\ &= 1.819 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (635.0 - 355.6) \cdot 16.0 \cdot 1.0 \\ &= 44.704 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.3523 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 12.9207 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 12.9207 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.331, \max(12.9207, 5.5)] \\ &= 8.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(1.3523, 8.3312) \\ &= 8.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 19.050 = 16.669 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	17.7,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	277.3 N./mm ²	Passed
Occasional	:	3.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	10.7,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 16.669, tr = 1.352, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.081$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	7 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 16.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.902$, Temp. Reduction = 5 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
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Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.902$, Temp. Reduction = 5 °C
 Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-20 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14.0, tr = 9.921, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.902$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 9.921, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.902$, Temp. Reduction = 5 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-24 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $1000.00 / 1960.01 = 0.510$

Weld Size Calculations, Description: Tube inlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * W_o \text{ mm}$
Pad Weld	$5.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t-tr))Sv)$
 $= \max(0, (32.046 - 3.3672 + 2 \cdot 19.05 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 9.9207))138)$
 $= 40850.41 \text{ Kg}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$



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$$= (8.323 + 44.704 + 1.8192 - 0.0 \cdot 0.86) \cdot 138$$

$$= 77177.22 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (8.323 + 0.0 + 1.9239 + (3.5836)) \cdot 138$$

$$= 19461.77 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (8.323 + 0.0 + 1.8192 + 44.704 + (3.5836)) \cdot 138$$

$$= 82219.92 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 355.6 \cdot 15.0 \cdot 0.49 \cdot 118$$

$$= 49399. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 755.6 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 98205. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 168.275) \cdot (19.05 - 0.0) \cdot 0.7 \cdot 118$$

$$= 84822. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 355.6 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 93063. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 355.6 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 63981. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (98205 + 84822) = 183026 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (49399 + 93063 + 63981 + 0) = 206442 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (98205 + 63981 + 0) = 162185 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 183026 Kgf, must exceed W = 40850 Kgf	or W1 = 77177 Kgf
Path 2-2 = 206442 Kgf, must exceed W = 40850 Kgf	or W2 = 19461 Kgf
Path 3-3 = 162185 Kgf, must exceed W = 40850 Kgf	or W3 = 82219 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1082 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



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 Nozzle Calcs.: Tube inlet Nozl: 9 10:19am Sep 9,2026

Nozzle is O.K. for the External Pressure: 1 KPa.

The Drop for this Nozzle is : 11.6724 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 375.6725 mm

Input Echo, WRC297 Item 1, Description: Tube inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2720.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	317.500	mm
Nozzle Thickness	19.0500	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	755.600	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1000.00	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1376.5(14.0 - 3.0)}$$

$$= 761.668 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2753.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	27.000	mm
Nozzle Thickness used	(t) =	19.050	mm



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T / t = 1.417
 d / t = 18.667

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.304$$

Nr/P = 0.087
 Mr/P = 0.139
 M0/P = 0.060
 N0/P = 0.138
 MrD/Mc = 0.245
 NrDL/Mc = 0.127
 M0d/Mc = 0.120
 N0DL/Mc = 0.093
 MrD/M1 = 0.162
 NrDL/M1 = 0.094
 M0D/M1 = 0.078
 N0DL/M1 = 0.193

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	49	50	49	50
Outplane Stress Summary	24	61	111	24

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-64	64	64	-64
Inplane Pressure Stress	24	24	24	24
Inplane Stress Summary	-12	56	128	-60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	38	64	129	85



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	24	24	24	24
Outplane Stress Summary	-10	58	96	-24

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-97	97	97	-97
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	49	50	49	50
Inplane Stress Summary	-22	113	188	-64

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	23	113	189	65

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	1	2	27	28

Nozzle Stresses (N./mm²):



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LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	41	-41	41	-41
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-91	91	91	-91
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-55	45	150	-116

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	58	47	150	145

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	8	9	20	21

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	41	-41	41	-41
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-145	145	145	-145
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-108	98	203	-169

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0



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Shear due to (V1)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	8	8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	116	99	203	192

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1368.5(14.0 - 3.0)}$
 $= 760.486 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2737.000 mm
Nozzle Outside Diameter	(d) =	755.600 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	219.050 mm
T / t	=	0.050
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.355$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.078
M0/P	=	0.021
N0/P	=	0.024
MrD/Mc	=	0.247
NrDL/Mc	=	0.108
M0d/Mc	=	0.074
N0DL/Mc	=	0.047
MrD/Ml	=	0.054
NrDL/Ml	=	0.060
M0D/Ml	=	0.016
N0DL/Ml	=	0.027

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	123	124	123	124
Outplane Stress Summary	132	115	179	88

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	101	-101	101	-101
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-60	60	60	-60
Inplane Pressure Stress	61	61	61	61
Inplane Stress Summary	102	20	246	-78

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	133	115	246	166

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-82	82	82	-82
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	61	61	61	61
Outplane Stress Summary	2	113	186	-34

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	101	-101	101	-101
Inplane Membrane (Mc)	-20	-20	20	20
Inplane Bending (Mc)	-277	277	277	-277
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	123	124	123	124
Inplane Stress Summary	-61	291	533	-223

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1



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Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	64	291	533	223

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	49	50	49	50	49	50	49	50
Circ. Pl (SUS)	-8	-8	16	16	-5	-5	11	11
Circ. Q (SUS)	-18	18	42	-42	-67	67	127	-127
Long. Pm (SUS)	24	24	24	24	24	24	24	24
Long. Pl (SUS)	-3	-3	9	9	-2	-2	10	10
Long. Q (SUS)	-34	34	94	-94	-35	35	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	49.0	50.0	49.0	50.0	49.0	50.0	49.0	50.0
Pm+Pl (SUS)	41.0	42.0	65.0	66.0	44.0	45.0	60.0	61.0
Pm+Pl+Q (Total)	37.4	63.1	127.4	85.2	23.8	112.2	187.3	66.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	50.00	138.00	Passed
Pm+Pl (SUS)	66.03	207.00	Passed
Pm+Pl+Q (TOTAL)	187.27	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	123	124	123	124	123	124	123	124
Circ. Pl (SUS)	0	0	10	10	-10	-10	30	30
Circ. Q (SUS)	9	-9	45	-45	-176	176	378	-378
Long. Pm (SUS)	61	61	61	61	61	61	61	61
Long. Pl (SUS)	-1	-1	21	21	-3	-3	13	13
Long. Q (SUS)	41	-41	161	-161	-55	55	109	-109



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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	123.0	124.0	123.0	124.0	123.0	124.0	123.0	124.0
Pm+P1 (SUS)	123.0	124.0	133.0	134.0	113.0	114.0	153.0	154.0
Pm+P1+Q (Total)	132.3	115.1	243.0	168.0	66.0	290.0	531.0	224.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	124.00	138.00	Passed
Pm+P1 (SUS)	154.01	207.00	Passed
Pm+P1+Q (TOTAL)	531.03	413.99	Failed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. P1 (SUS)	-8	-8	16	16	-2	-2	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. P1 (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-50	50	132	-132	-104	104	186	-186
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+P1 (SUS)	6.3	7.3	25.1	26.1	12.2	13.2	19.2	20.2
Pm+P1+Q (Total)	57.7	46.1	147.2	143.3	116.4	99.3	201.3	191.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	118.00	Passed
Pm+P1 (SUS)	26.09	177.00	Passed
Pm+P1+Q (TOTAL)	201.27	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Input, Nozzle Desc: Tube outlet

From: 30

Pressure for Reinforcement Calculations	P	1026.661	KPa.
Temperature for Internal Pressure	Temp	82	°C
Design External Pressure	Pext	1.01	KPa.
Temperature for External Pressure	Tempex	82	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2720.00	mm
Design Length of Section	L	1500.0000	mm
Shell Finished (Minimum) Thickness	t	14.0000	mm
Shell Internal Corrosion Allowance	c	3.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1146.35	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

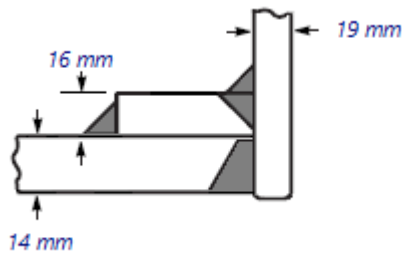
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		14.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-182 F22	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	15.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	755.6000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		200.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	12.500 in.
Actual Thickness Used in Calculation	0.750 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026.66 \cdot 1363.0 / (138 \cdot 1.0 - 0.6 \cdot 1026.66)$
 $= 10.1864 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1026.66 \cdot 158.75 / (118 \cdot 1.0 - 0.6 \cdot 1026.66)$
 $= 1.3886 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 0.1791 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	635.0000 mm
Parallel to Vessel Wall, opening length	d	317.5000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		27.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

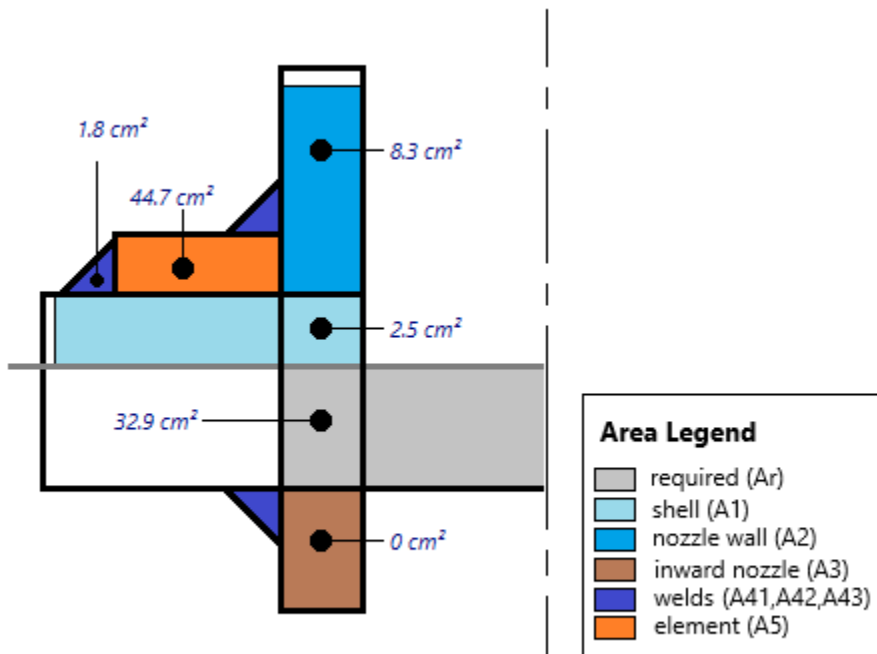


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	32.904	1.752	NA
Area in Shell	A1	2.538	30.933	NA
Area in Nozzle Wall	A2	8.306	8.875	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.819	1.819	NA
Area in Element	A5	44.704	44.704	NA
TOTAL AREA AVAILABLE	Atot	57.367	86.331	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 482.1049 16.0000 mm
 Based on given Pad Diameter: 755.6000 7.2444 mm
 Based on Shell or Nozzle Thickness: 500.1770 14.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$
 $= (317.5 \cdot 10.1864 \cdot 1.0 + 2 \cdot 19.05 \cdot 10.1864 \cdot 1.0 \cdot (1 - 0.86))$
 $= 32.904 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 317.5(1.0 \cdot 11.0 - 1.0 \cdot 10.1864) - 2 \cdot 19.05$
 $(1.0 \cdot 11.0 - 1.0 \cdot 10.1864) \cdot (1 - 0.855)$



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$$= 2.538 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 27.5) \cdot (19.05 - 1.39) \cdot 0.8551 \\ &= 8.306 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= (Wo^2 - Ar \text{ Lost}) \cdot Fr3 + ((Wi - can / 0.707)^2 - Ar \text{ Lost}) \cdot fr2 + Wp^2 \cdot fr4 \\ &= (2.1275) \cdot 0.86 + (0.0) \cdot 0.86 + 0.0^2 \cdot 1.0 \\ &= 1.819 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (635.0 - 355.6) \cdot 16.0 \cdot 1.0 \\ &= 44.704 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.3886 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 13.1864 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 13.1864 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 8.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[8.331, \max(13.1864, 5.5)] \\ &= 8.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(1.3886, 8.3312) \\ &= 8.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 19.050 = 16.669 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 17.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 277.2 N./mm ²	Passed
Occasional	: 4.0,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 10.7,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: B

Govrn. thk, tg = 16.669, tr = 1.389, c = 0.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.083$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-13 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 16.0, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.926$, Temp. Reduction = 4 °C
 Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 16.0, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.926$, Temp. Reduction = 4 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-18 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 14.0, tr = 10.186, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.926$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 14.0, tr = 10.186, c = 3.0 mm, E* = 1.0

Thickness Ratio = $tr * E* / (tg - c) = 0.926$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-22 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-29 °C
Gov. MDMT of the Reinforcement Pad	:	-29 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-29 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-47 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1026.66 / 1960.01 = 0.524$

Weld Size Calculations, Description: Tube outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	11.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$10.6050 = 0.7 * Wo \text{ mm}$
Pad Weld	$5.5000 = 0.5 * TminPad$	$8.4840 = 0.7 * Wp \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) Sv)$$

$$= \max(0, (32.9042 - 2.5383 + 2 \cdot 19.05 \cdot 0.8551 \cdot (1.0 \cdot 11.0 - 10.1864)) 138)$$

$$= 43102.64 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



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$$= (A2 + A5 + A4 - (W_i - C_{an} / .707)^2 \cdot fr2) \cdot S_v$$

$$= (8.306 + 44.704 + 1.8192 - 0.0 \cdot 0.86) \cdot 138$$

$$= 77153.24 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (8.306 + 0.0 + 1.9239 + (3.5836)) \cdot 138$$

$$= 19437.79 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (8.306 + 0.0 + 1.8192 + 44.704 + (3.5836)) \cdot 138$$

$$= 82195.94 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 355.6 \cdot 15.0 \cdot 0.49 \cdot 118$$

$$= 49399. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 755.6 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 98205. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo}) / 4) \cdot (Thk - C_{an}) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 168.275) \cdot (19.05 - 0.0) \cdot 0.7 \cdot 118$$

$$= 84822. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 355.6 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 93063. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 355.6 \cdot (14.0 - 3.0) \cdot 0.74 \cdot 138$$

$$= 63981. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (98205 + 84822) = 183026 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (49399 + 93063 + 63981 + 0) = 206442 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (98205 + 63981 + 0) = 162185 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 183026 Kgf, must exceed W = 43102 Kgf	or W1 = 77153 Kgf
Path 2-2 = 206442 Kgf, must exceed W = 43102 Kgf	or W2 = 19437 Kgf
Path 3-3 = 162185 Kgf, must exceed W = 43102 Kgf	or W3 = 82195 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1082 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle is O.K. for the External Pressure: 1 KPa.

The Drop for this Nozzle is : 11.6724 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 375.6725 mm

Input Echo, WRC297 Item 1, Description: Tube outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2720.000	mm
Shell Thickness	14.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	0.0000	mm
Nozzle Diameter	317.500	mm
Nozzle Thickness	19.0500	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	755.600	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1026.66	KPa.
Radial Load	P	-2676.93	Kgf
Circumferential Shear	Vc	1784.62	Kgf
Longitudinal Shear	Vl	1784.62	Kgf
Circumferential Moment	Mc	1748.26	Kg-m.
Longitudinal Moment	Ml	1748.26	Kg-m.
Torsional Moment	Mt	2497.50	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)} \\
 &= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1376.5(14.0 - 3.0)} \\
 &= 761.668 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2753.000	mm
Nozzle Outside Diameter	(d) =	355.600	mm
Vessel Thickness used	(T) =	27.000	mm



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Nozzle Thickness used (t) = 19.050 mm
 T / t = 1.417
 d / t = 18.667

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.304$$

Nr/P = 0.087
 Mr/P = 0.139
 M0/P = 0.060
 N0/P = 0.138
 MrD/Mc = 0.245
 NrDL/Mc = 0.127
 M0d/Mc = 0.120
 N0DL/Mc = 0.093
 MrD/Ml = 0.162
 NrDL/Ml = 0.094
 M0D/Ml = 0.078
 N0DL/Ml = 0.193

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-30	30	30	-30
Normal Pressure Stress	51	52	51	52
Outplane Stress Summary	25	62	112	26

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-64	64	64	-64
Inplane Pressure Stress	25	25	25	25
Inplane Stress Summary	-11	56	129	-59

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	4	4	4	4
Shear Stress Summary	5	5	3	3

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	38	66	130	85
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	12	-12	12	-12
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-47	47	47	-47
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	25	25	25	25
Outplane Stress Summary	-10	59	97	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	30	-30	30	-30
Inplane Membrane (Mc)	-8	-8	8	8
Inplane Bending (Mc)	-97	97	97	-97
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	52	51	52
Inplane Stress Summary	-21	114	190	-63

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	4	4	4	4
Shear Stress Summary	3	3	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	22	114	190	64

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	1	3	27	28



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	41	-41	41	-41
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-10	-10	10	10
Axial Bending (ML)	-91	91	91	-91
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-54	45	150	-116

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	6	6	6	6
Shear Stress Summary	8	8	4	4

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	59	47	150	145

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	9	10	9	10
Hoop Stress Summary	8	9	20	22

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	41	-41	41	-41
Axial Membrane (Mc)	-10	-10	10	10
Axial Bending (Mc)	-145	145	145	-145
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	4	4	4	4
Axial Stress Summary	-108	99	203	-169

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	6	6	6	6
Shear Stress Summary	4	4	8	8

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	116	99	203	192

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$
 $= 355.6 + 2 \cdot 1.65 \cdot \sqrt{1368.5(14.0 - 3.0)}$
 $= 760.486 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2737.000 mm
Nozzle Outside Diameter	(d) =	755.600 mm
Vessel Thickness used	(T) =	11.000 mm
Nozzle Thickness used	(t) =	219.050 mm
T / t	=	0.050
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.355$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.051
Mr/P	=	0.078
M0/P	=	0.021
N0/P	=	0.024
MrD/Mc	=	0.247
NrDL/Mc	=	0.108
M0d/Mc	=	0.074
N0DL/Mc	=	0.047
MrD/Ml	=	0.054
NrDL/Ml	=	0.060
M0D/Ml	=	0.016
N0DL/Ml	=	0.027

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-5	-5	5	5
Outplane Bending (ML)	-18	18	18	-18
Normal Pressure Stress	126	127	126	127
Outplane Stress Summary	136	118	183	91

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	101	-101	101	-101
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-11	-11	11	11
Inplane Bending (ML)	-60	60	60	-60
Inplane Pressure Stress	63	63	63	63
Inplane Stress Summary	103	22	247	-76

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Shear stress normal to longitudinal plane)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE
 (Stress Intensities)

	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	136	118	247	168

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses Normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	27	-27	27	-27
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-82	82	82	-82
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	63	63	63	63

Outplane Stress Summary	4	114	188	-33
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stresses parallel to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	10	10	10	10
Inplane Bending (P)	101	-101	101	-101
Inplane Membrane (Mc)	-20	-20	20	20
Inplane Bending (Mc)	-277	277	277	-277
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	126	127	126	127

Inplane Stress Summary	-58	294	537	-220
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Shear stress normal to circumferential plane)

	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0



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Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	63	294	537	220

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	51	52	51	52	51	52	51	52
Circ. Pl (SUS)	-8	-8	16	16	-5	-5	11	11
Circ. Q (SUS)	-18	18	42	-42	-67	67	127	-127
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. Pl (SUS)	-3	-3	9	9	-2	-2	10	10
Long. Q (SUS)	-34	34	94	-94	-35	35	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	51.0	52.0	51.0	52.0	51.0	52.0	51.0	52.0
Pm+Pl (SUS)	43.0	44.0	67.0	68.0	46.0	47.0	62.0	63.0
Pm+Pl+Q (Total)	38.3	64.8	128.5	86.2	21.9	114.2	189.3	64.6

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	52.00	138.00	Passed
Pm+Pl (SUS)	68.03	207.00	Passed
Pm+Pl+Q (TOTAL)	189.26	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	126	127	126	127	126	127	126	127
Circ. Pl (SUS)	0	0	10	10	-10	-10	30	30
Circ. Q (SUS)	9	-9	45	-45	-176	176	378	-378
Long. Pm (SUS)	63	63	63	63	63	63	63	63
Long. Pl (SUS)	-1	-1	21	21	-3	-3	13	13
Long. Q (SUS)	41	-41	161	-161	-55	55	109	-109



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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	126.0	127.0	126.0	127.0	126.0	127.0	126.0	127.0
Pm+P1 (SUS)	126.0	127.0	136.0	137.0	116.0	117.0	156.0	157.0
Pm+P1+Q (Total)	135.3	118.1	245.0	169.0	65.0	293.0	534.0	221.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	127.00	138.00	Passed
Pm+P1 (SUS)	157.01	207.00	Passed
Pm+P1+Q (TOTAL)	534.03	413.99	Failed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. P1 (SUS)	-8	-8	16	16	-2	-2	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. P1 (SUS)	-9	-9	11	11	-9	-9	11	11
Long. Q (SUS)	-50	50	132	-132	-104	104	186	-186
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	6	6	6	6	6	6	6	6
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+P1 (SUS)	6.3	7.3	25.1	26.1	12.2	13.2	19.2	20.2
Pm+P1+Q (Total)	57.7	46.1	147.2	143.3	116.4	99.3	201.3	191.5

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	118.00	Passed
Pm+P1 (SUS)	26.09	177.00	Passed
Pm+P1+Q (TOTAL)	201.27	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Input, Nozzle Desc: Shell inlet

From: 60

Pressure for Reinforcement Calculations	P	2500.000	KPa.
Temperature for Internal Pressure	Temp	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2720.00	mm
Shell Finished (Minimum) Thickness	t	26.5000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5041.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

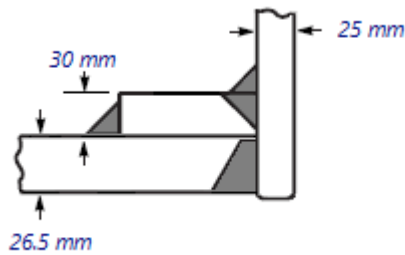
Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		30.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	25.0000	mm
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1382.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	22.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	30.0000	mm
Reinforcing Pad Width		285.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	
Flange Series		B	

The Pressure Design option was Design Pressure + static head.



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Reinforcement CALCULATION, Description: Shell inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.984 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2500.0 \cdot 1361.0 / (138 \cdot 1.0 - 0.6 \cdot 2500.0)$
 $= 24.9287 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2500.0 \cdot 384.0 / (118 \cdot 1.0 - 0.6 \cdot 2500.0)$
 $= 8.2410 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1536.0000	mm
Parallel to Vessel Wall, opening length	d	768.0000	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		63.7500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

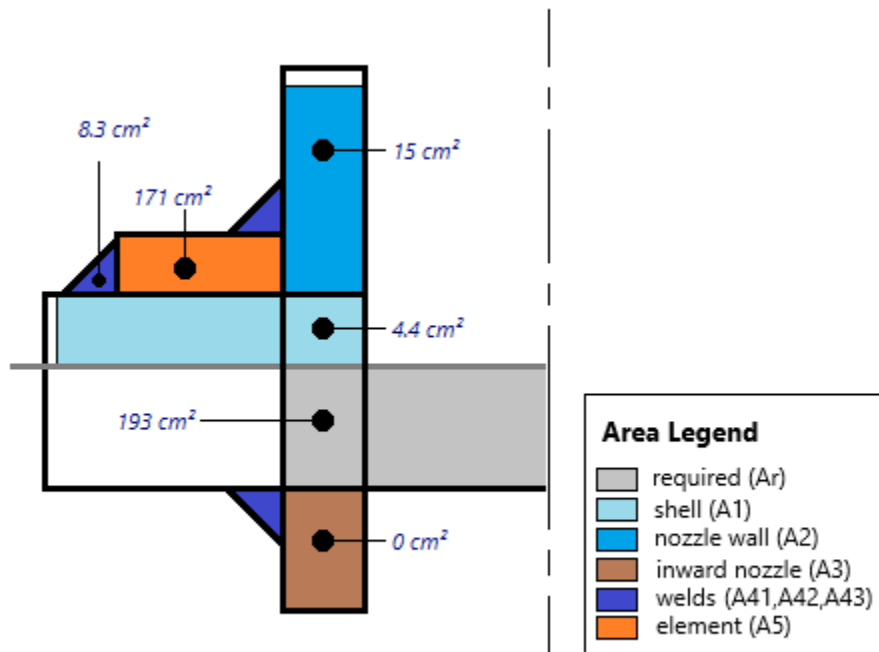


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	193.042	NA	NA
Area in Shell	A1	4.351	NA	NA
Area in Nozzle Wall	A2	15.000	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds A41+A42+A43		8.260	NA	NA
Area in Element	A5	171.000	NA	NA
TOTAL AREA AVAILABLE	Atot	198.612	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1363.4325 30.0000 mm
 Based on given Pad Diameter: 1382.0001 29.0228 mm
 Based on Shell or Nozzle Thickness: 1473.7190 25.0000 mm

Area Required [A]:
 $= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1))$ UG-37(c)
 $= (768.0 \cdot 24.9287 \cdot 1.0 + 2 \cdot 22.0 \cdot 24.9287 \cdot 1.0 \cdot (1 - 0.86))$
 $= 193.042 \text{ cm}^2$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:
 $= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$
 $= 768.0(1.0 \cdot 25.5 - 1.0 \cdot 24.929) - 2 \cdot 22.0$
 $(1.0 \cdot 25.5 - 1.0 \cdot 24.9287) \cdot (1 - 0.855)$



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= 4.351 cm²

Area Available in Nozzle Wall Projecting Outward [A2]:

= (2 • Tlwp) • (tn - trn) • fr2
 = (2 • 63.75) • (22.0 - 8.24) • 0.8551
 = 15.000 cm²

Area Available in Welds [A41 + A42 + A43]:

= Wo² • fr3 + (Wi-can/0.707)² • fr2 + Wp² • fr4
 = 20.0² • 0.86 + (0.0)² • 0.86 + 22.0² • 1.0
 = 8.260 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4
 = (1382.0 - 812.0) • 30.0 • 1.0
 = 171.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 11.2410 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 25.9287 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 25.9287 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]
 = min[11.331, max(25.9287, 5.5)]
 = 11.3312 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(11.241, 11.3312)
 = 11.3312 mm

Available Nozzle Neck Thickness = 25.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 30.8,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 264.2 N./mm ²	Passed
Occasional	: 21.2,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 7.5,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 25.0, tr = 8.241, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr • E*/(tg - c) = 0.375, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-41 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 25.0, tr = 8.241, c = 3.0 mm, E* = 1.0
 Thickness Ratio = tr • E*/(tg - c) = 0.375, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-1 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 °C



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Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 25.0, tr = 8.241, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.375$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 26.5, tr = 24.929, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.978$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -1 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 25.0, tr = 8.241, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.375$, Temp. Reduction = 60 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -41 °C
 Gov. MDMT of the Reinforcement Pad : -1 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -1 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2500.00 / 5100.00 = 0.490$

Weld Size Calculations, Description: Shell inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	14.1400 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5*TminPad	15.5540 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (193.0418 - 4.3514 + 2 \cdot 22.0 \cdot 0.8551 \cdot (1.0 \cdot 25.5 - 24.9287)) \cdot 138)$
 $= 265819.41 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$
 $= (15.0003 + 171.0 + 8.2603 - 0.0 \cdot 0.86) \cdot 138$
 $= 273355.22 \text{ Kgf}$



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Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (15.0003 + 0.0 + 3.4203 + (9.5939)) \cdot 138$$

$$= 39420.83 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (15.0003 + 0.0 + 8.2603 + 171.0 + (9.5939)) \cdot 138$$

$$= 286855.38 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 812.0 \cdot 20.0 \cdot 0.49 \cdot 118$$

$$= 150400. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1382.0 \cdot 22.0 \cdot 0.49 \cdot 138$$

$$= 329298. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 395.0) \cdot (25.0 - 3.0) \cdot 0.7 \cdot 118$$

$$= 229939. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 812.0 \cdot 30.0 \cdot 0.74 \cdot 138$$

$$= 398448. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot (T - cas) \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 812.0 \cdot (26.5 - 1.0) \cdot 0.74 \cdot 138$$

$$= 338681. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (329298 + 229939) = 559237 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (150400 + 398448 + 338681 + 0) = 887528 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (329298 + 338681 + 0) = 667979 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 559237 Kgf, must exceed W = 265819 Kgf or W1 = 273355 Kgf
 Path 2-2 = 887528 Kgf, must exceed W = 265819 Kgf or W2 = 39420 Kgf
 Path 3-3 = 667978 Kgf, must exceed W = 265819 Kgf or W3 = 286855 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2530 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 62.0154 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 438.5154 mm



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This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 768.00 mm

Input Echo, WRC297 Item 1, Description: Shell inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2720.000	mm
Shell Thickness	26.5000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	762.000	mm
Nozzle Thickness	25.0000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1382.000	mm
Thickness of Reinforcing Pad	30.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	2500.00	KPa.
Radial Load	P	-5736.27	Kgf
Circumferential Shear	Vc	3824.19	Kgf
Longitudinal Shear	Vl	3824.19	Kgf
Circumferential Moment	Mc	8027.71	Kg-m.
Longitudinal Moment	Ml	8027.71	Kg-m.
Torsional Moment	Mt	11468.14	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 812.0 + 2 \cdot 1.65 \cdot \sqrt{1388.75(26.5 - 1.0)}$$

$$= 1433.006 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2777.500	mm
Nozzle Outside Diameter	(d) =	812.000	mm
Vessel Thickness used	(T) =	55.500	mm
Nozzle Thickness used	(t) =	22.000	mm
T / t	=	2.523	



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$$d / t = 36.909$$

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.068$$

Nr/P	=	0.070
Mr/P	=	0.074
M0/P	=	0.054
N0/P	=	0.172
MrD/Mc	=	0.165
NrDL/Mc	=	0.100
M0d/Mc	=	0.139
N0DL/Mc	=	0.213
MrD/Ml	=	0.094
NrDL/Ml	=	0.063
M0D/Ml	=	0.074
N0DL/Ml	=	0.318

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-10	-10	10	10
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	60	62	60	62
Outplane Stress Summary	45	63	93	55

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-17	17	17	-17
Inplane Pressure Stress	30	30	30	30
Inplane Stress Summary	19	38	59	7

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	45	63	93	55



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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	5	-5	5	-5
Outplane Membrane (Mc)	-6	-6	6	6
Outplane Bending (Mc)	-26	26	26	-26
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	30	30	30	30
Outplane Stress Summary	5	46	72	7

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	8	-8	8	-8
Inplane Membrane (Mc)	-3	-3	3	3
Inplane Bending (Mc)	-31	31	31	-31
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	60	62	60	62
Inplane Stress Summary	35	83	103	27

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	35	83	103	28

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-10	-10	10	10
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	50	53	50	53
Hoop Stress Summary	44	47	64	67

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-74	74	74	-74
Axial Pressure Stress	25	25	25	25
Axial Stress Summary	-28	66	136	-68

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	6	6	3	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	73	68	136	135

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-6	-6	6	6
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	50	53	50	53
Hoop Stress Summary	47	50	60	63

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-138	138	138	-138
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	25	25	25	25
Axial Stress Summary	-91	129	199	-131

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1



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Shear due to Torsion	4	4	4	4
Shear Stress Summary	3	3	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	139	130	199	195

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 812.0 + 2 \cdot 1.65 \cdot \sqrt{1373.75(26.5 - 1.0)}$$

$$= 1429.644 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) =	2747.500 mm
Nozzle Outside Diameter (d) =	1382.000 mm
Vessel Thickness used (T) =	25.500 mm
Nozzle Thickness used (t) =	307.000 mm
T / t =	0.083
d / t =	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.221$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P =	0.045
Mr/P =	0.070
M0/P =	0.019
N0/P =	0.028
MrD/Mc =	0.239
NrDL/Mc =	0.098
M0d/Mc =	0.071
N0DL/Mc =	0.053
MrD/Ml =	0.048
NrDL/Ml =	0.046
M0D/Ml =	0.015
N0DL/Ml =	0.026

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	10	-10	10	-10
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	132	134	132	134
Outplane Stress Summary	134	132	154	121

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3



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Inplane Bending (P)	36	-36	36	-36
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3
Inplane Bending (ML)	-25	25	25	-25
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	77	54	135	12

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)	0	0	0	0	0
Outplane Shear (Vl)	0	0	0	0	0
Outplane Shear (Mt)	1	1	1	1	1
Shear Stress Summary	2	2	0	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities)		Top Outside	Top Inside	Bottom Outside	Bottom Inside
Two * Max Shear Stress	134	132	154	121	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Membrane (P)	2	2	2	2	2
Outplane Bending (P)	10	-10	10	-10	
Outplane Membrane (Mc)	-4	-4	4	4	
Outplane Bending (Mc)	-37	37	37	-37	
Outplane Membrane (ML)	0	0	0	0	
Outplane Bending (ML)	0	0	0	0	
Normal Pressure Stress	66	66	66	66	
Outplane Stress Summary	36	91	120	25	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	3	3	3	3	
Inplane Bending (P)	36	-36	36	-36	
Inplane Membrane (Mc)	-8	-8	8	8	
Inplane Bending (Mc)	-125	125	125	-125	
Inplane Membrane (ML)	0	0	0	0	
Inplane Bending (ML)	0	0	0	0	
Inplane Pressure Stress	132	134	132	134	
Inplane Stress Summary	38	219	306	-15	

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)		Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0	
Outplane Shear (Vl)	0	0	0	0	
Torsional Shear (Mt)	1	1	1	1	



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Shear Stress Summary 0 0 2 2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	38	219	306	41

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	60	62	60	62	60	62	60	62
Circ. Pl (SUS)	-7	-7	13	13	-2	-2	4	4
Circ. Q (SUS)	-8	8	18	-18	-23	23	39	-39
Long. Pm (SUS)	30	30	30	30	30	30	30	30
Long. Pl (SUS)	0	0	2	2	-3	-3	9	9
Long. Q (SUS)	-9	9	25	-25	-21	21	31	-31
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	60.0	62.0	60.0	62.0	60.0	62.0	60.0	62.0
Pm+Pl (SUS)	53.0	55.0	73.0	75.0	58.0	60.0	64.0	66.0
Pm+Pl+Q (Total)	45.0	63.0	91.0	57.0	35.0	83.0	103.0	27.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	62.00	138.00	Passed
Pm+Pl (SUS)	75.00	207.00	Passed
Pm+Pl+Q (TOTAL)	103.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	132	134	132	134	132	134	132	134
Circ. Pl (SUS)	0	0	4	4	-5	-5	11	11
Circ. Q (SUS)	3	-3	17	-17	-89	89	161	-161
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	0	0	6	6	-2	-2	6	6
Long. Q (SUS)	11	-11	61	-61	-27	27	47	-47
Shear Pm (SUS)	0	0	0	0	0	0	0	0



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Shear P1 (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	132.0	134.0	132.0	134.0	132.0	134.0	132.0	134.0
Pm+P1 (SUS)	132.0	134.0	136.0	138.0	127.0	129.0	143.0	145.0
Pm+P1+Q (Total)	135.0	131.0	153.0	121.0	38.6	218.0	304.0	41.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	134.00	138.00	Passed
Pm+P1 (SUS)	145.00	207.00	Passed
Pm+P1+Q (TOTAL)	304.01	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	50	53	50	53	50	53	50	53
Circ. P1 (SUS)	-7	-7	13	13	-3	-3	9	9
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	25	25	25	25	25	25	25	25
Long. P1 (SUS)	-6	-6	8	8	-6	-6	8	8
Long. Q (SUS)	-47	47	101	-101	-111	111	165	-165
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	50.0	53.0	50.0	53.0	50.0	53.0	50.0	53.0
Pm+P1 (SUS)	43.0	46.0	63.0	66.0	47.0	50.0	59.0	62.0
Pm+P1+Q (Total)	71.7	67.2	134.1	134.1	139.1	130.1	198.2	194.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	53.00	118.00	Passed
Pm+P1 (SUS)	66.03	177.00	Passed
Pm+P1+Q (TOTAL)	198.18	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Input, Nozzle Desc: Shell outlet

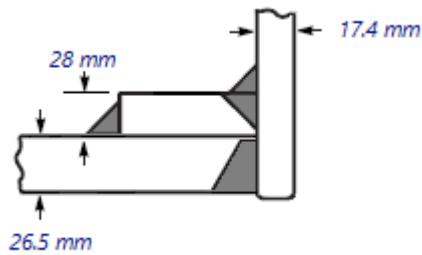
From: 60

Pressure for Reinforcement Calculations	P	2526.662	KPa.
Temperature for Internal Pressure	Temp	230	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	2720.00	mm
Shell Finished (Minimum) Thickness	t	26.5000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		5041.70	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		24.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	350.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	20.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	28.5000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1109.6000	mm
Thickness of Pad	te	28.0000	mm
Weld leg size between Pad and Shell	Wp	22.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	28.0000	mm
Reinforcing Pad Width		250.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: Shell outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	22.626 in.
Actual Thickness Used in Calculation	0.687 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2526.66 \cdot 1361.0 / (138 \cdot 1.0 - 0.6 \cdot 2526.66)$
 $= 25.1975 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 2526.66 \cdot 290.35 / (118 \cdot 1.0 - 0.6 \cdot 2526.66)$
 $= 6.2985 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1161.4008	mm
Parallel to Vessel Wall, opening length	d	580.7004	mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		63.7500	mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$

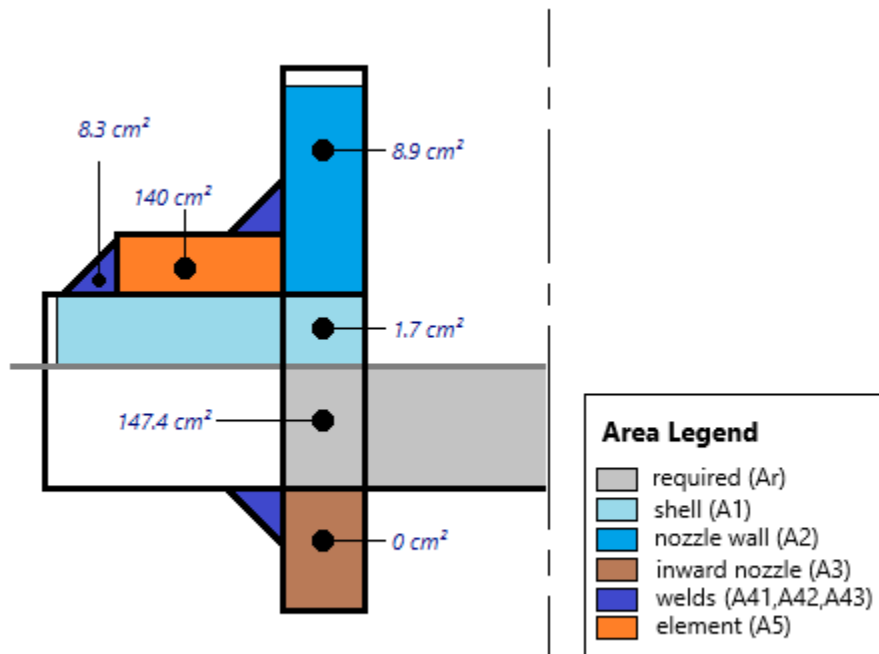


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	147.377	NA	NA
Area in Shell	A1	1.744	NA	NA
Area in Nozzle Wall	A2	8.887	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	8.260	NA	NA
Area in Element	A5	140.000	NA	NA
TOTAL AREA AVAILABLE	Atot	158.891	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Insufficient.
 The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 1068.4785 28.0000 mm
 Based on given Pad Diameter: 1109.6000 25.6972 mm
 Based on the Estimated Diameter Limit: 1159.8134 23.3520 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (580.7004 \cdot 25.1975 \cdot 1.0 + 2 \cdot 14.4498 \cdot 25.1975 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 147.377 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 580.7(1.0 \cdot 25.5 - 1.0 \cdot 25.197) - 2 \cdot 14.45$$

$$(1.0 \cdot 25.5 - 1.0 \cdot 25.1975) \cdot (1 - 0.855)$$



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$$= 1.744 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned} &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\ &= (2 \cdot 63.75) \cdot (14.45 - 6.3) \cdot 0.8551 \\ &= 8.887 \text{ cm}^2 \end{aligned}$$

Area Available in Welds [A41 + A42 + A43]:

$$\begin{aligned} &= Wo^2 \cdot fr3 + (Wi-can/0.707)^2 \cdot fr2 + Wp^2 \cdot fr4 \\ &= 20.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 22.0^2 \cdot 1.0 \\ &= 8.260 \text{ cm}^2 \end{aligned}$$

Area Available in Element [A5]:

$$\begin{aligned} &= (\min(Dp, DL) - (\text{Nozzle OD})) \cdot (\min(tp, Tlwp, te)) \cdot fr4 \\ &= (1109.6 - 609.6) \cdot 28.0 \cdot 1.0 \\ &= 140.000 \text{ cm}^2 \end{aligned}$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 9.2985 mm
Wall Thickness per UG16(b),	tr16b = 5.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 26.1975 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 26.1975 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 5.5000 mm
Wall Thickness per table UG-45	tb3 = 11.3312 mm

Determine Nozzle Thickness candidate [tb]:

$$\begin{aligned} &= \min[tb3, \max(tb1, tb2)] \\ &= \min[11.331, \max(26.1975, 5.5)] \\ &= 11.3312 \text{ mm} \end{aligned}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$\begin{aligned} &= \max(ta, tb) \\ &= \max(9.2985, 11.3312) \\ &= 11.3312 \text{ mm} \end{aligned}$$

Available Nozzle Neck Thickness = $0.875 \cdot 17.450 = 15.269 \text{ mm}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 41.2,	Allowable	: 118.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 253.7 N./mm ²	Passed
Occasional	: 24.8,	Allowable	: 156.9 N./mm ²	Passed
Shear	: 12.6,	Allowable	: 82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 15.269, tr = 6.298, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.513$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve A	4 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-26 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 15.269, tr = 6.298, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot E*/(tg - c) = 0.513$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-16 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-46 °C



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Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 15.269, tr = 6.298, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.513$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -16 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 26.5, tr = 25.197, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.988$, Temp. Reduction = 1 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 1 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 15.269, tr = 6.298, c = 3.0 mm, E* = 1.0
 Thickness Ratio = $tr * E* / (tg - c) = 0.513$, Temp. Reduction = 31 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -16 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -46 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Gov. MDMT of the Nozzle : -26 °C
 Gov. MDMT of the Reinforcement Pad : 0 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : 0 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -48 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $2526.66 / 5100.00 = 0.495$

Weld Size Calculations, Description: Shell outlet

Intermediate Calc. for nozzle/shell Welds Tmin 14.4498 mm
 Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$14.1400 = 0.7 * W_o \text{ mm}$
Pad Weld	$9.5000 = 0.5 * \text{TminPad}$	$15.5540 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (147.3772 - 1.7441 + 2 \cdot 14.4498 \cdot 0.8551 \cdot (1.0 \cdot 25.5 - 25.1975)) \cdot 138)$
 $= 205033.70 \text{ Kgf}$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A5+A4-(W_i - Can/.707)^2 \cdot f_r2) \cdot Sv$
 $= (8.8867 + 140.0 + 8.2603 - 0.0 \cdot 0.86) \cdot 138$
 $= 221130.50 \text{ Kgf}$



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Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S_v$$

$$= (8.8867 + 0.0 + 3.4203 + (6.3014)) \cdot 138$$

$$= 26184.88 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (8.8867 + 0.0 + 8.2603 + 140.0 + (6.3014)) \cdot 138$$

$$= 229997.52 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot 20.0 \cdot 0.49 \cdot 118$$

$$= 112911. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW}$$

$$= (3.1416/2.0) \cdot 1109.6 \cdot 22.0 \cdot 0.49 \cdot 138$$

$$= 264392. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - Can) \cdot 0.7 \cdot S_n$$

$$= (3.1416 \cdot 297.5751) \cdot (17.4498 - 3.0) \cdot 0.7 \cdot 118$$

$$= 113776. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg}$$

$$= (3.1416/2) \cdot 609.6 \cdot 28.0 \cdot 0.74 \cdot 138$$

$$= 279188. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot D_{lo} \cdot (T - cas) \cdot 0.74 \cdot S_{ng}$$

$$= (3.1416/2.0) \cdot 609.6 \cdot (26.5 - 1.0) \cdot 0.74 \cdot 138$$

$$= 254261. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (264392 + 113776) = 378168 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw})$$

$$= (112911 + 279188 + 254261 + 0) = 646360 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (S_{pew} + T_{ngw} + S_{inw})$$

$$= (264392 + 254261 + 0) = 518652 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 378167 Kgf, must exceed W = 205033 Kgf or W1 = 221130 Kgf
 Path 2-2 = 646360 Kgf, must exceed W = 205033 Kgf or W2 = 26184 Kgf
 Path 3-3 = 518652 Kgf, must exceed W = 205033 Kgf or W3 = 229997 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 2530 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 34.5956 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 411.0956 mm



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Input Echo, WRC297 Item 1, Description: Shell outlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	2720.000	mm
Shell Thickness	26.5000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	574.700	mm
Nozzle Thickness	17.4498	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	1109.600	mm
Thickness of Reinforcing Pad	28.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	2526.66	KPa.
Radial Load	P	-4589.03	Kgf
Circumferential Shear	Vc	3059.35	Kgf
Longitudinal Shear	Vl	3059.35	Kgf
Circumferential Moment	Mc	5137.73	Kg-m.
Longitudinal Moment	Ml	5137.73	Kg-m.
Torsional Moment	Mt	7339.62	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$\begin{aligned}
 &= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)} \\
 &= 609.6 + 2 \cdot 1.65 \cdot \sqrt{1387.75(26.5 - 1.0)} \\
 &= 1230.383 \text{ mm}
 \end{aligned}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2775.500	mm
Nozzle Outside Diameter	(d) =	609.600	mm
Vessel Thickness used	(T) =	53.500	mm
Nozzle Thickness used	(t) =	14.450	mm
T / t	=	3.702	
d / t	=	42.187	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.582$$



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Nr/P = 0.055
 Mr/P = 0.065
 MØ/P = 0.073
 NØ/P = 0.231
 MrD/Mc = 0.131
 NrDL/Mc = 0.092
 MØd/Mc = 0.190
 NØDL/Mc = 0.266
 MrD/Ml = 0.090
 NrDL/Ml = 0.069
 MØD/Ml = 0.118
 NØDL/Ml = 0.434

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-12	-12	12	12
Outplane Bending (ML)	-20	20	20	-20
Normal Pressure Stress	63	65	63	65
Outplane Stress Summary	40	70	106	54

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-15	15	15	-15
Inplane Pressure Stress	31	31	31	31
Inplane Stress Summary	20	39	56	12

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	41	70	106	54

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-32	32	32	-32
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	31	31	31	31

Outplane Stress Summary	1	53	82	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	6	-6	6	-6
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-22	22	22	-22
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	63	65	63	65

Inplane Stress Summary	44	80	95	40
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	2	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	44	80	95	40

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses in the hoop direction)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-12	-12	12	12
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	59	62	59	62

Hoop Stress Summary	50	53	75	78
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	1	1	1	1



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Axial Bending (P)	48	-48	48	-48
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-12	-12	12	12
Axial Bending (ML)	-130	130	130	-130
Axial Pressure Stress	29	29	29	29
Axial Stress Summary	-64	101	223	-135

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	8	8	8	8
Shear Stress Summary	10	10	6	6

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Outside	Inside	Outside	Inside
Two * Max Shear Stress	116	103	223	214

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside
Hoop Membrane (P)	3	3	3	3
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-7	-7	7	7
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	59	62	59	62
Hoop Stress Summary	55	58	70	73

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	1	1	1	1
Axial Bending (P)	48	-48	48	-48
Axial Membrane (Mc)	-12	-12	12	12
Axial Bending (Mc)	-201	201	201	-201
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	29	29	29	29
Axial Stress Summary	-134	171	293	-205

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	8	8	8	8
Shear Stress Summary	6	6	10	10



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	190	171	294	280

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 609.6 + 2 \cdot 1.65 \cdot \sqrt{1373.75(26.5 - 1.0)}$$

$$= 1227.244 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	2747.500 mm
Nozzle Outside Diameter	(d) =	1109.600 mm
Vessel Thickness used	(T) =	25.500 mm
Nozzle Thickness used	(t) =	264.450 mm
T / t	=	0.096
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.192$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.050
Mr/P	=	0.080
MØ/P	=	0.023
NØ/P	=	0.033
MrD/Mc	=	0.250
NrDL/Mc	=	0.103
MØd/Mc	=	0.075
NØDL/Mc	=	0.059
MrD/Ml	=	0.059
NrDL/Ml	=	0.057
MØD/Ml	=	0.018
NØDL/Ml	=	0.036

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-2	-2	2	2
Outplane Bending (ML)	-7	7	7	-7
Normal Pressure Stress	133	136	133	136
Outplane Stress Summary	135	133	155	123

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-3	-3	3	3



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Inplane Bending (ML)	-24	24	24	-24
Inplane Pressure Stress	66	66	66	66
Inplane Stress Summary	74	57	132	16

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)				
	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	1	1	1	1
Shear Stress Summary	2	2	0	0

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)				
	Au	Al	Bu	Bl
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	135	133	155	123

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)				
	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-4	-4	4	4
Outplane Bending (Mc)	-31	31	31	-31
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	66	66	66	66
Outplane Stress Summary	43	86	114	32

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)				
	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	3	3	3	3
Inplane Bending (P)	33	-33	33	-33
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-104	104	104	-104
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	133	136	133	136
Inplane Stress Summary	58	204	282	8

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)				
	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	1	1	1	1
Shear Stress Summary	0	0	2	2

Vessel Stresses (N./mm²):



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CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	58	204	282	32

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	63	65	63	65	63	65	63	65
Circ. Pl (SUS)	-9	-9	15	15	-2	-2	2	2
Circ. Q (SUS)	-14	14	26	-26	-16	16	28	-28
Long. Pm (SUS)	31	31	31	31	31	31	31	31
Long. Pl (SUS)	-1	-1	1	1	-4	-4	10	10
Long. Q (SUS)	-9	9	21	-21	-26	26	38	-38
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	63.0	65.0	63.0	65.0	63.0	65.0	63.0	65.0
Pm+Pl (SUS)	54.0	56.0	78.0	80.0	61.0	63.0	65.0	67.0
Pm+Pl+Q (Total)	40.2	70.1	104.1	54.1	45.1	79.2	93.3	39.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	65.00	138.00	Passed
Pm+Pl (SUS)	80.00	207.00	Passed
Pm+Pl+Q (TOTAL)	104.08	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	133	136	133	136	133	136	133	136
Circ. Pl (SUS)	0	0	4	4	-4	-4	10	10
Circ. Q (SUS)	2	-2	16	-16	-71	71	137	-137
Long. Pm (SUS)	66	66	66	66	66	66	66	66
Long. Pl (SUS)	0	0	6	6	-2	-2	6	6
Long. Q (SUS)	9	-9	57	-57	-22	22	40	-40
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	1	1	1	1	1	1	1	1
Pm (SUS)	133.0	136.0	133.0	136.0	133.0	136.0	133.0	136.0



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Pm+Pl (SUS)	133.0	136.0	137.0	140.0	129.0	132.0	143.0	146.0
Pm+Pl+Q (Total)	135.0	134.0	153.0	124.0	58.1	203.0	280.0	32.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	136.00	138.00	Passed
Pm+Pl (SUS)	146.00	207.00	Passed
Pm+Pl+Q (TOTAL)	280.01	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	59	62	59	62	59	62	59	62
Circ. Pl (SUS)	-9	-9	15	15	-4	-4	10	10
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	29	29	29	29	29	29	29	29
Long. Pl (SUS)	-11	-11	13	13	-11	-11	13	13
Long. Q (SUS)	-82	82	178	-178	-153	153	249	-249
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	8	8	8	8	8	8	8	8
Pm (SUS)	59.0	62.0	59.0	62.0	59.0	62.0	59.0	62.0
Pm+Pl (SUS)	50.1	53.1	74.1	77.1	55.1	58.1	69.1	72.1
Pm+Pl+Q (Total)	115.7	102.0	220.2	213.3	190.4	171.3	291.4	279.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	62.00	118.00	Passed
Pm+Pl (SUS)	77.11	177.00	Passed
Pm+Pl+Q (TOTAL)	291.45	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.



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Input Echo, Tubesheet Number 1, Description: Tubesheet

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	2500.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2500.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	1.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2720.000 mm
Shell Temperature for Internal Pressure	Ts	230.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2720.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5600
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5350.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5000.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	437.34 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1262.920 mm
Largest Center-to-Center Tube Distance	U1	15.8750 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	48355.9 cm ²
Perimeter of Tube Layout	Cp	8128.3 mm

Tubesheet Data:

Tubesheet TYPE: Stationary, Gasketed both Sides, Conf. d
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	175.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2806.000 mm
Dimension G for the Channel Side	Gc	2777.669 mm
Area of the Untubed Lanes	AL	435.8 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Both
Flange Outside Diameter	A	3144.000 mm
Flange Inside Diameter	B	2724.000 mm
Flange Face Outside Diameter	Fod	2809.000 mm
Flange Face Inside Diameter	Fid	2724.000 mm
Gasket Outside Diameter	Go	2798.000 mm
Gasket Inside Diameter	Gi	2756.000 mm
Small end Hub thk.	g0	28.0000 mm
Large end Hub thk.	g1	42.0000 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	6.3500 mm
Full face Gasket Flange Option		Program Selects
Length of Partition Gasket	lp	2725.000 mm
Width of Partition Gasket	wp	10.0000 mm
Partition Gasket Factor,	mPart	3.0000
Partition Gasket Design Seating Stress	yPart	68.95 N./mm ²
Partition Gasket Facing Sketch		Code Sketch 1a
Partition Gasket Column for Gasket Seating		Code Column II

Bolting Information:

Diameter of Bolt Circle	C	3004.000 mm
Nominal Bolt Diameter	dB	69.8500 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	32
Bolt Material		SA-193 B7
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	159.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	159.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm



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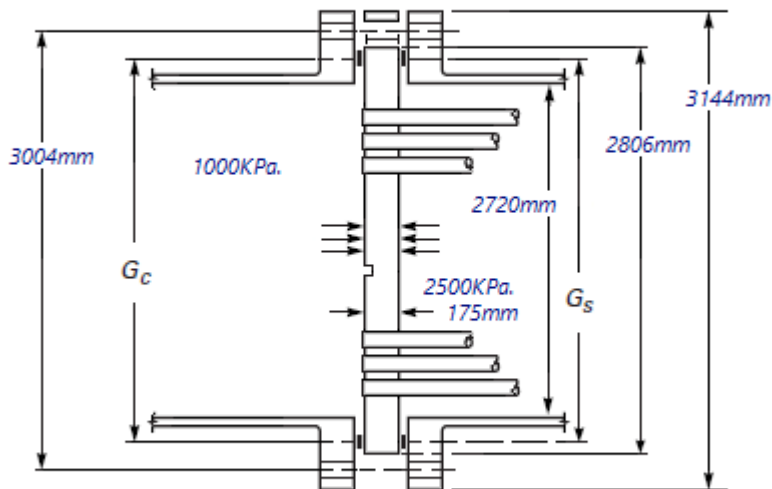
Alternate Flange Operating Bolt Load, Wm1	1697411.12 Kg
Alternate Flange Seating Bolt Load, Wm2	597492.62 Kg
Alternate Flange Design Bolt Load, W	1760277.62 Kg

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with	None
Tubesheet Extended as Flange	No

Is Exchanger in Creep range (skip EP, Use 3S for Sps)	No
---	----

Tubesheet Gasketed w/Shell and Channel



Configuration d:

Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
 Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
 Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1



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Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1
 TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

Note:

The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 1000.0)$
 $= 0.0678 + 0.0000 = 0.0678 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-5 at 82.00 °C
 Elastic Modulus for Material 190996000.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5000.00	12.04	50.0000	0.0075908	182.68

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 182.675) / (3 \cdot 12.0379) = 20232.123 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6638	25.40	5000.00	38.26	50.0000	0.0007513	71.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 71.7506) / (3 \cdot 38.2643) = 2500.0347 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6638 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	20232.12 KPa.
Required External Design Pressure	2500.00 KPa.
Required Thickness due to Shell Side pressure	0.6638 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Gasketed Tubesheets:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	21.000 mm
Basic Gasket Width,	$b0 = N / 2$	10.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	8.162 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2781.675 mm

Results for ASME Stationary Tubesheet Calculations for Configuration d:



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Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \quad (\text{modified for corrosion if present})$$

$$= 0/175.0$$

$$= 0.0000 \quad (\text{must be } 0 \leq \rho \leq 1)$$

Effective tube hole diameter [d^*]:

$$= \max(d_t - 2 \cdot t_t (E_t/E) \cdot (S_t/S) \cdot (\rho); d_t - 2 \cdot t_t)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.19529E+09/0.19529E+09) \cdot (186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [D_o]:

$$= 2 \cdot r_o + d_t$$

$$= 2 \cdot 1262.92 + 25.4$$

$$= 2551.240 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - d_t) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$

Equivalent outer tube limit radius [a_o]:

$$= D_o / 2$$

$$= 2551.2402/2$$

$$= 1275.6201 \text{ mm}$$

Effective tube pitch [p^*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 31.75 / \sqrt{1 - 4 \cdot \min(435.77 \cdot 100.0; 4 \cdot 2551.24 \cdot 31.75) / (\pi \cdot 2551.24^2)}$$

$$= 31.8862 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.8862 - 25.4) / 31.8862$$

$$= 0.2034$$

Ratio [ρ_s]:

$$= a_s / a_o$$

$$= 1390.8345/1275.6201$$

$$= 1.0903$$

Ratio [ρ_c]:

$$= a_c / a_o$$

$$= 1388.8345/1275.6201$$

$$= 1.0888$$

Compute parameter [x_t]:

$$= 1 - Nt((d_t - 2 \cdot t_t) / (2 \cdot a_o))^2$$

$$= 1 - 5600((25.4 - 2 \cdot 2.11) / (2 \cdot 1275.6201))^2$$

$$= 0.6140$$

Determine parameter [x_s]:

$$= 1 - Nt(d_t / (2 \cdot a_o))^2$$

$$= 1 - 5600(25.4 / (2 \cdot 1275.6201))^2$$

$$= 0.4449$$

Determine the value [$h'g$]:

$$= \max(h_g - CATC; 0) \quad (\text{For pressure only cases})$$

$$= \max(0 - 0; 0)$$

$$= 0.000 \text{ mm}$$



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ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 $\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 $\beta_C, k_C, \lambda_C, \delta_C$ are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:

$h/p = 5.511811$; $\mu^* = 0.203417$; $E^* = 31127258$ KPa.
 $E^*/E = 0.159387$; $\nu^* = 0.453616$

Note: As h/p (5.512) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor $[X_a]$:
 $= ((24(1 - \nu^{*2})Nt \cdot Et \cdot tt(dt - tt)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$
 $= ((24(1 - 0.454^2)5600 \cdot 0.19529E+09 \cdot 2.11$
 $(25.4 - 2.11)1275.6201^2) / (31127258 \cdot 5000.0 \cdot 175.0^3))^{1/4}$
 $= 6.6862$

Values from Table 4.18.3

$Z_d = 0.004955$; $Z_v = 0.022169$; $Z_m = 0.211858$
 $Z_a = 0.198592E+03$; $Z_w = 0.022169$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio $[K]$:
 $= A / D_o$
 $= 2806.0/2551.2402$
 $= 1.0999$

Compute coefficient $[F]$:
 $= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_C + E \cdot \ln(K))$
 $= (1 - 0.45) / 31127258 \cdot (0 + 0 +$
 $0.19529E+09 \cdot \ln(1.1))$
 $= 0.3263$

Compute parameter $[\phi]$:
 $= (1 + \nu^*) F$
 $= (1 + 0.4536) 0.3263$
 $= 0.4743$

Compute parameter $[Q_1]$:
 $= (\rho_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$
 $= (1.0903 - 1 - 0.4743 \cdot 0.0222) / (1 + 0.4743 \cdot 0.2119)$
 $= 0.072519$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor $[\gamma_b]$:
 $= (G_c - G_s) / D_o$ (config d)
 $= (2777.6689 - 2781.6689) / 2551.2402 = -0.00157$

Calculate parameter $[\omega_s]$:
 $= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$
 $= 1.0903 \cdot 0 \cdot 0 \cdot 0 (1 + 175.0 \cdot 0)$
 $= 0.0000 \text{ mm}^2$

Calculate parameter $[\omega^*s]$:
 $= ao^2(\rho_s^2 - 1) \cdot (\rho_s - 1) / 4 - \omega_s$
 $= 1275.62^2(1.09^2 - 1) \cdot (1.09 - 1) / 4 - 0$
 $= 6936.9199 \text{ mm}^2$

Parameter $[\omega_C]$:



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$$= \rho c \cdot k c [\delta C (C m \theta + \beta C + C q \Delta \cdot h \cdot \beta C^2) - \zeta c (C m \theta + h \cdot C q \theta + h \cdot \beta C / 2)]$$

$$= 1.089 \cdot 0 [0 (0 + 0 \cdot 0 \cdot 175.0) - 0 (0/175.0 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [$\omega \cdot c$]:

$$= a o^2 [(\rho c^2 + 1) \cdot (\rho c - 1) / 4 - (\rho s - 1) / 2] - \omega c$$

$$= 1275.62012 [(1.08875^2 + 1) \cdot (1.08875 - 1) / 4 - (1.09032 - 1) / 2] - 0$$

$$= 5417.4453 \text{ mm}^2$$

Pressure [$P \cdot S$]:

$$= 0 \text{ For Pressure only cases or Configurations d,e,f,A,B,C,D}$$

Pressure [$P \cdot C$]:

$$= 0 \text{ For Pressure only cases or Configurations b,c,d,B,C,D}$$

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t$$

$$= 2500.0 - 1000.0 = 1500.0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega \cdot s \cdot P_s - \omega \cdot c \cdot P_t) - (\omega \cdot s \cdot P \cdot s - \omega \cdot c \cdot P \cdot c)) + W \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Z_m)$$

$$= [((6936.92 \cdot 2500.0 - 5417.445 \cdot 1000.0) - (0 \cdot 0 - 0 \cdot 0)) \cdot 0. + 1760277 \cdot -0.002 / (2 \cdot \pi)] / (1 + 0.47429 \cdot 0.21186)$$

$$= 705.8749 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q_1 + 2 \cdot Q_2 / (P_e \cdot a o^2)$$

$$= 0.073 + 2 \cdot 705.875 / (1500.0 \cdot 1275.62^2)$$

$$= 0.07819$$

Fm Value from Table 4.18.3 = 0.065823

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a o / (H - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0658 / 0.2034) \cdot (2 \cdot 1275.6201 / (175.0 - 0))^2 \cdot 1500.0$$

$$= 154.7471 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 187.0 = 280.49 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0603 / 0.2034) \cdot (2 \cdot 1275.6201 / (124.389 - 0))^2 \cdot 1500.0$$

$$= 280.4949 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_a t_s + C_a t_c$$

$$= 124.3894 + 0 + 0$$

$$= 124.3894 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 175.0 / 1275.62) \cdot 149.597$$

$$= 8.2092 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ limit



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Tubesheet Shear Stress - final thickness [tf]:

$$= (A_p \cdot P_e) / (\mu \cdot h \cdot C_p)$$

$$= (48355.9 \cdot 1500.0) / (0.2 \cdot 29.827 \cdot 8128.3)$$

$$= 149.5973 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Shear Stress (including CA) [HreqS]:

$$= H + C_{ats} + C_{atc}$$

$$= 29.8273 + 0 + 0$$

$$= 29.8273 \text{ mm}$$

Reqd Tubesheet Thickness for Given Loadings (including CA) [Hreqd]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(124.3894; 29.8273)$$

$$= 124.3894 \text{ mm}$$

ASME Section VIII-2 Part 4.18.9.4 Step 9:

The Ftmin and Ftmax Coefficients from Table 4.18.4:

$$F_{tmin} = -0.883, F_{tmax} = 6.6834$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((2500.0 \cdot 0.4449 - 1000.0 \cdot 0.614) - (1500.0) \cdot -0.883) / (0.614 - 0.4449)$$

$$= 10.7784 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((2500.0 \cdot 0.4449 - 1000.0 \cdot 0.614) - (1500.0) \cdot 6.683) / (0.614 - 0.4449)$$

$$= -56.3345 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 56.334 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:

$$= S_{ot} = 186.9966 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(0.005 + 0.078 \cdot 0.022)6.686^4); 1.25)$$

$$= 1.2500 \text{ (Should be } \leq 2)$$

Determine the factor [rt]:

$$= (\sqrt{d_t^2 + (d_t - 2 \cdot t_t)^2}) / 4$$

$$= (\sqrt{25.4^2 + (25.4 - 2 \cdot 2.11)^2}) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{(2 \cdot \pi^2 \cdot E_t / S_{yt})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19529E+09 / 437)}$$

$$= 93.8884$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 0 \cdot 0 / 8.268$$

$$= 0.0000$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (437 / 1.25) \cdot (1 - 0 / (2 \cdot 93.888))$$

$$= 186.997 \text{ N./mm}^2 \text{ (Never greater than } S_{ot})$$



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Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 56.33 \cdot 1.5438$$

$$= 886.84 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 186.997$$

$$= 2943.791 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 1.651 (25.4 + 0.67 \cdot 1.651) 187.0$$

$$= 2228.2708 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 186.997 / 186.997$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 1.651 (25.4 + 0.67 \cdot 1.651) 187.0$$

$$= 1441.8221 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) \cdot f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1.0 \cdot 1.0)} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 2943.79 is $\geq$ tube strength 2943.79 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Computations completed for ASME tubesheet configuration: d

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	154.7	$\leq 280.5 \text{ N./mm}^2$	Pass
Tubesheet Shear Stress	25.5	$\leq 149.6 \text{ N./mm}^2$	Pass
Maximum Stress in the Tubes	56.3	$\leq 187.0 \text{ N./mm}^2$	Pass
Minimum Tube Stress (Buckling)	-56.3	$\leq -187.0 \text{ N./mm}^2$	Pass
Maximum Force on any one Tube	886.8	$\leq 2943.8 \text{ Kg}$	Pass



Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	124.389	175.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	1.651	Ok

Stationary Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet Stresses				Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	101.462	4.606	106	280	17	150	Ps+Pt-Th D1	Ok
D2uc	166.782	11.515	258	280	42	150	Ps+Pt-Th D2	Ok
D3uc	124.389	6.909	155	280	25	150	Ps+Pt-Th D3	Ok
D1c	101.462	4.606	106	280	17	150	Ps+Pt-Th-c D1	Ok
D2c	166.782	11.515	258	280	42	150	Ps+Pt-Th-c D2	Ok
D3c	124.389	6.909	155	280	25	150	Ps+Pt-Th-c D3	Ok
Max:	166.7824	11.515 mm		0.920		0.284	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses						Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	37	186	-9	-186	576	2943	...	...	...	...	...	...	Ok
D2uc	92	186	-92	-186	1453	2943	...	...	...	...	...	...	Ok
D3uc	56	186	-56	-186	887	2943	...	...	...	...	...	...	Ok
D1c	37	186	-9	-186	576	2943	...	...	...	...	...	...	Ok
D2c	92	186	-92	-186	1453	2943	...	...	...	...	...	...	Ok
D3c	56	186	-56	-186	887	2943	...	...	...	...	...	...	Ok
Max RATIO	0.493			0.493		0.493	...				...		

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
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Tubesheet thickness :	166.782	175.000	Ok
Tube thickness :	0.664	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	1.651	Ok
Tube-tubesheet groove weld leg :	1.333	1.651	Ok

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	2682.15	1.000	2714.85	1.000
Tubesheet Shear Stress	8800.65	1.000	8800.65	1.000
Tube Tensile Stress	5168.65	1.000	5064.76	1.000
Tube Compressive Stress	20961.52	1.000	5064.76	1.000
Tube-Tubesheet Joint load	5168.65	1.000	5064.76	1.000
Tube Pressure Stress	33277.65	1.000	20232.11	1.000
Minimum MAWP	2682.15		2714.85	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	2682.15	1.000	2714.85	1.000
Tubesheet Shear Stress	8800.65	1.000	8800.65	1.000
Tube Tensile Stress	5168.65	1.000	5064.76	1.000
Tube Compressive Stress	20961.52	1.000	5064.76	1.000
Tube-Tubesheet Joint load	5168.65	1.000	5064.76	1.000
Tube Pressure Stress	33277.65	1.000	22169.73	1.000
Minimum MAPnc	2682.15		2714.85	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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Input Echo, Tubesheet Number 1, Description: Floating tubesh

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	2500.00 KPa.
Shell Maximum Operating Pressure	Psox,max	2500.00 KPa.
Shell Minimum Operating Pressure	Psox,min	0.00 KPa.
Shell Thickness	ts	28.0000 mm
Shell Internal Corrosion Allowance	cas	1.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	2720.000 mm
Shell Temperature for Internal Pressure	Ts	230.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: Ch.shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	1000.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	1000.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	3.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	2720.000 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02700
Channel Allowable Stress at Temperature	Sc	138.00 N./mm ²
Channel Allowable Stress at Ambient		138.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	5600
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	5350.00 mm
Straight Tube Length (bet. inner tubsht faces) L		5000.00 mm
Design Temperature of the Tubes		82.00 °C
Tube Material		SA-789
Tube Material UNS Number		S32205
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		187.00 N./mm ²
Tube Allowable Stress At Ambient		187.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	437.34 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	1262.920 mm
Largest Center-to-Center Tube Distance	U1	15.8750 mm



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Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	48355.9 cm ²
Perimeter of Tube Layout	Cp	8128.3 mm

Tubesheet Data:

Tubesheet TYPE: Floating, Tubesheet Gasketed, Conf. B
 Exchanger TYPE: With an Immersed Floating Head, Conf. a

Tubesheet Design Metal Temperature	T	82.00 °C
Tubesheet Material		SA-182 F60
Tubesheet Material UNS Number		S32205
Tubesheet Allowable Stress at Temperature	S	187.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	187.00 N./mm ²
Thickness of Tubesheet	h	175.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm
Tubesheet Outside Diameter	A	2710.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	435.8 cm ²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	0.000 mm
Tube End condition corresponding to Span (l)	k	0.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Junction Stress Reduction option		Perform Plastic Calculation

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel
Flange Outside Diameter	A	2710.000 mm
Flange Inside Diameter	B	2568.000 mm
Flange Face Outside Diameter	Fod	2600.000 mm
Flange Face Inside Diameter	Fid	2568.000 mm
Gasket Outside Diameter	Go	2597.000 mm
Gasket Inside Diameter	Gi	2571.000 mm
Gasket Factor,	m	3.75
Gasket Design Seating Stress	y	52.40 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	4.5000 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	2650.000 mm
Nominal Bolt Diameter	dB	38.1000 mm
Thread Series		UNC Thread Series
Number of Bolts	n	52
Bolt Material		SA-193 B7M
Bolt Material UNS Number		G41400
Bolt Allowable Stress At Temperature	Sb	138.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

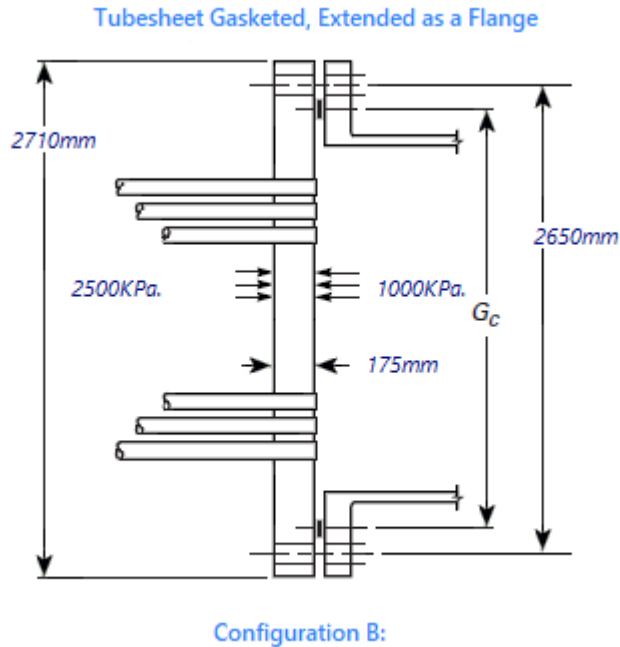
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	159.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes



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Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



Messages regarding the Tubesheet & Exchanger Input Data:

Warning - Tube end condition k should be bet 0 - 1 (.00).

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	230.0 °C	0.19046E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Carbon & Low Alloy Steels, Group 1
Channel - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TE-1 Low Alloy Steels, Group 2 Group 1
Tubes - TM-1 Material Group H

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000125990 /°C
Elastic Mod. at Design Temperature	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20179E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	82.0 °C	0.19529E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19994E+09 KPa.

TubeSheet - TE-1 Low Alloy Steels, Group 2 Group 1



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TubeSheet - TM-1 Material Group H

Elastic Mod. at Design Temperature 82.0 °C 0.19529E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.19994E+09 KPa.

Note:
 The Elasticity and Alpha values are taken from Tables in ASME II D.
 Please insure these properties are consistent with the
 type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (1000.0 \cdot (25.4/2 - 0)) / (187.0 \cdot 1.0 + 0.4 \cdot 1000.0)$
 $= 0.0678 + 0.0000 = 0.0678 \text{ mm}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-5 at 82.00 °C
 Elastic Modulus for Material 190996000.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	5000.00	12.04	50.0000	0.0075908	182.68

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 182.675) / (3 \cdot 12.0379) = 20232.123 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6638	25.40	5000.00	38.26	50.0000	0.0007513	71.75

EMAWP = $(4 \cdot B) / (3 \cdot (D/T)) = (4 \cdot 71.7506) / (3 \cdot 38.2643) = 2500.0347 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6638 mm
Allowable Internal Pressure at Corroded thickness	33277.66 KPa.
Required Internal Design Pressure	1000.00 KPa.
Allowable External Pressure at Corroded thickness	20232.12 KPa.
Required External Design Pressure	2500.00 KPa.
Required Thickness due to Shell Side pressure	0.6638 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	13.000 mm
Basic Gasket Width,	$b0 = N / 2$	6.500 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) \cdot 2.5$	6.422 mm
Gasket Reaction Diameter,	$G = Go - 2.0 \cdot b$	2584.156 mm

Results for ASME Floating Tubesheet Calculations for Configuration B:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.9.4 Step 1:

Tube expansion depth ratio $[\rho]$:
 $= ltx / h$ (modified for corrosion if present)
 $= 0/175.0$
 $= 0.0000$ (must be $0 \leq \rho \leq 1$)



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Effective tube hole diameter [d*]:
 $= \max(dt - 2 \cdot tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$
 $= \max(25.4 - 2 \cdot 2.11 (0.19529E+09/.19529E+09) \cdot$
 $(186/186) \cdot (0); 25.4 - 2 \cdot 2.11)$
 $= 25.4000 \text{ mm}$

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot ro + dt$
 $= 2 \cdot 1262.92 + 25.4$
 $= 2551.240 \text{ mm}$

Determine the basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (31.75 - 25.4) / 31.75$
 $= 0.200$

Equivalent outer tube limit radius [ao]:
 $= Do / 2$
 $= 2551.2402/2$
 $= 1275.6201 \text{ mm}$

Effective tube pitch [p*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) }$
 $= 31.75 / \sqrt{ (1 - 4 \cdot \min(435.77 \cdot 100.0 ; 4 \cdot 2551.24 \cdot 31.75) / (\pi \cdot 2551.24^2)) }$
 $= 31.8862 \text{ mm}$

Effective ligament efficiency for bending [μ *]:
 $= (p* - d*) / p*$
 $= (31.8862 - 25.4) / 31.8862$
 $= 0.2034$

Ratio [ps]:
 $= as / ao$
 $= 1292.0778/1275.6201$
 $= 1.0129$

Ratio [pc]:
 $= ac / ao$
 $= 1292.0778/1275.6201$
 $= 1.0129$

Compute parameter [xt]:
 $= 1 - Nt((dt - 2 \cdot tt) / (2 \cdot ao))^2$
 $= 1 - 5600 ((25.4 - 2 \cdot 2.11) / (2 \cdot 1275.6201))^2$
 $= 0.6140$

Determine parameter [xs]:
 $= 1 - Nt(dt / (2 \cdot ao))^2$
 $= 1 - 5600 (25.4 / (2 \cdot 1275.6201))^2$
 $= 0.4449$

Determine the value [h'g]:
 $= \max((hg - CATC); 0)$ (For pressure only cases)
 $= \max((0 - 0); 0)$
 $= 0.000 \text{ mm}$

ASME Section VIII-2 Part 4.18.9.4 Step 2:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:
 $\beta_s, k_s, \lambda_s, \delta_s$ are all = 0

Intermediate parameters for Tubesheet Gasketed on the Channel Side:
 $\beta_C, k_C, \lambda_C, \delta_C$ are all = 0

ASME Section VIII-2 Part 4.18.9.4 Step 3:



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$h/p = 5.511811$; $\mu^* = 0.203417$; $E^* = 31127258 \text{ KPa}$.
 $E^*/E = 0.159387$; $\nu^* = 0.453616$

Note: As h/p (5.512) is > 2 , data values for $h/p = 2$ were used.

Tube Bundle Stiffness Factor [Xa]:

$$= \left(\left(24(1 - \nu^*)Nt \cdot Et \cdot tt(dt - tt)ao^2 \right) / (E^* \cdot L \cdot H^3) \right)^{1/4}$$

$$= \left(\left(24(1 - 0.454^2)5600 \cdot 0.19529E+09 \cdot 2.11 \right. \right. \\ \left. \left. (25.4 - 2.11)1275.6201^2 \right) / (31127258 \cdot 5000.0 \cdot 175.0^3) \right)^{1/4}$$

$$= 6.6862$$

Values from Table 4.18.3

$Z_d = 0.004955$; $Z_v = 0.022169$; $Z_m = 0.211858$
 $Z_a = 0.198592E+03$; $Z_w = 0.022169$

ASME Section VIII-2 Part 4.18.9.4 Step 4:

Diameter ratio [K]:

$$= A / Do$$

$$= 2710.0/2551.2402$$

$$= 1.0622$$

Compute coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.45) / 31127258 \cdot (0 + 0 +$$

$$0.19529E+09 \cdot \ln(1.06))$$

$$= 0.2069$$

Compute parameter [ϕ]:

$$= (1 + \nu^*)F$$

$$= (1 + 0.4536)0.2069$$

$$= 0.3008$$

Compute parameter [Q1]:

$$= (ps - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.0129 - 1 - 0.3008 \cdot 0.0222) / (1 + 0.3008 \cdot 0.2119)$$

$$= 0.005859$$

ASME Section VIII-2 Part 4.18.9.4 Step 5:

Determine factor [γ_b]:

$$= (G_c - C) / Do \text{ (config b)}$$

$$= (2584.1555 - 2650.0) / 2551.2402 = -0.02581$$

Calculate parameter [ω_s]:

$$= ps \cdot ks \cdot \beta_s \cdot \delta S(1 + h \cdot \beta_s)$$

$$= 1.0129 \cdot 0 \cdot 0 \cdot 0(1 + 175.0 \cdot 0)$$

$$= 0.0000 \text{ mm}^2$$

Calculate parameter [ω^*s]:

$$= ao^2(ps^2 - 1) \cdot (ps - 1) / 4 - \omega_s$$

$$= 1275.62^2(1.013^2 - 1) \cdot (1.013 - 1) / 4 - 0$$

$$= 136.3001 \text{ mm}^2$$

Parameter [ω_c]:

$$= pc \cdot kc[\delta C(CmTheta \cdot \beta C + Cq\Delta \cdot h \cdot \beta C^2) -$$

$$\zeta c(CmTheta + h \cdot CqTheta \cdot h \cdot \beta C / 2))$$

$$= 1.013 \cdot 0 [0(0 + 0 \cdot 0 \cdot 175.0) -$$

$$0(0/175.0 + 0 \cdot 0/2)]$$

$$= 0.0000 \text{ mm}^2$$

Parameter [ω^*c]:

$$= ao^2[(pc^2 + 1) \cdot (pc - 1) / 4 - (ps - 1) / 2] - \omega_c$$

$$= 1275.62012^2[(1.0129^2 + 1) \cdot (1.0129 - 1) / 4 - (1.0129 - 1) / 2] - 0$$



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$$= 136.3000 \text{ mm}^2$$

Pressure [P*S]:
 = 0 For Pressure only cases or Configurations d,e,f,A,B,C,D

Pressure [P*C]:
 = 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.9.4 Step 6:

$$P_e = P_s - P_t \\ = 2500.0 - 1000.0 = 1500.0 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.9.4 Step 7:

Determine factor [Q2]:

$$= [((\omega * s * P_s - \omega * c * P_t) - (\omega s * P * s - \omega c * P * c)) + \\ W * \gamma b / (2 * \pi)] / (1 + \phi * Z_m) \\ = [((136.3 * 2500.0 - 136.3 * 1000.0) - \\ (0 * 0 - 0 * 0)) * 0. + \\ 610868 * -0.026 / (2 * \pi)] / (1 + 0.30082 * 0.21186) \\ = -2339.2737 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q_1 + 2 * Q_2 / (P_e * a o^2) \\ = 0.006 + 2 * -2339.274 / (1500.0 * 1275.62^2) \\ = -0.01294$$

Fm Value from Table 4.18.3 = 0.032337

Tubesheet Bending Stress - original thickness [σ]:

$$= (1.5 * F_m / \mu *) * (2 * a o / (H - h'g))^2 * P_e \\ = (1.5 * 0.0323 / 0.2034) * (2 * 1275.6201 / (175.0 - 0))^2 * 1500.0 \\ = 76.0238 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 * S = 1.5 * 187.0 = 280.49 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 * F_m / \mu *) * (2 * a o / (h - h'g))^2 * P_e \\ = (1.5 * 0.0131 / 0.2034) * (2 * 1275.6201 / (58.063 - 0))^2 * 1500.0 \\ = 280.4759 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc} \\ = 58.0626 + 0 + 0 \\ = 58.0626 \text{ mm}$$

$$H \text{ reqd.} = H_{ReqB} = 58.0626 \text{ mm}$$

Computations completed for ASME tubesheet configuration: B

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	76.0 <=	280.5 N./mm ²	Pass

Floating Tubesheet results per ASME Div. 2 Part 4.18.9 2025

Results for 6 Load Cases:

Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Case Type	Pass/Fail



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D1uc	76.342	49.320	76	280	...	...	Ps+Pt-Th	D1	Ok
D2uc	110.851	...	151	280	...	...	Ps+Pt-Th	D2	Ok
D3uc	58.063	...	76	280	...	...	Ps+Pt-Th	D3	Ok
D1c	76.342	49.320	76	280	...	...	Ps+Pt-Th-c	D1	Ok
D2c	110.851	...	151	280	...	...	Ps+Pt-Th-c	D2	Ok
D3c	58.063	...	76	280	...	...	Ps+Pt-Th-c	D3	Ok

Max: 110.8512 49.320 mm 0.537 ... (Str. Ratio)

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[(+)-Th]:
 With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
 Operating Load Cases using the Maximum and Minimum Operating Pressures and
 Operating Temperatures

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	110.851	175.000	Ok
Tubesheet thickness flanged extension :	49.320	159.000	Ok
Tube thickness :	0.664	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	1.651	Ok
Tube-tubesheet groove weld leg :	0.000	1.651	Ok

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	0 shellside Stress Rat.	Shellside MAWP	0 tubeside Stress Rat.
Tubesheet Bending Stress	3731.96	1.000	4654.54	1.000
Tube Pressure Stress	33277.65	1.000	20232.11	1.000
Tubesheet Extension Stress	1062.90	...	No Calc	No Calc
Minimum MAWP	1062.90		4654.54	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	0 shellside Stress Rat.	Shellside MAPnc	0 tubeside Stress Rat.
Tubesheet Bending Stress	3731.96	1.000	4654.54	1.000
Tube Pressure Stress	33277.65	1.000	22169.73	1.000
Tubesheet Extension Stress	1062.90	...	No Calc	No Calc



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Minimum MAPnc	1062.90		4654.54
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(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

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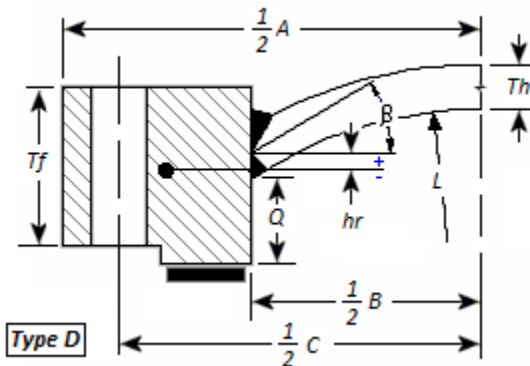
Input Echo, Floating Head Item 1, Description: Operating

Floating Head Type		D	
Tube Side (Internal) Design Pressure	Pts	1000.00	KPa.
Shell Side (External) Design Pressure	Pss	2500.00	KPa.
Design Temperature for Spherical Head	Temp	230.00	°C
Head Material		SA-240	
Head Material UNS Number		S32205	
Head Allowable Stress at Temperature	Sho	171.60	N./mm ²
Head Allowable Stress at Ambient	Sha	187.00	N./mm ²
Crown Radius for Spherical Head	L	2316.000	mm
Head Thickness	Th	50.000	mm
Tube Side (Internal) Corrosion Allowance	Cats	0.0000	mm
Shell Side (External) Corrosion Allowance	Cass	0.0000	mm
Flange Material		SA-182 F60	
Flange Material UNS Number		S32205	
Flange Allowable Stress at Temperature	Sfo	171.60	N./mm ²
Flange Allowable Stress at Ambient	Sfa	187.00	N./mm ²
Flange Outside Diameter	Fod	2710.000	mm
Flange Inside Diameter	Fid	2568.000	mm
Flange Thickness	Tf	280.0000	mm
Bolt Material		SA-193 B7M	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	138.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	138.00	N./mm ²
Diameter of Bolt Circle	C	2650.000	mm
Nominal Bolt Diameter	dB	38.1000	mm
Type of Threads		UNC Thread Series	
Number of Bolts		52	
Full face Gasket Flange Option		Program Selects	
Flange Face Outside Diameter	Ffod	2600.000	mm
Flange Face Inside Diameter	Ffid	2568.000	mm
Gasket Outside Diameter	Go	2597.000	mm
Gasket Inside Diameter	Gi	2571.000	mm
Gasket Factor,	m	3.7500	
Gasket Design Seating Stress	y	52.40	N./mm ²
Flange Facing Sketch		Code Sketch 1a	
Column for Gasket Seating		Code Column I	
Gasket Thickness		4.5000	mm
Width of raised face		0.0000	mm
Dist. from Head Centerline to Flange Centroid hr		2.1689	mm
The Flange is not Slotted.			
This unit does not have a Backing Ring.			
Use Full Bolt Load in Flange Design		Yes	



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Internal Pressure results for Spherical Heads:

ASME Code, Section VIII, Division 1, 2025, App. 1-6

Thickness Due to Internal Pressure[thr]:
 $= 5PL / 6Sho$
 $= (5 \cdot 1000.0 \cdot 2316.0) / (6 \cdot 171)$
 $= 11.2478 \text{ mm}$

Maximum Allowable Working Pressure at Given Thickness[Pa]:
 $= 6Sho \cdot (Th - Cass - Cats) / 5L$
 $= (6 \cdot 171 \cdot 50.0) / (5 \cdot 2316.0)$
 $= 4445.30 \text{ KPa.}$

Maximum Allowable Pressure, New and Cold[Pnc]:
 $= 6Sha / 5L$
 $= (6 \cdot 186 \cdot 50.0) / (5 \cdot 2316.0)$
 $= 4844.19 \text{ KPa.}$

Actual Stress at given Pressure and Thickness[Sact]:
 $= 5PL / 6(Th - Cass - Cats)$
 $= (5 \cdot 1000.0 \cdot 2316.0) / (6 \cdot 50.0)$
 $= 39. \text{ N./mm}^2$

Spherical head Percent Elongation $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$: 1.602 %

External Pressure results, Spherical Head ASME Code, Section VIII Division 1, 2025

External Pressure Chart HA-5 at 230.00 °C
 Elastic Modulus for Material 180805456.00 KPa.

Results for Maximum Allowable External Pressure:

Corroded Thickness of Shell	TCA	50.0000	mm
Outside Crown Radius of Shell	OD/2	2366.0000	mm
Crown Radius / Thickness Ratio	$(OD/2)/T$	47.3200	
Geometry Factor, A	f(D/T, LD)	0.0026416	
Materials Factor, B	f(A, Chart)	123.6042	N./mm ²
Maximum Allowable Working Pressure		2611.94	KPa.
EMAWP = $B / ((OD/T)/2) = 123.6042 / (94.64/2.0) = 2611.9395$			

Results for Req'd Thickness for Ext. Pressure (Tca):

Corroded Thickness of Head	TCA	48.2109	mm
Outside Diameter of Shell	ODCA	4732.000	mm
Diameter / Thickness Ratio	(D/T)	98.1521	
Geometry Factor, A	f(DT, LD)	0.0025471	
Materials Factor, B	f(A, Chart)	122.7097	N./mm ²
Maximum Allowable Working Pressure		2500.25	KPa.
EMAWP = $B / ((D/T)/2) = 122.7097 / (98.1521/2) = 2500.2527$			



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Intermediate Calculations for Flanged portion:

ASME Code, Section VIII, Div. 1, 2025 App. 2

Gasket Contact Width, $N = (Goc - Gic) / 2$ 13.000 mm
 Basic Gasket Width, $b0 = N / 2$ 6.500 mm
 Effective Gasket Width, $b = \text{sqrt}(b0) * 2.5$ 6.422 mm
 Gasket Reaction Diameter, $G = Go - 2.0 * b$ 2584.156 mm

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:
 $= 2a + 6t / (m + 0.5)$
 $= 2 * 38.1 + 6 * 280.0 / (3.75 + 0.5)$
 $= 471.494 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C * \sin(\pi / n)$
 $= 2650.0 * \sin(\pi / 52)$
 $= 160.003 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:
 $= \max(\sqrt{Bs / (2a + t)}, 1)$
 $= \max(\sqrt{160.003 / (2 * 38.1 + 280.0)}, 1)$
 $= 1.0000$

Bolting Information for UNC Thread Series (Non Mandatory):

Total Area of Bolts 434.115 cm²

	Minimum	Actual	Maximum
<i>Radial Distance between Bolts and Edge:</i>	<i>38.100</i>	<i>30.000</i>	
Circ. Spacing between the Bolts:	82.550	160.003	471.494

Results for the Internal Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	534839.8	Kgf
Contact Load on Gasket Surfaces	Hp	39875.9	Kgf
Hydrostatic End Load at Flange ID	Hd	528173.3	Kgf
Pressure Force on Flange Face	Ht	6666.5	Kgf
Radial Component of Head Membrane Force	Hr	804758.4	Kgf
Operating Bolt Load:	Wm1	574715.8	Kgf
Gasket Seating Bolt Load	Wm2	278594.4	Kgf
Required Bolt Area	Am	408.423	cm ²

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:
 $= Ab * Sb / (y * PI * (Go + Gi))$
 $= 434.115 * 138.0 / (52.4 * 3.14 * (2597.0 + 2571.0))$
 $= 7.041 \text{ mm}$

Note: Recommended Min. Width for Sheet and Composite Gaskets per table 2-4 :
 = 38.100 mm [Note: Exceeds actual gasket width, 13.000]

Flange Design Bolt Load (Seating)	W	610868.3	Kgf
Gasket Seating Force (Operating)	Hg	39875.9	Kgf
Distance to Gasket Load Reaction	hg	32.9223	mm
Distance to Face Pressure Reaction	ht	36.9612	mm
Distance to End Pressure Reaction	hd	41.0000	mm

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure,	Md	528173.	41.0000	1.0000
				21656. Kg-m.



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Face Pressure,	Mt	6667.	36.9612	1.0000	246. Kg-m.
Gasket Load,	Mg	39876.	32.9223	1.0000	1313. Kg-m.
Floating Hd. Load,	Mh	804758.	2.1689	1.0000	-1745. Kg-m.
Gasket Seating,	Ma	610868.	32.9223	1.0000	20112. Kg-m.

Total Moment for Operation (Internal Pressure) 21469.322 Kg-m.
 Total Moment for Gasket Seating (Int. Pressure) 20111.596 Kg-m.

Results for the External Pressure case:

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure	H	1337099.5	Kgf
Hydrostatic End Load at Flange ID	Hd	1320433.2	Kgf
Pressure Force on Flange Face	Ht	16666.2	Kgf
Radial Component of Head Membrane Force	Hr	2011896.0	Kgf
Gasket Seating Bolt Load	Wm2	278594.4	Kgf
Required Bolt Area	Am	197.984	cm ²

Flange Design Bolt Load (Seating)	W	610868.3	Kgf
Distance to Gasket Load Reaction	hg	32.9223	mm
Distance to Face Pressure Reaction	ht	36.9612	mm
Distance to End Pressure Reaction	hd	41.0000	mm

Summary of Moments for External Pressure:

Loading	Force	Distance	Bolt Corr	Moment	
End Pressure,	Md	1320433.	8.0777	1.0000	10666. Kg-m.
Face Pressure,	Mt	16666.	4.0389	1.0000	67. Kg-m.
Floating Hd. Load,	Mh	2011896.	2.1689	1.0000	-4364. Kg-m.
Gasket Seating,	Ma	610868.	32.9223	1.0000	20112. Kg-m.

Total Moment for Operation (External Pressure) 6369.947 Kg-m.
 Total Moment for Gasket Seating (Ext. Pressure) 20111.596 Kg-m.

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 1000.0 \cdot 2568.0 \cdot \sqrt{(4 \cdot 2316.0 \cdot 2316.0 - 2568.0 \cdot 2568.0) / (8 \cdot 171.598(2710.0 - 2568.0))}$$

$$= 50.7866 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 21469(2710.0 + 2568.0) / (171.598 \cdot 2568.0)(2710.0 - 2568.0)$$

$$= 17758.6914 \text{ mm}^2$$

Required thickness for Main Flange, Internal Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 50.787 + \sqrt{(50.787 \cdot 50.787 + 17758.691)}$$

$$= 193.3979 \text{ mm}$$

Factor [J]:

$$= Mo(A + B) / (S \cdot B)(A - B)$$

$$= 20111(2710.0 + 2568.0) / (186.997 \cdot 2568.0)(2710.0 - 2568.0)$$

$$= 15265.7705 \text{ mm}^2$$

Required thickness for Main Flange, Internal Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 15265.771)}$$

$$= 123.5547 \text{ mm}$$

Factor for integral type flanges [F]:

$$= P \cdot B \cdot \sqrt{(4 \cdot L \cdot L - B \cdot B) / (8 \cdot S(A - B))}$$

$$= 2500.0 \cdot 2568.0 \cdot \sqrt{(4 \cdot 2316.0 \cdot 2316.0 - 2568.0 \cdot 2568.0) / (8 \cdot 171.598(2710.0 - 2568.0))}$$



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= 126.9666 mm

Factor [J]:

$$= Mo(A + B)/(S \cdot B)(A - B)$$

$$= 6369(2710.0+2568.0)/(171.598 \cdot 2568.0)(2710.0-2568.0)$$

$$= 5269.0015 \text{ mm}^2$$

Required thickness for Main Flange, External Operating Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 126.967 + \sqrt{(126.967 \cdot 126.967 + 5269.001)}$$

$$= 273.2181 \text{ mm}$$

Factor [J]:

$$= Mo(A + B)/(S \cdot B)(A - B)$$

$$= 20111(2710.0+2568.0)/(186.997 \cdot 2568.0)(2710.0-2568.0)$$

$$= 15265.7705 \text{ mm}^2$$

Required thickness for Main Flange, External Bolt-up Condition [T]:

$$= F + \sqrt{(F \cdot F + J)} \text{ per 1-6(g)}$$

$$= 0 + \sqrt{(0 \cdot 0 + 15265.771)}$$

$$= 123.5547 \text{ mm}$$

Summary of Required Thicknesses(mm):

	Head	Flange
Tubeside (Internal) Pressure	11.2478	193.3979
Shellside (External) Pressure	48.2109	273.2181
Tubeside Gasket Seating Load		123.5547
Shellside Gasket Seating Load		123.5547
Maximum + Corrosion Allowance	48.2109	273.2181
Actual Thickness as Given	50.0000	280.0000

Floating Head MAWP and MAPnc (KPa.):

Stress Condition	Tubeside		Shellside	
	MAWP	MAPnc	MAWP	MAPnc
Flange Stress	1696.88	1849.11	2566.51	2796.51
Floating head Stress	4445.20	4844.09	2611.84	3380.60
Bolt Stress	1062.81	1062.81	...	...
Minimum MAWP & MAPnc	1062.81	1062.81	2566.51	2796.51

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: UCS-57 Joint Efficiency Violation!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: